

香港中文大學
The Chinese University of Hong Kong

Formal Languages and Automata Theory

Lecture 7

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Outline



Nondeterministic Pushdown Automata



Pushdown Automata and Context-Free Languages



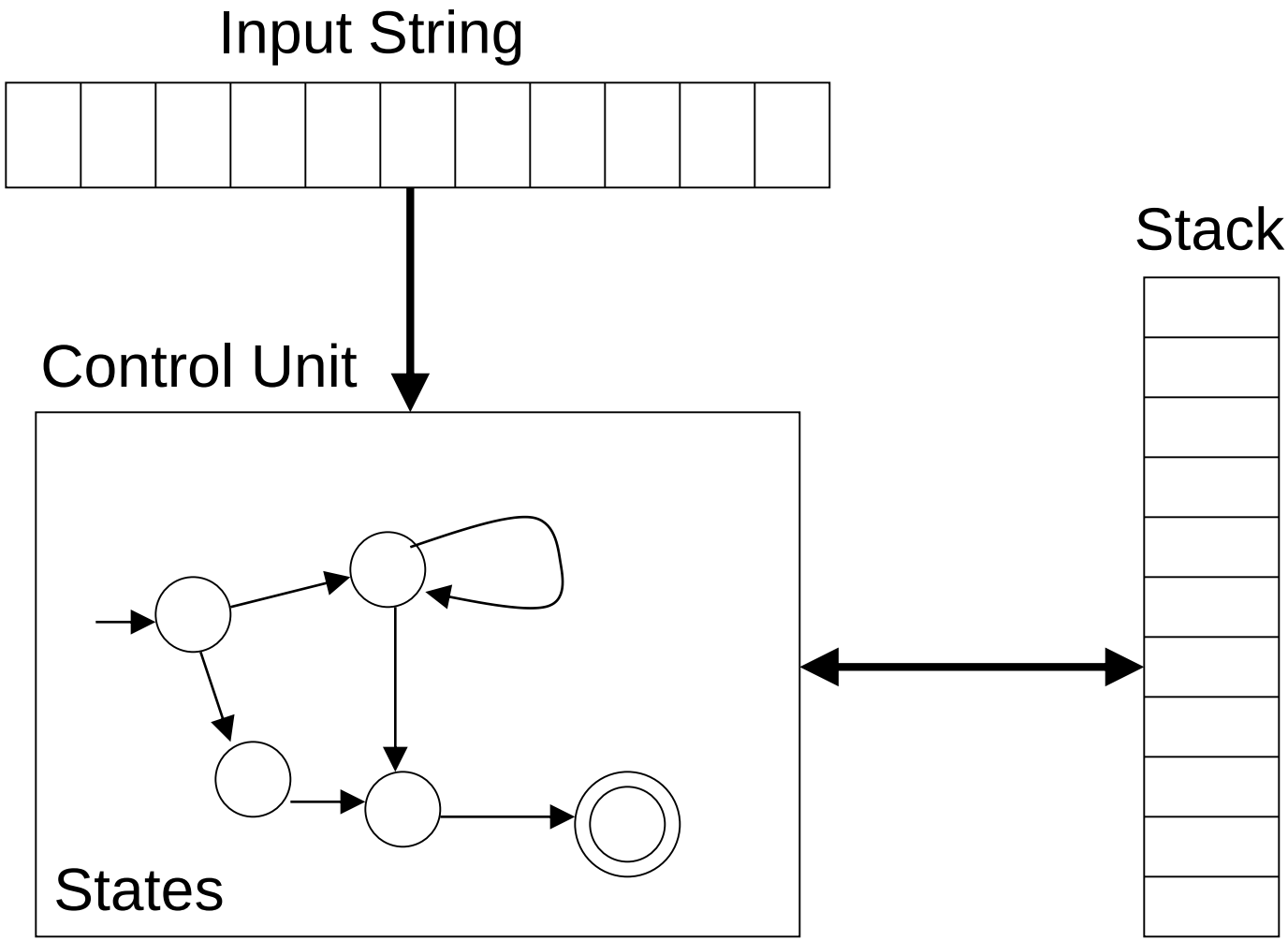
Deterministic Pushdown Automata and Deterministic CFLs

$$L = \{a^n b^n : n \geq 0\}$$

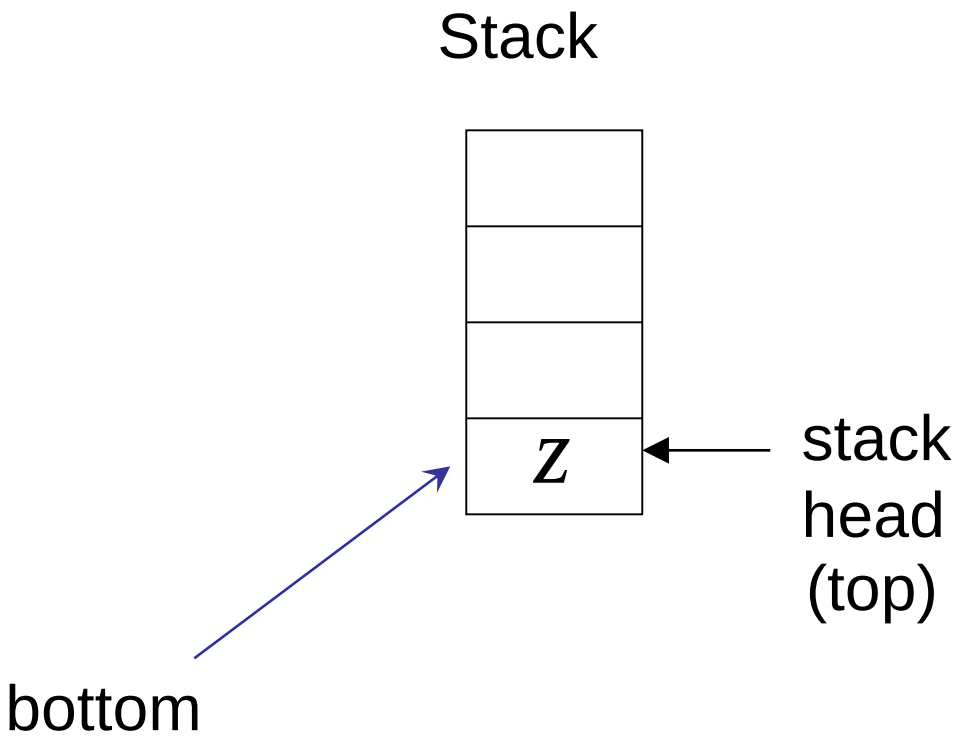
- Check all *a*'s precede the first *b*
- Count the number of *a*'s
- Count without limit
- Stack

PushDown Automata (PDA)

Pushdown Automaton -- PDA



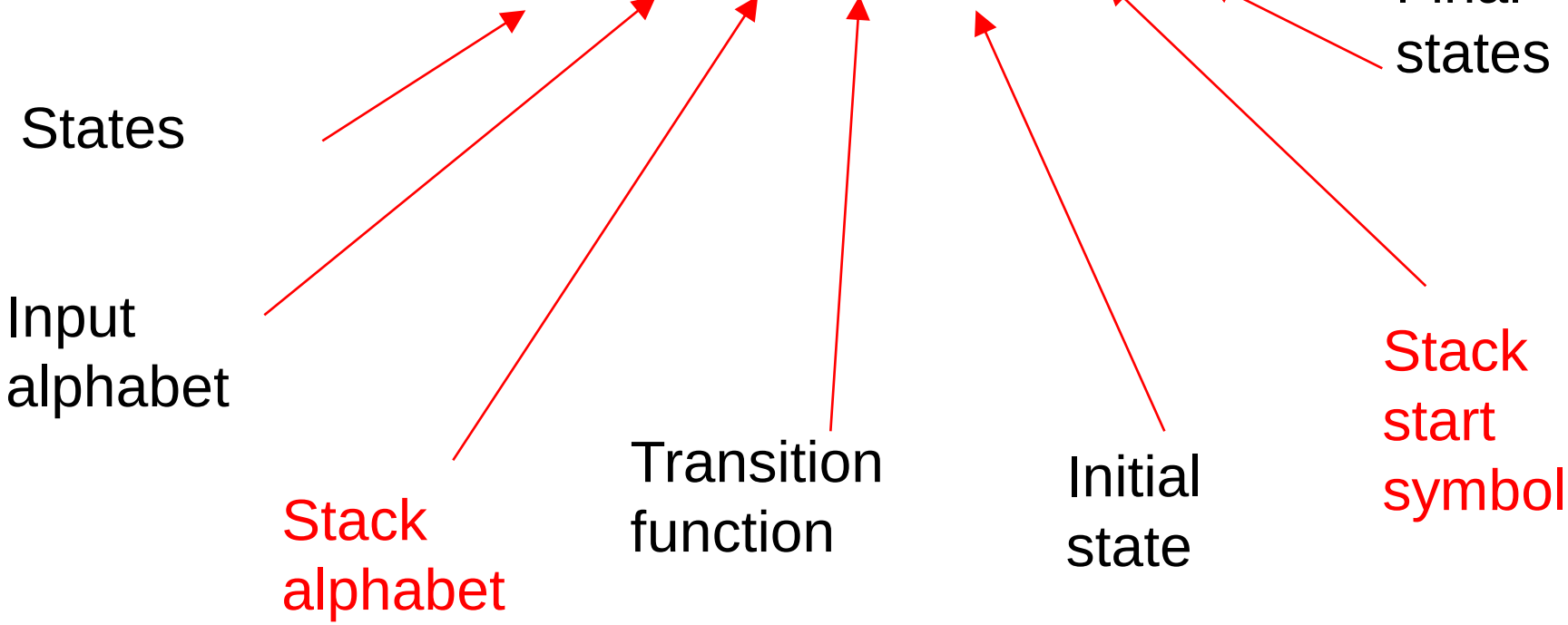
Initial Stack Symbol



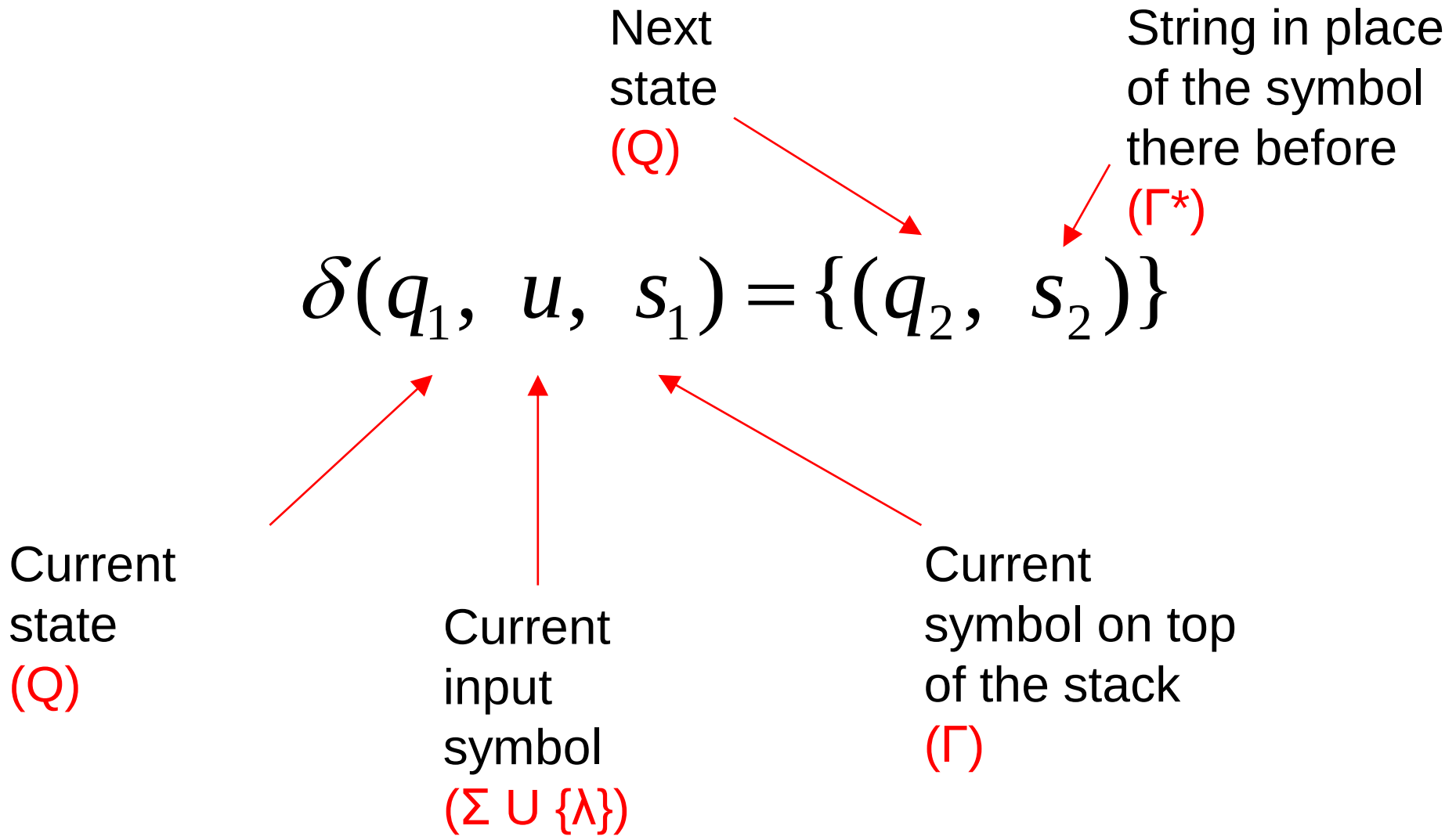
Definition 7.1

Non-Deterministic Pushdown Automaton (NPDA)

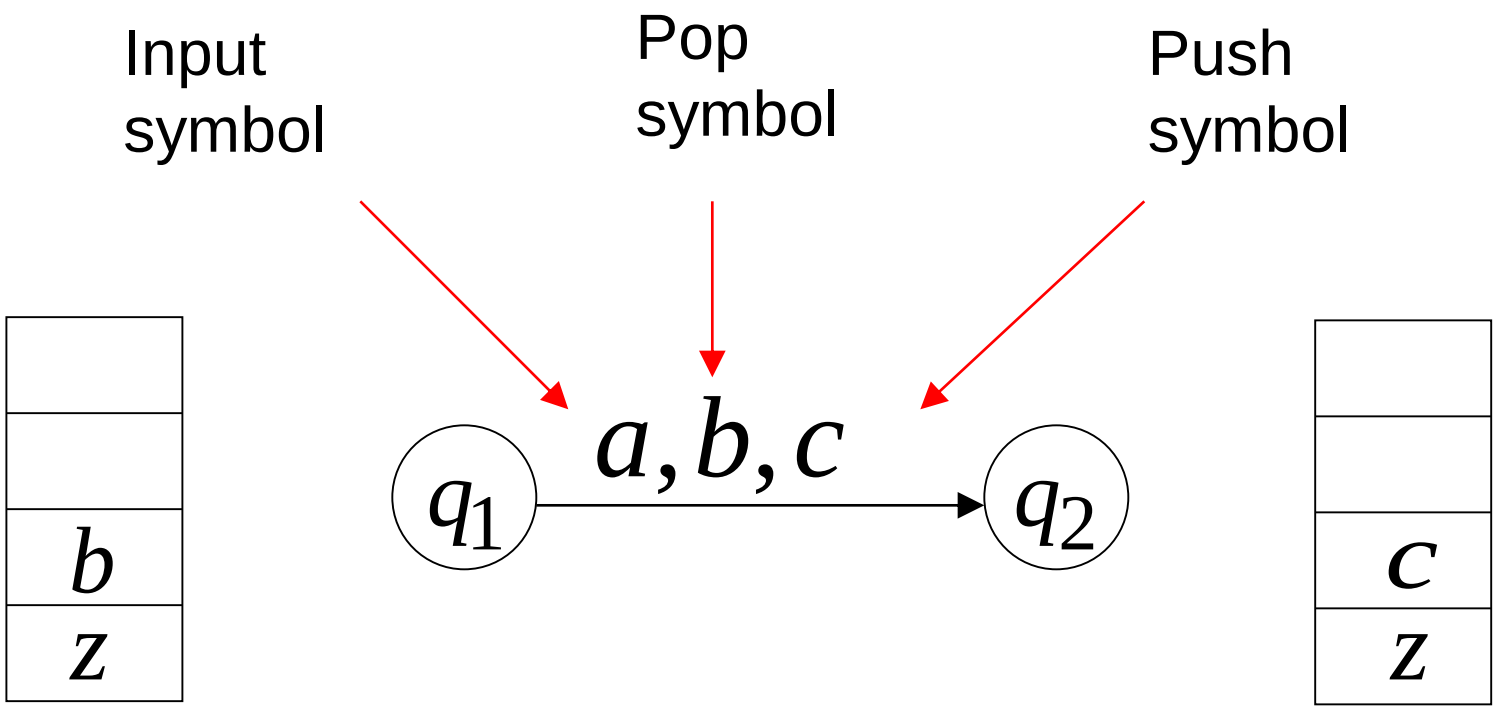
$$M = (Q, \Sigma, \Gamma, \delta, q_0, z, F)$$



Transition Function

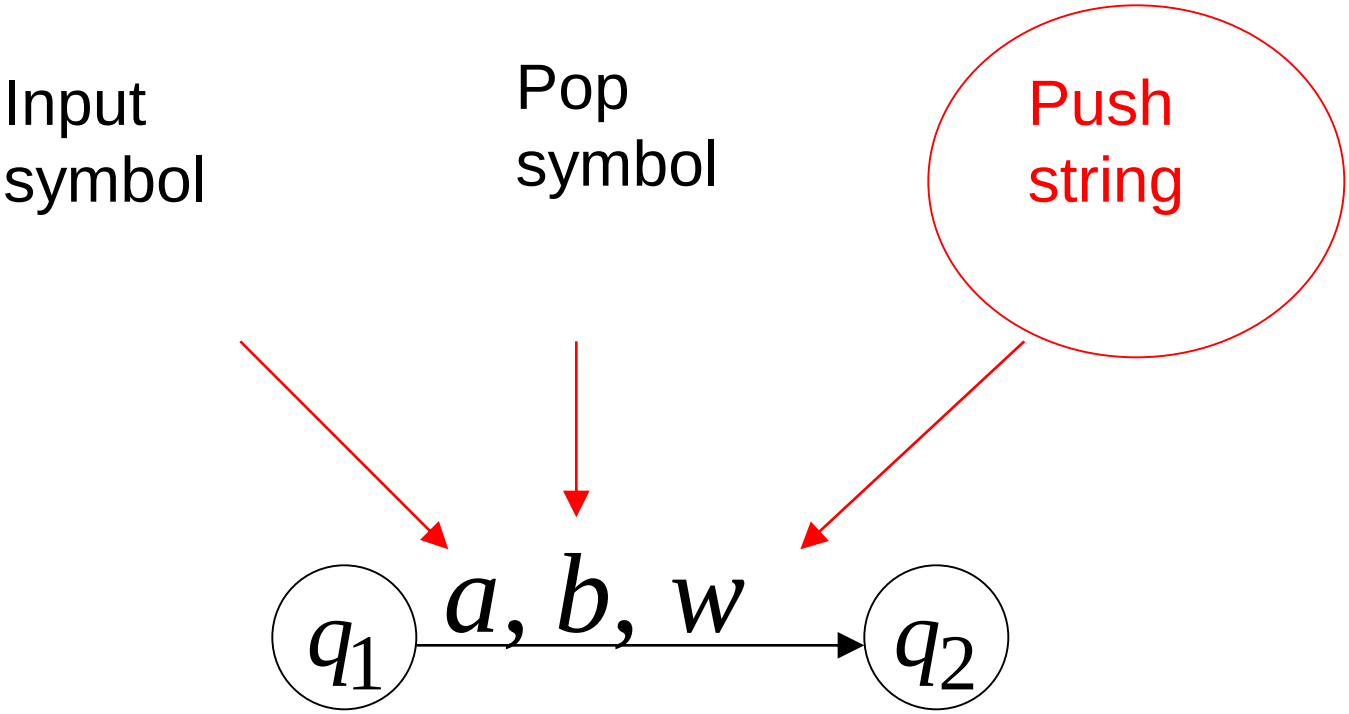


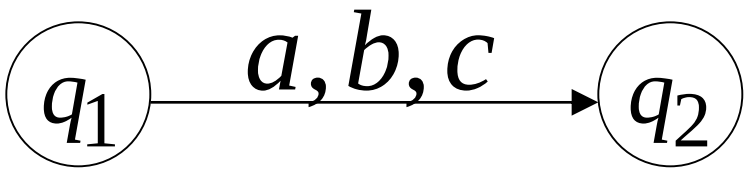
Labels on the Edges of Transition Graphs



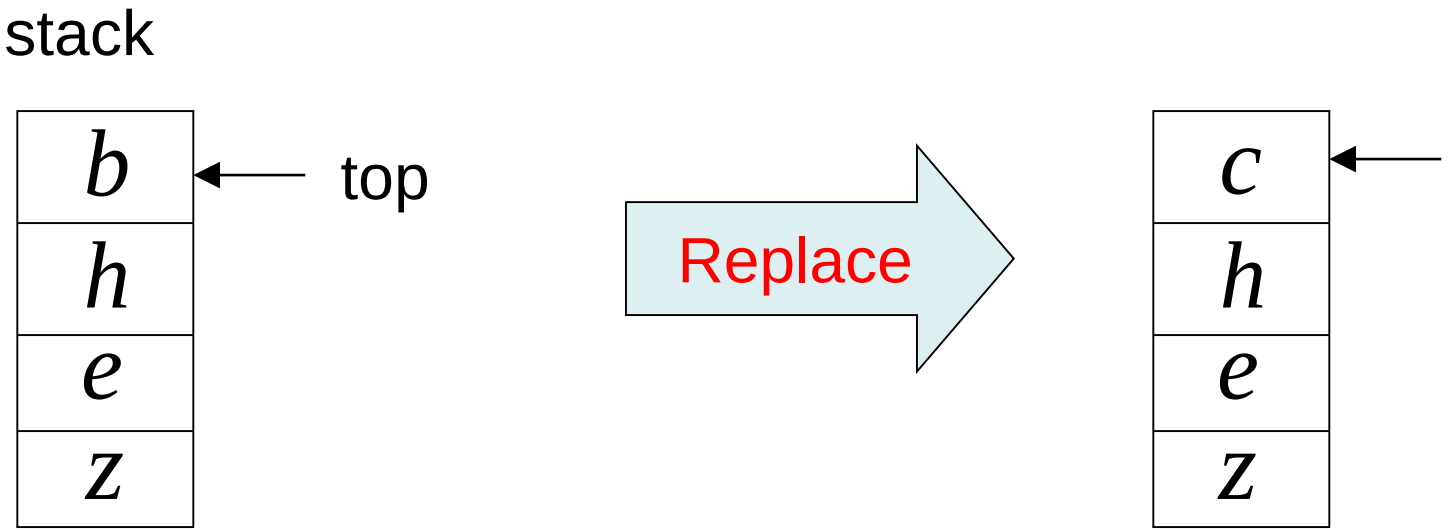
$$\delta(q_1,\ a,\ b) = \{(q_2,\ c)\}$$

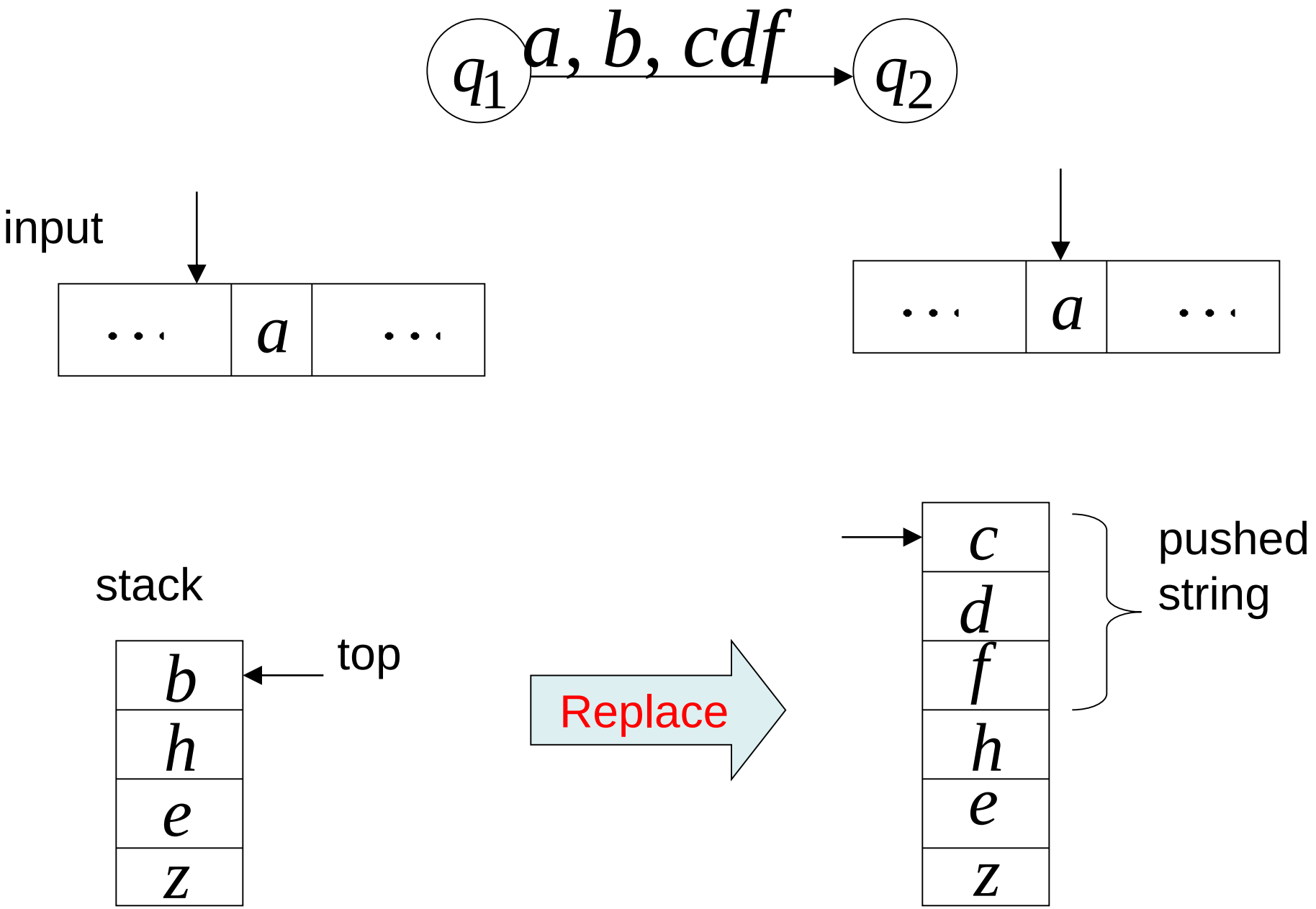
Pushing Strings

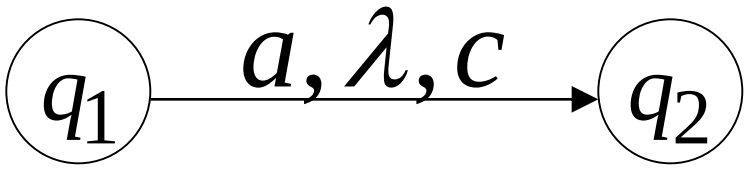



q_1
a, b, c
q_2

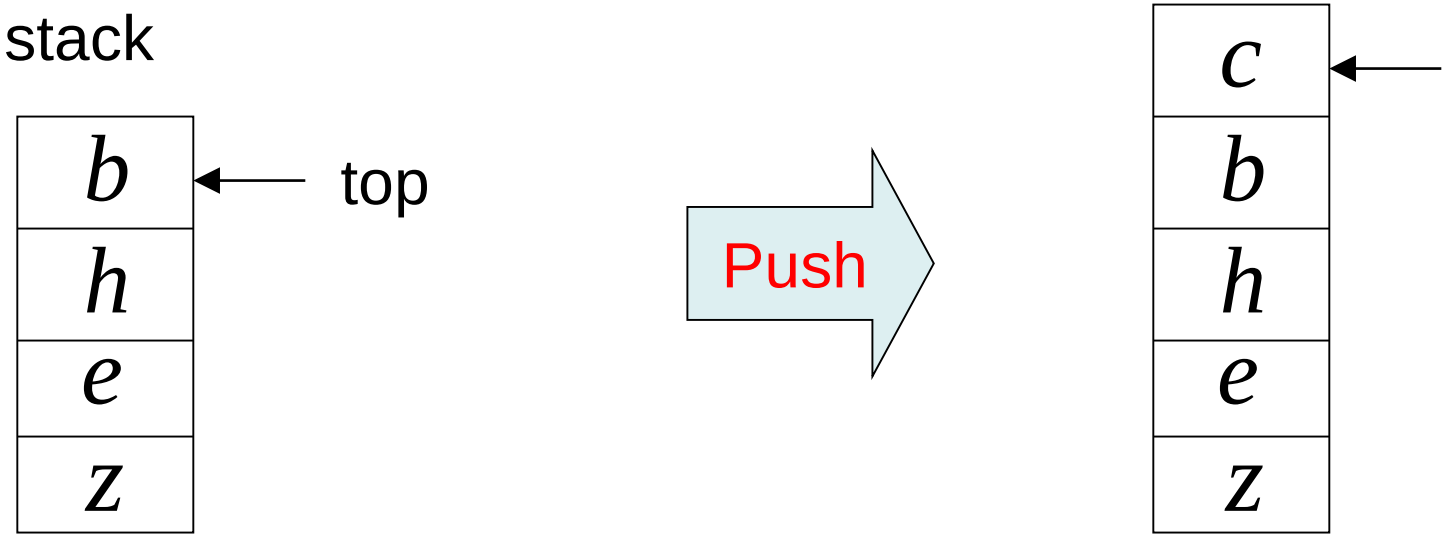

input
…
a
…
…
a
…

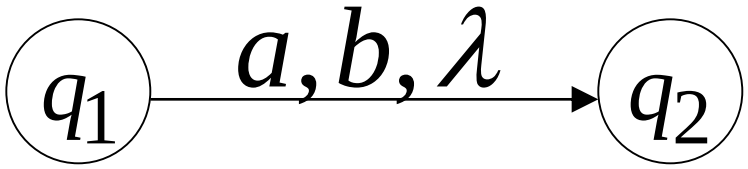

stack
b
top
h
e
z
Replace
c
h
e
z


q_1
a, b, cdf
q_2
input
a
a
stack
top
b
h
e
z
Replace
c
d
f
h
e
z
pushed
string

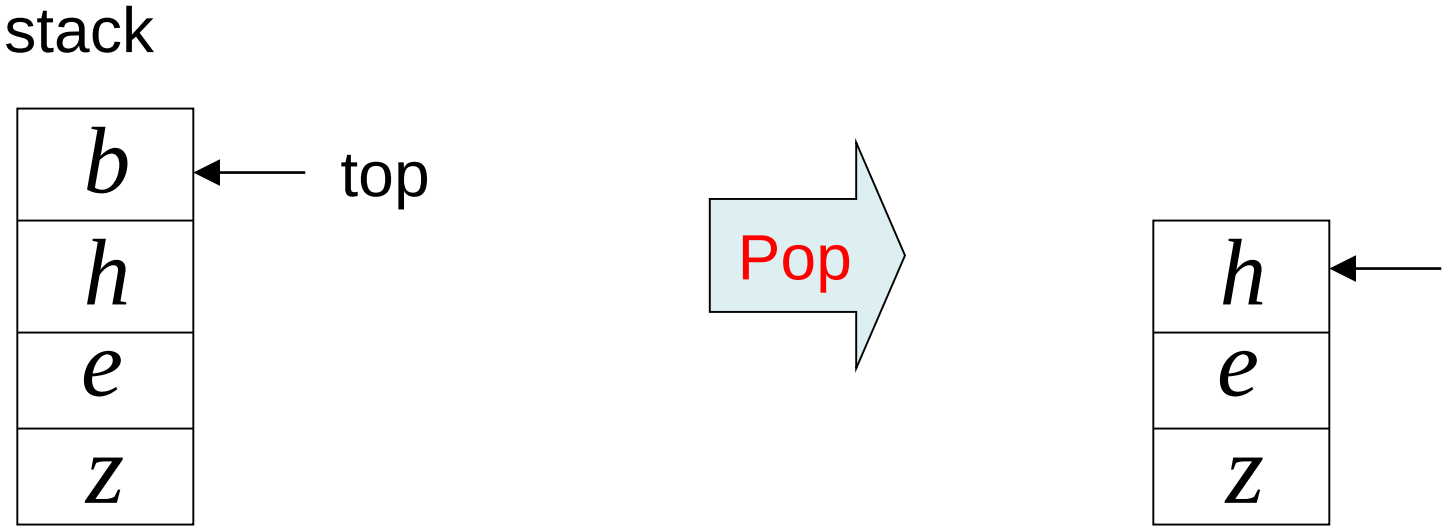

q_1
a, λ, c
q_2

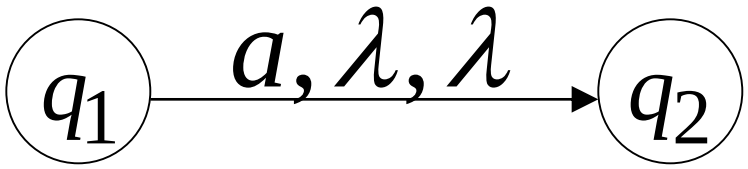

input
. . .
a
. . .
. . .
a
. . .


stack
b
top
h
e
z
Push
c
b
h
e
z

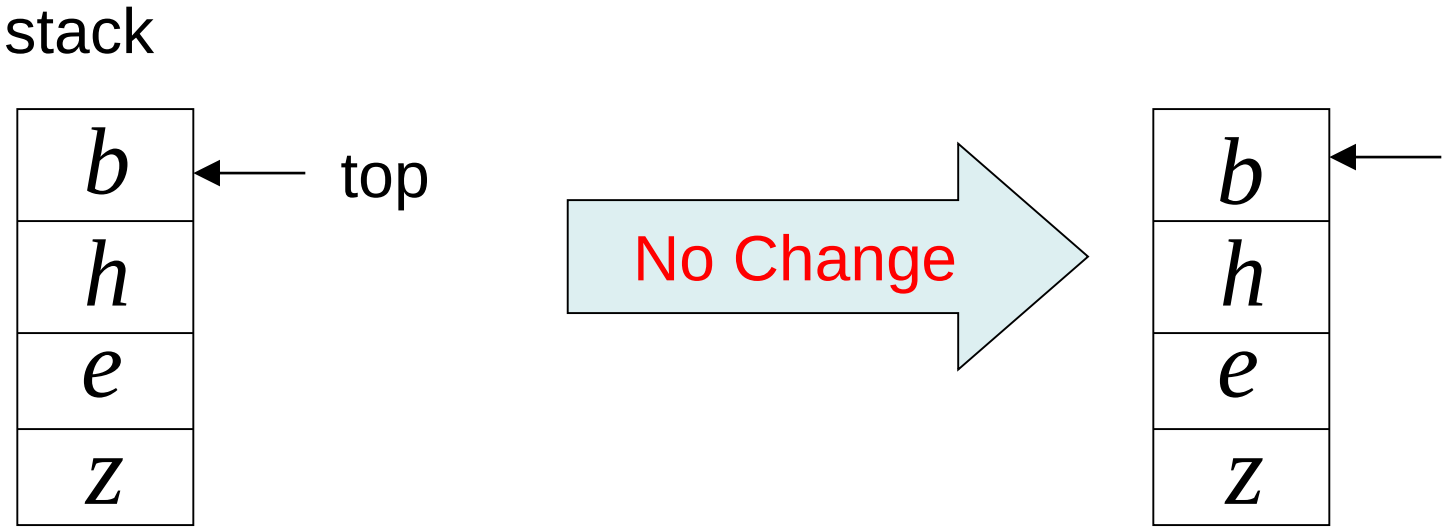

q_1
a, b, λ
q_2


input
…
a
…
…
a
…

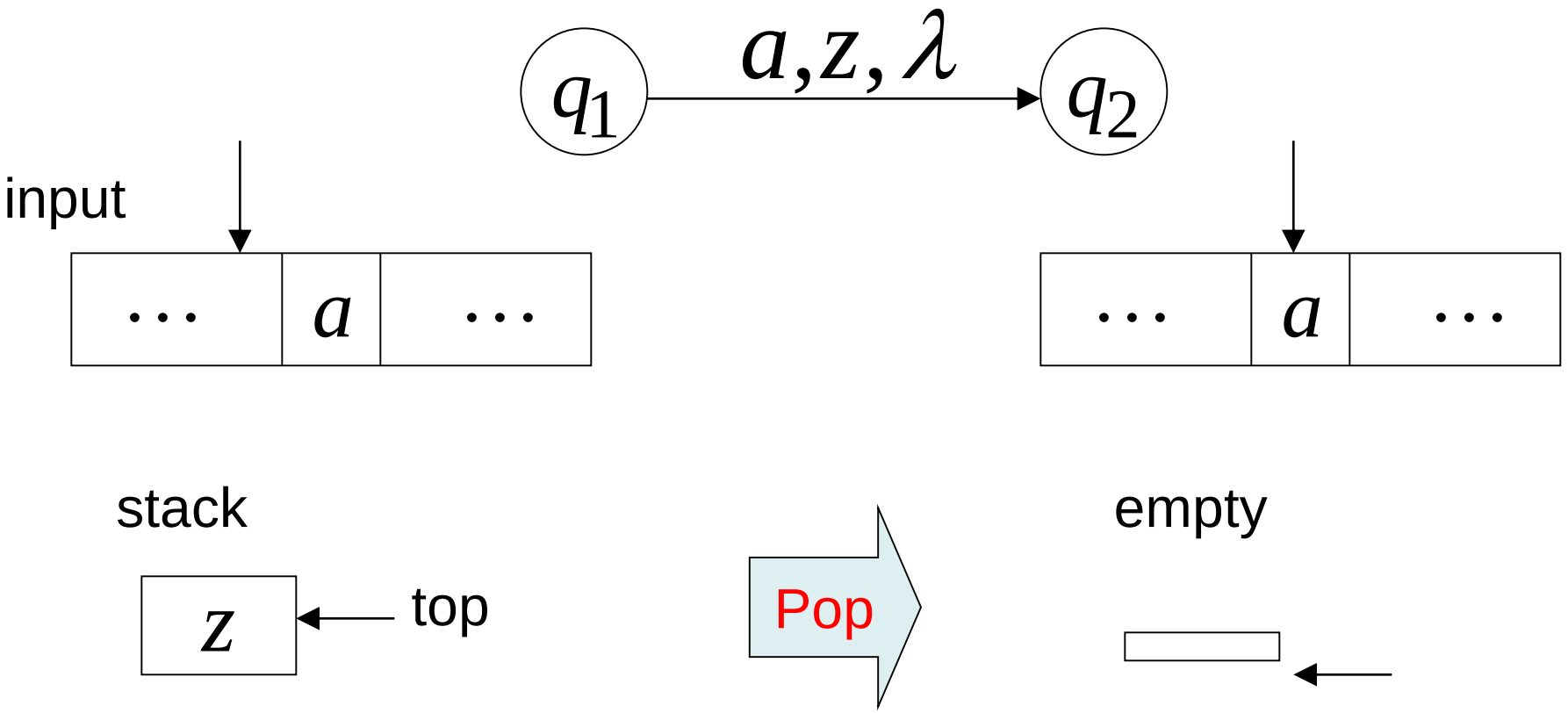

stack
b
top
h
e
z
Pop
h
e
z


q_1
a, λ, λ
q_2

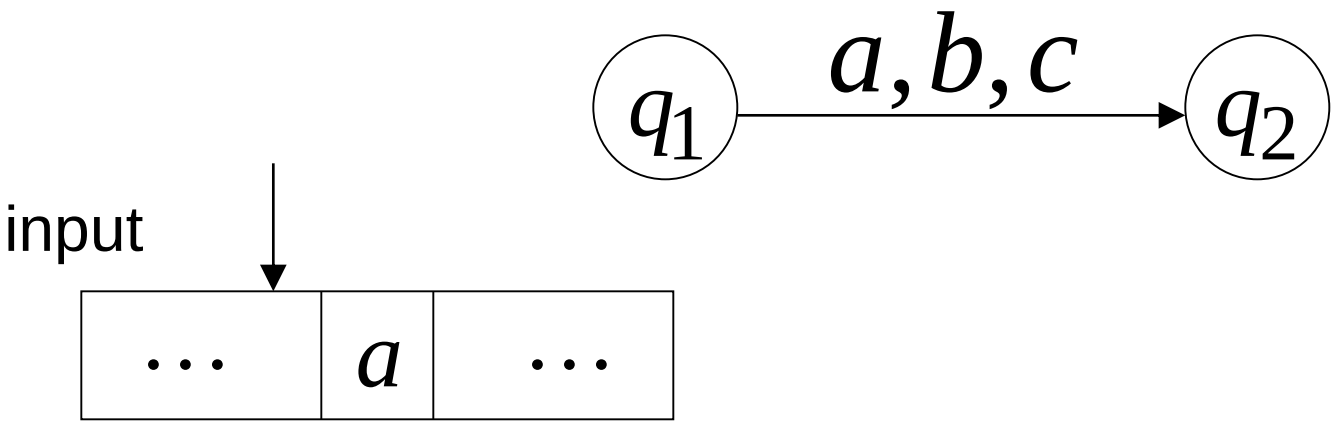

input
…
a
…
…
a
…


stack
b
top
h
e
z
No Change
b
h
e
z

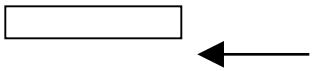
A Possible Transition

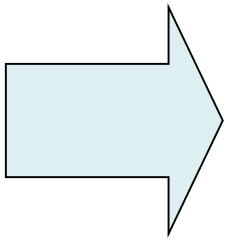


A Bad Transition



Empty stack

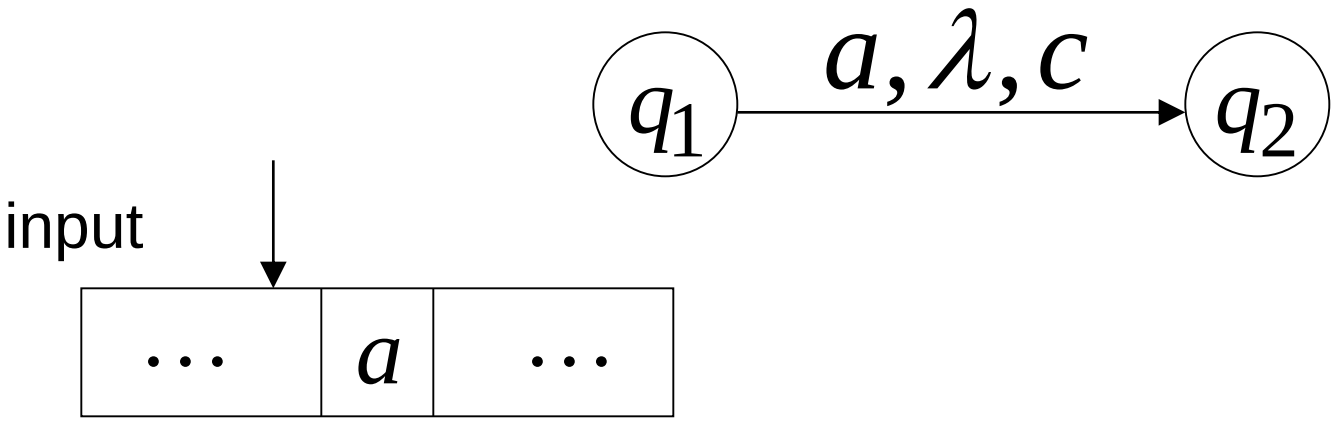




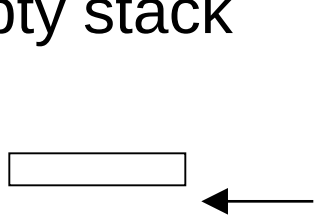
HALT

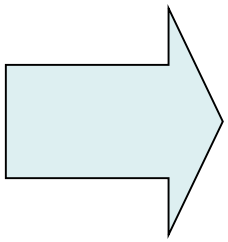
The automaton **Halts** in state q_1 and **Rejects** the input string

A Bad Transition



Empty stack

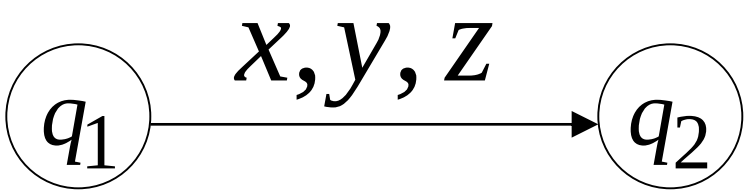




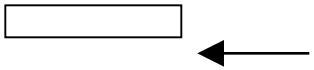
HALT

The automaton **Halts** in state q_1 and **Rejects** the input string

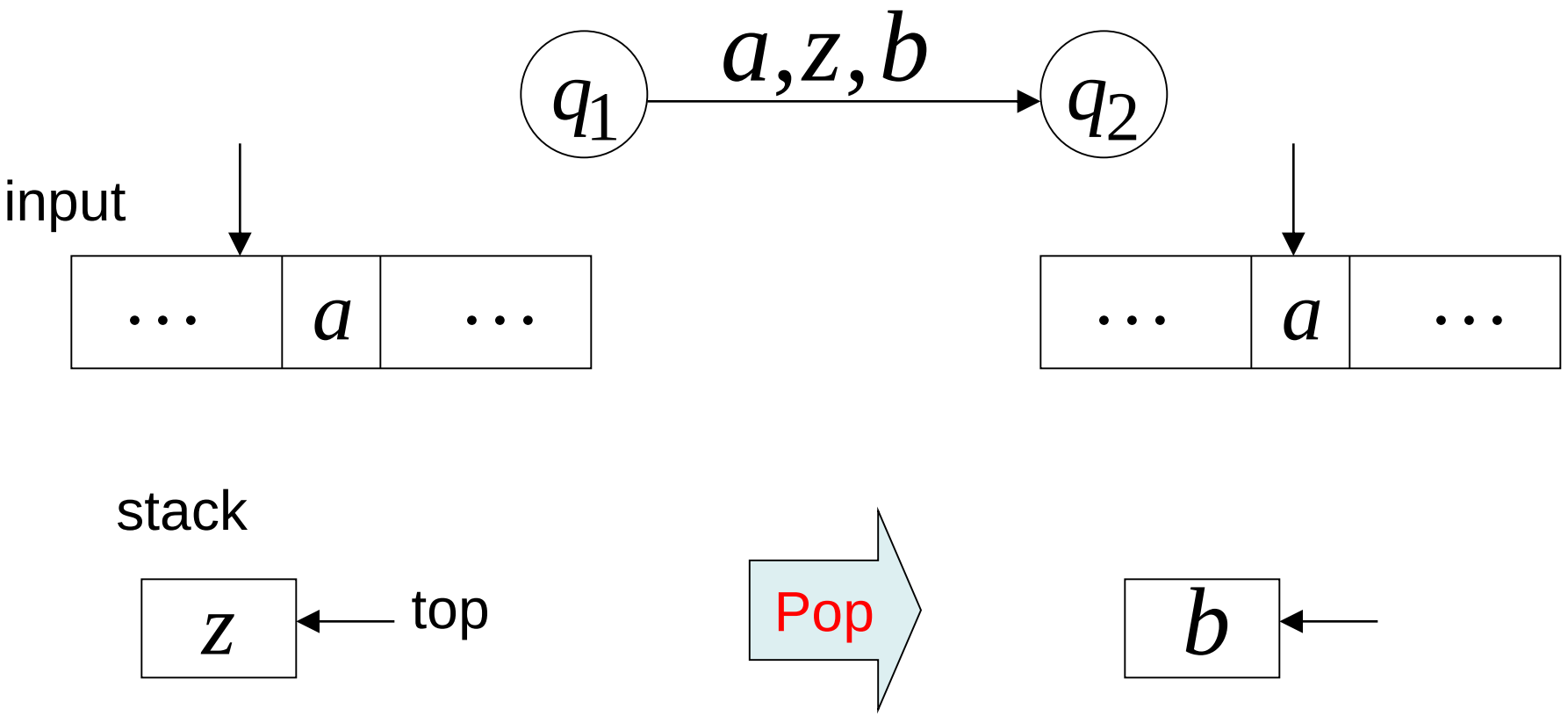
No transition is allowed to be followed
When the stack is empty



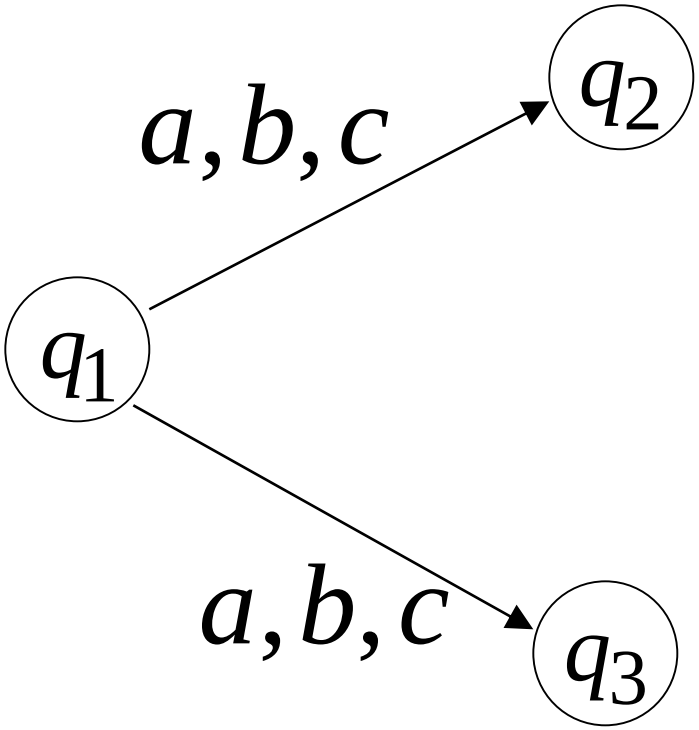
Empty stack

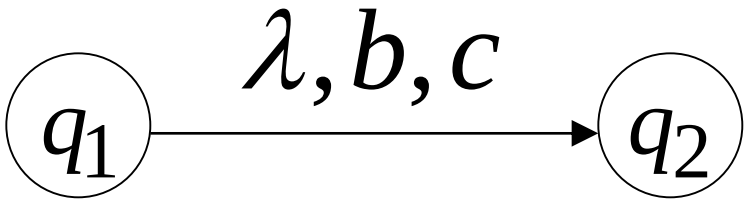


A Good Transition



Non-Determinism



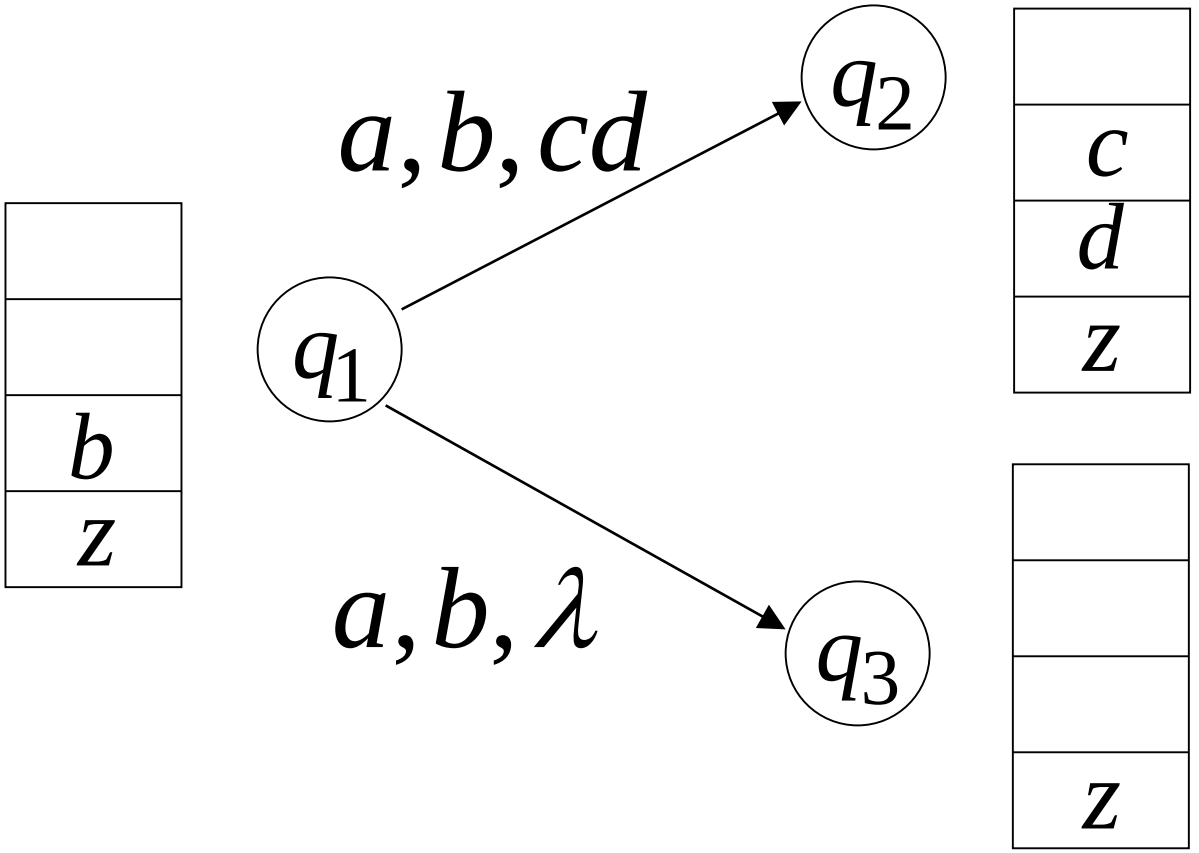


λ – transition

These are allowed transitions in a
Non-deterministic PDA (NPDA)

Example 7.1

$$\delta(q_1, a, b) = \{(q_2, cd), (q_3, \lambda)\},$$



Example 7.2

$$L = \{a^n b^n : n \geq 0\} \cup \{a\}$$

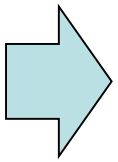
$Q = \{q_0, q_1, q_2, q_3\},$

$\Sigma = \{a, b\},$

$\Gamma = \{0,1\},$

$z = 0,$

$F = \{q_3\}$



$\delta(q_0, a, 0) = \{(q_1, 10), (q_3, \lambda)\}$

$\delta(q_0, \lambda, 0) = \{(q_3, \lambda)\}$

$\delta(q_1, a, 1) = \{(q_1, 11)\}$

$\delta(q_1, b, 1) = \{(q_2, \lambda)\}$

$\delta(q_2, b, 1) = \{(q_2, \lambda)\}$

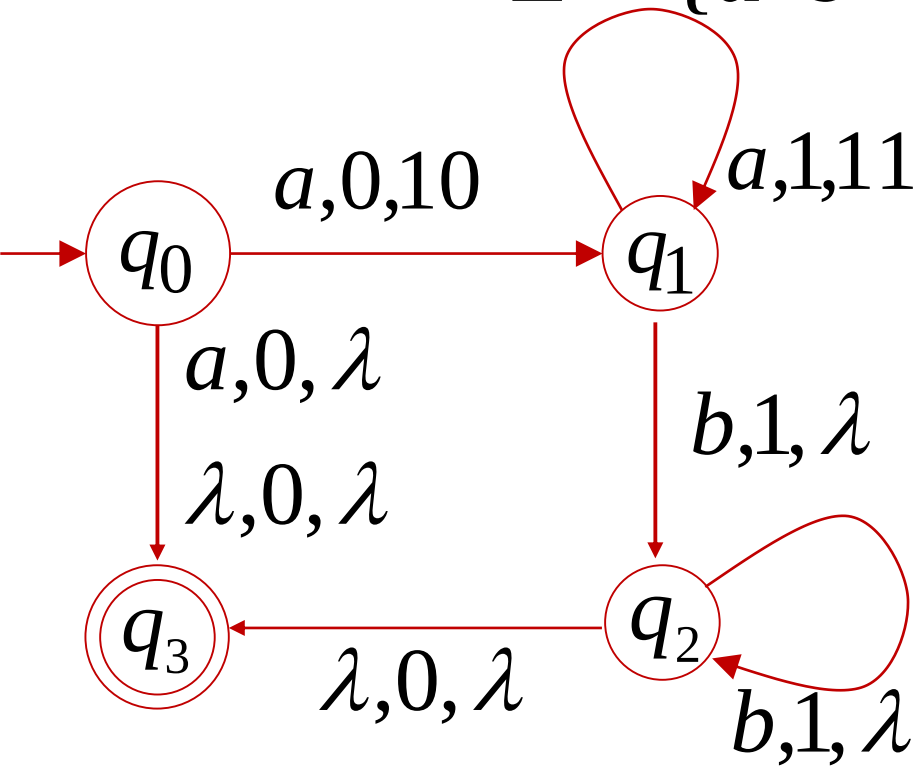
$\delta(q_2, \lambda, 0) = \{(q_3, \lambda)\}$

$\delta(q_0, b, 0)?$

Nondeterministic

Example 7.3

$$L = \{a^n b^n : n \geq 0\} \cup \{a\}$$



$$\delta(q_0, a, 0) = \{(q_1, 10), (q_3, \lambda)\}$$

$$\delta(q_0, \lambda, 0) = \{(q_3, \lambda)\}$$

$$\delta(q_1, a, 1) = \{(q_1, 11)\}$$

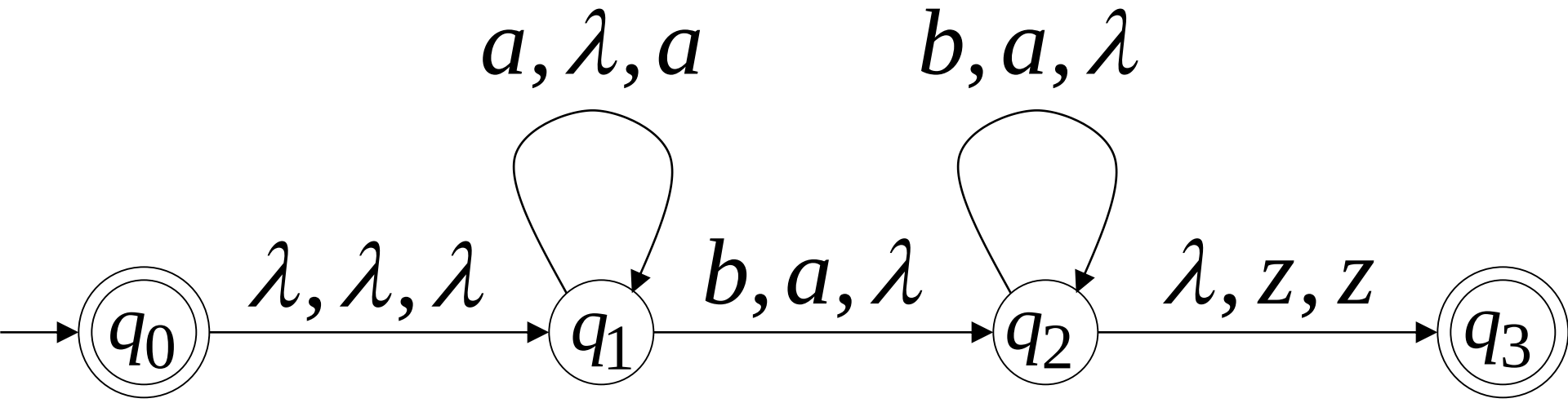
$$\delta(q_1, b, 1) = \{(q_2, \lambda)\}$$

$$\delta(q_2, b, 1) = \{(q_2, \lambda)\}$$

$$\delta(q_2, \lambda, 0) = \{(q_3, \lambda)\}$$

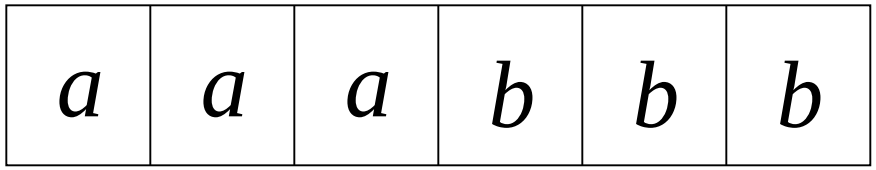
Another Example

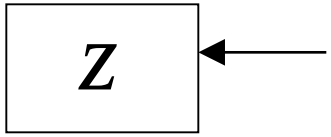
Example: $L = \{a^n b^n : n \geq 0\}$



Execution Example: Time 0

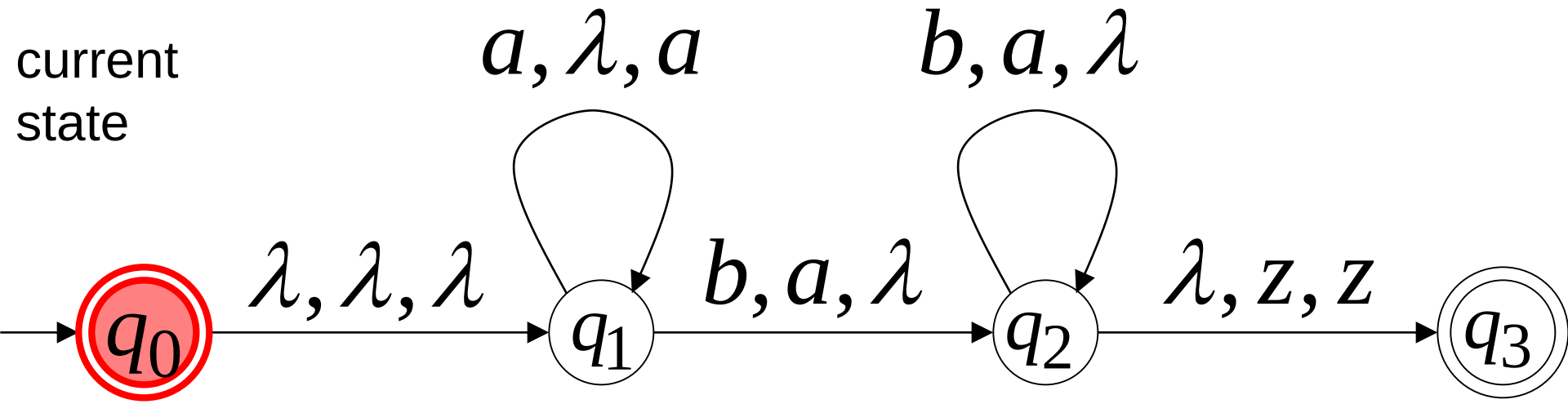
Input





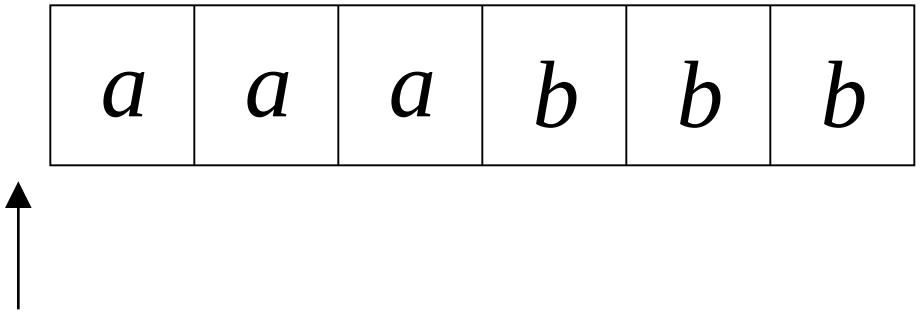
Stack

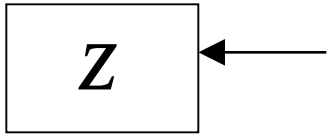
current state



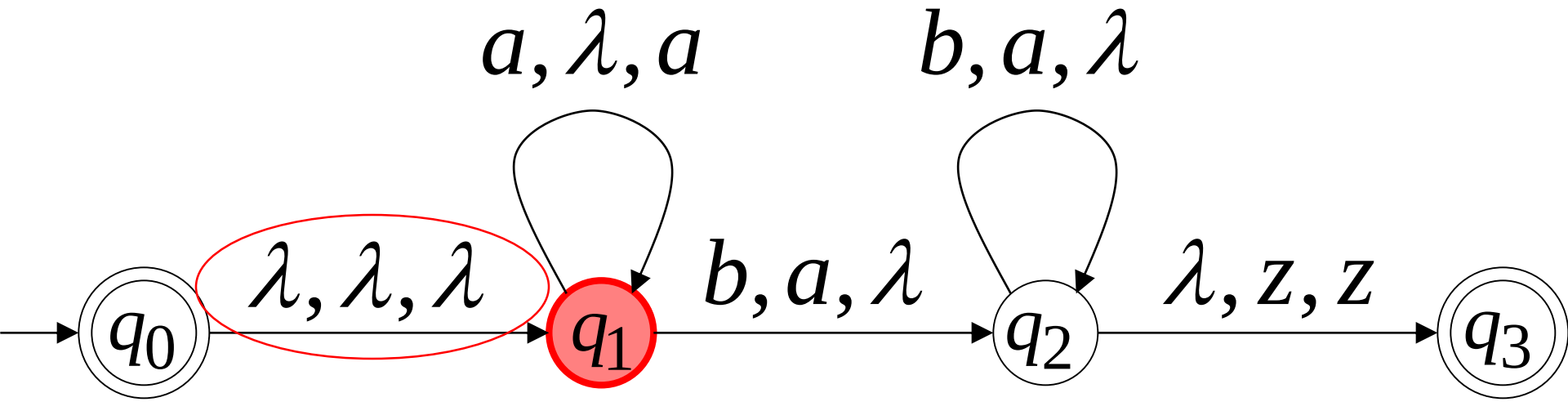
Time 1

Input



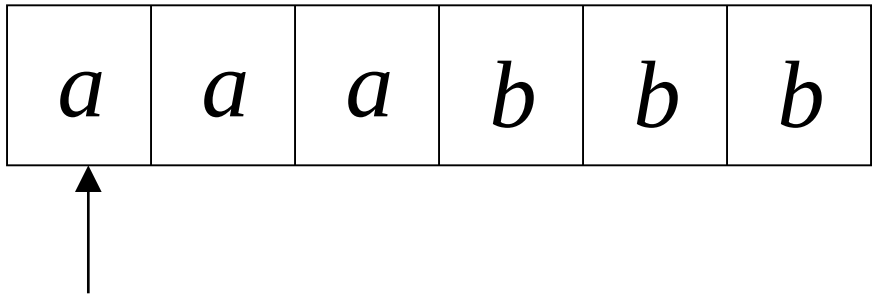


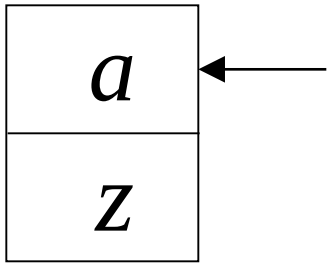
Stack



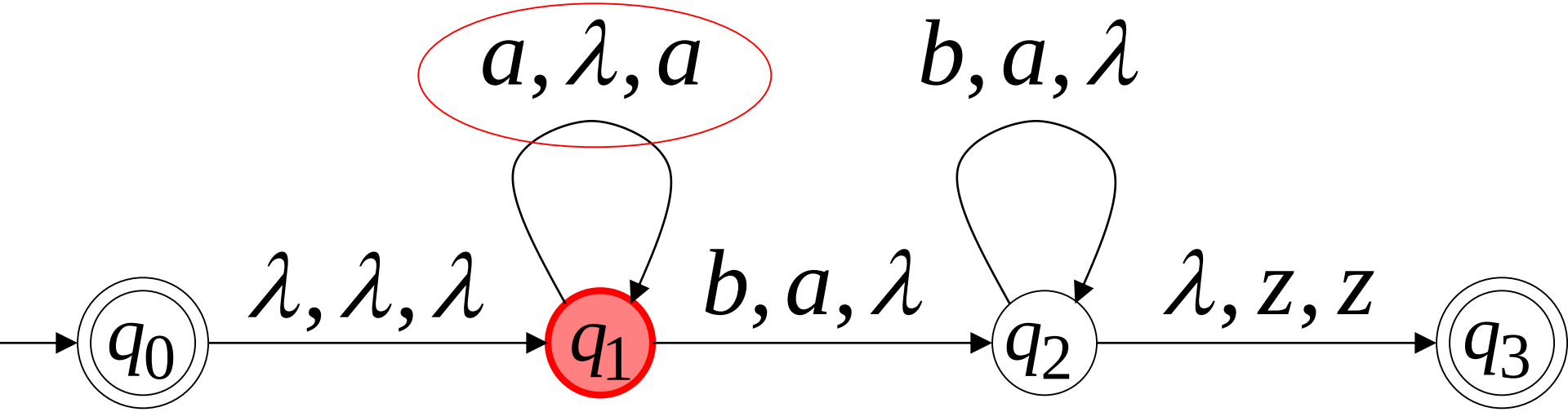
Time 2

Input



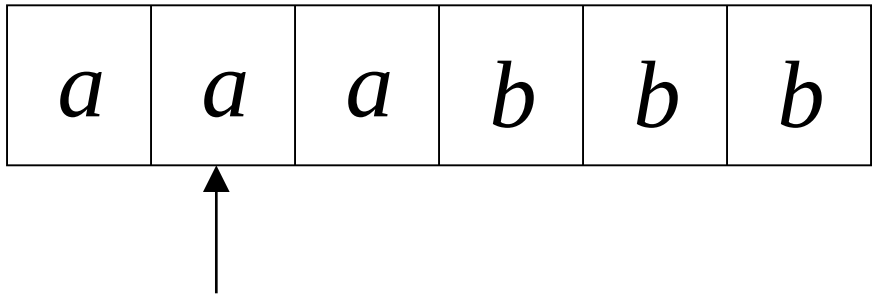


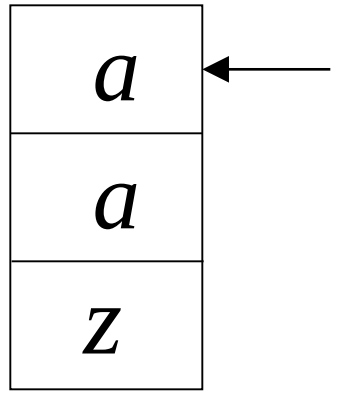
Stack



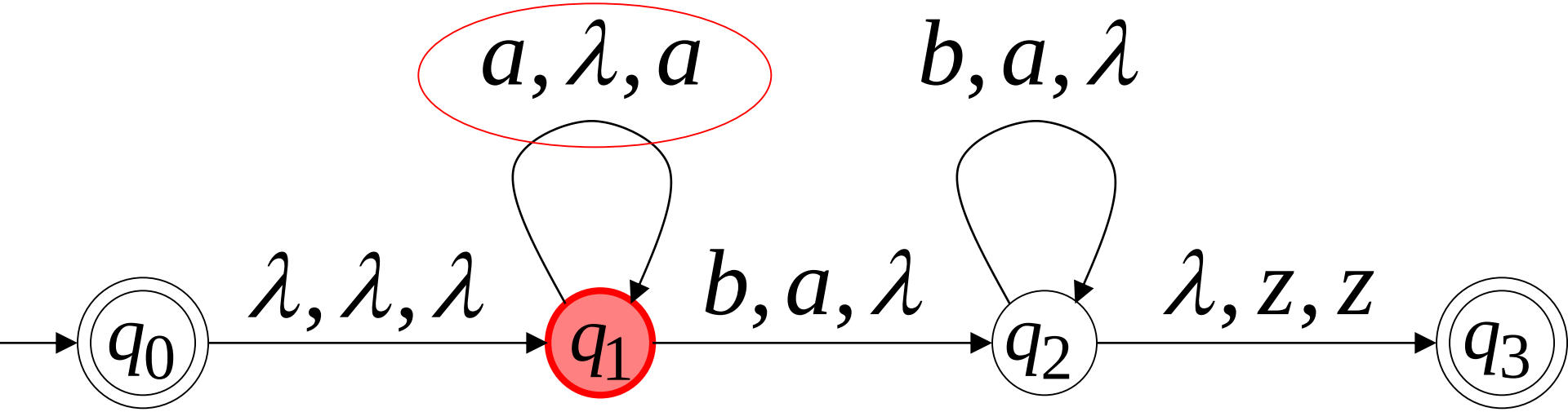
Time 3

Input



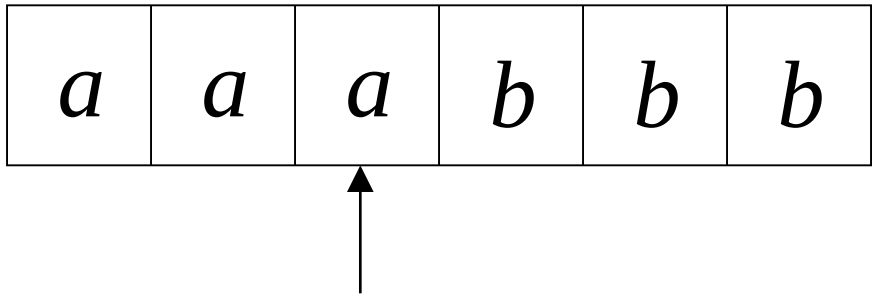


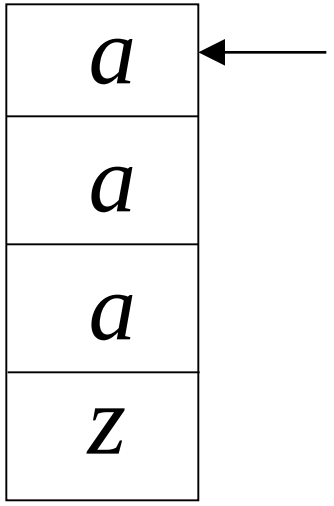
Stack



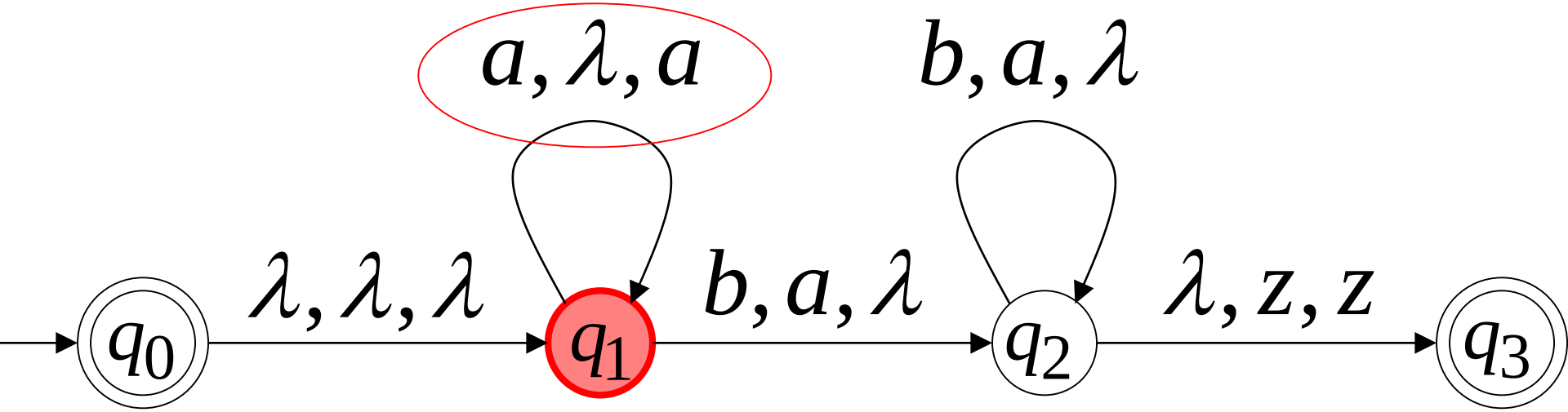
Time 4

Input



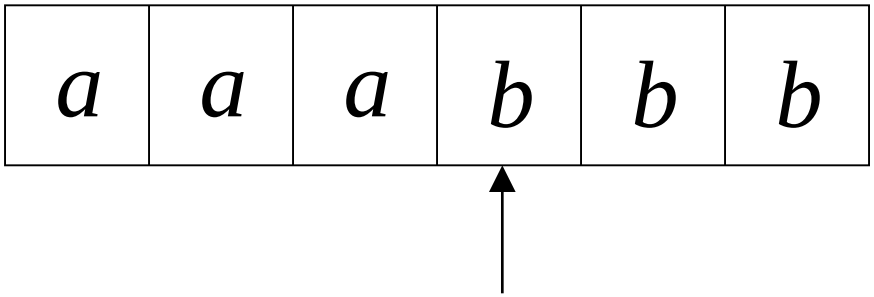


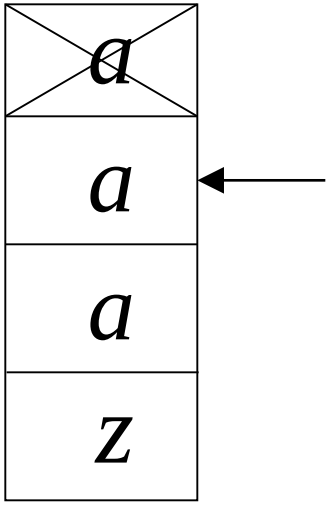
Stack



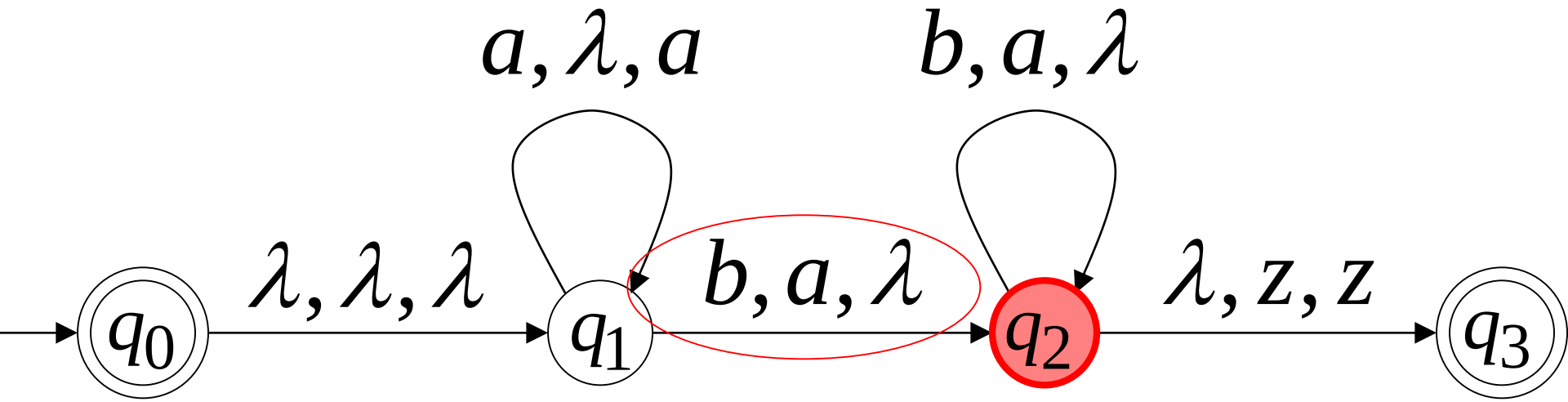
Time 5

Input



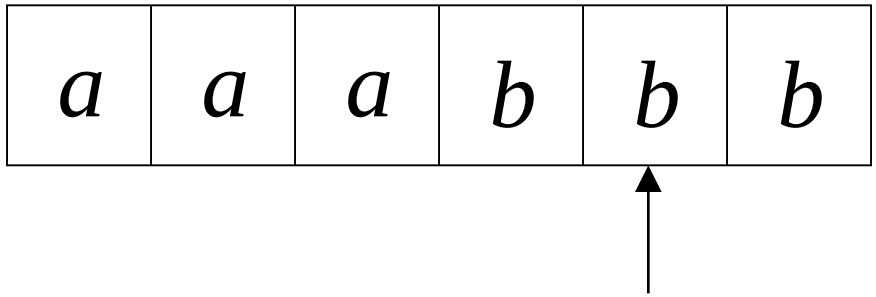


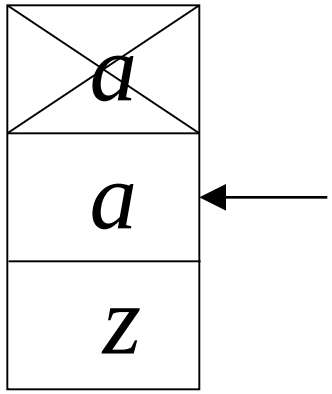
Stack



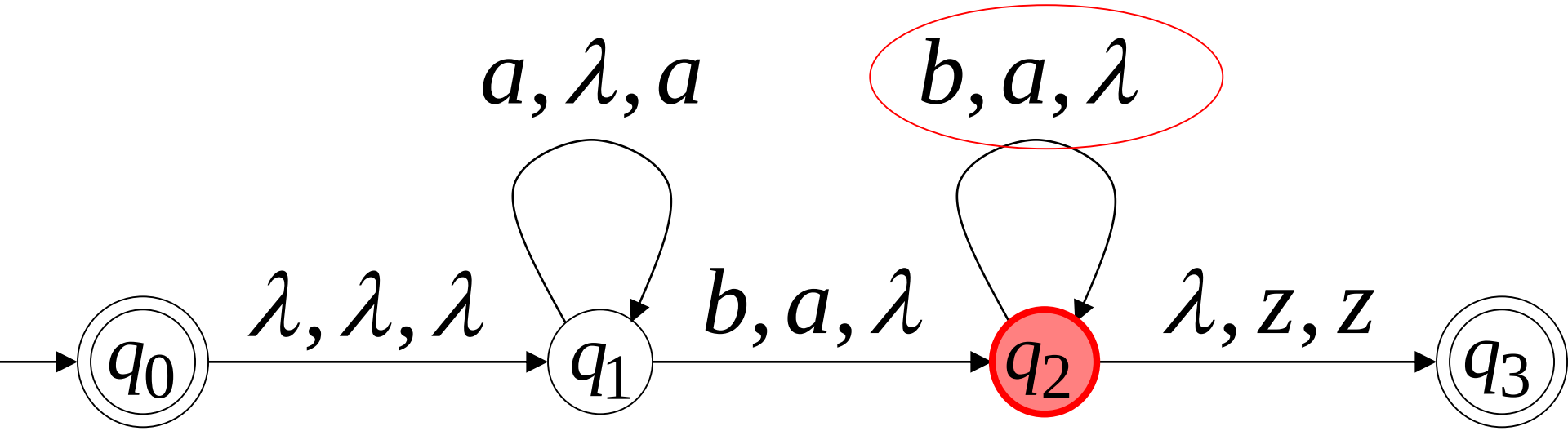
Time 6

Input



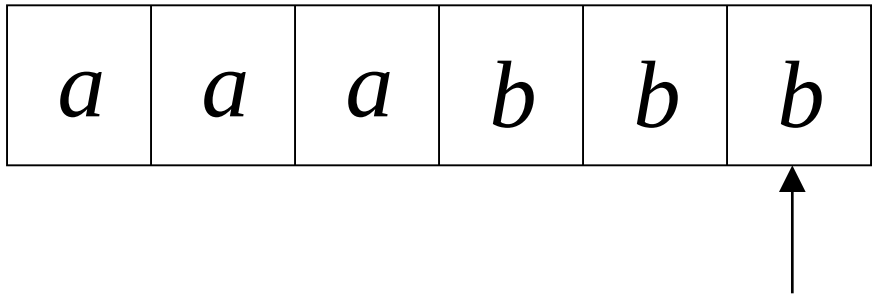


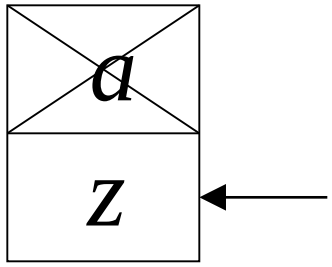
Stack



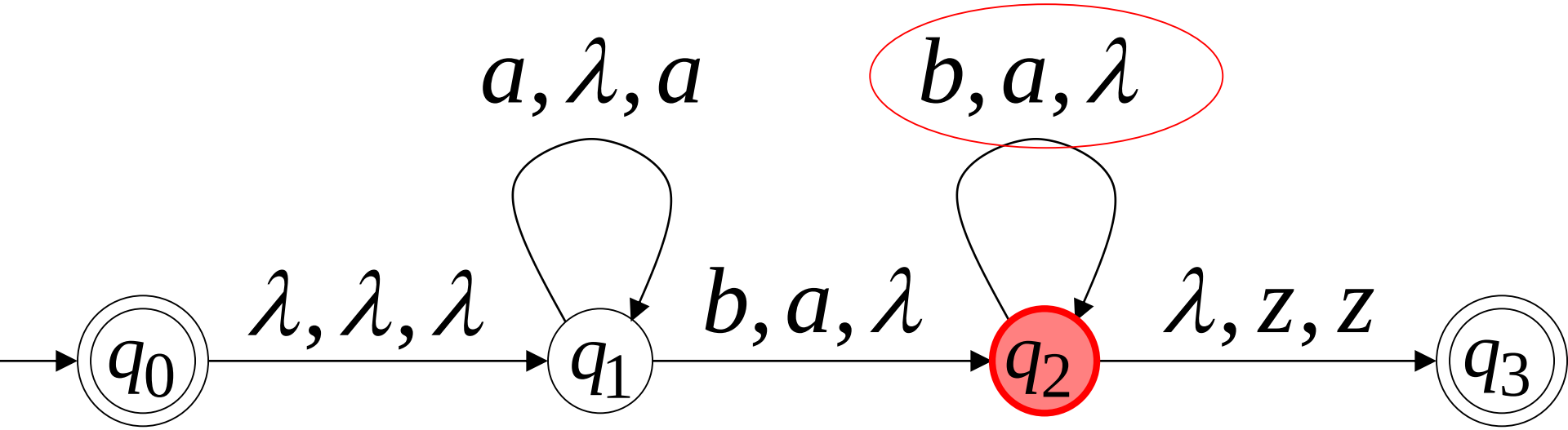
Time 7

Input



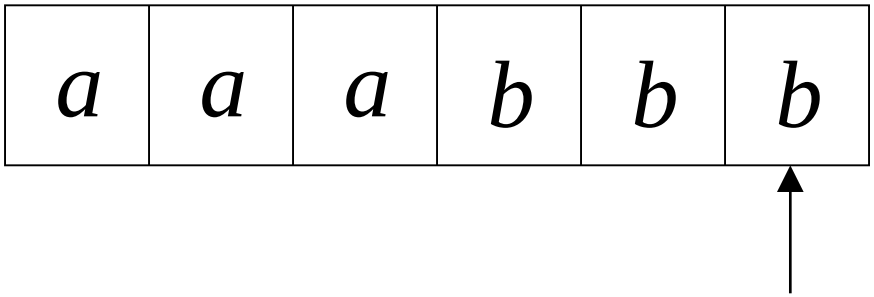


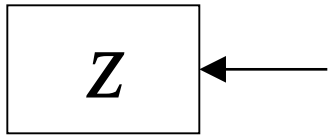
Stack



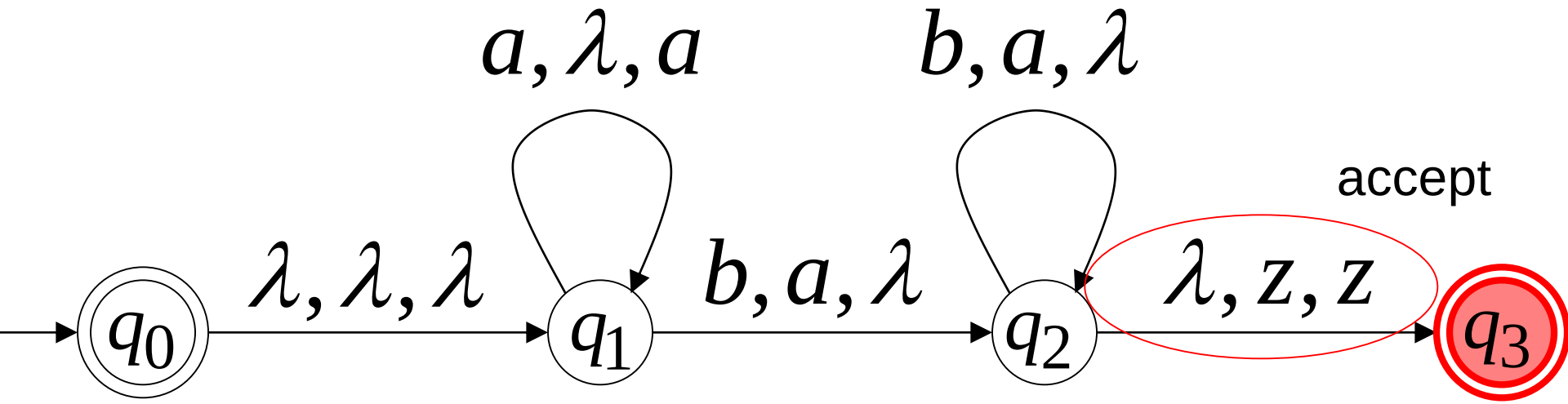
Time 8

Input





Stack



A string is accepted if there is a computation such that:

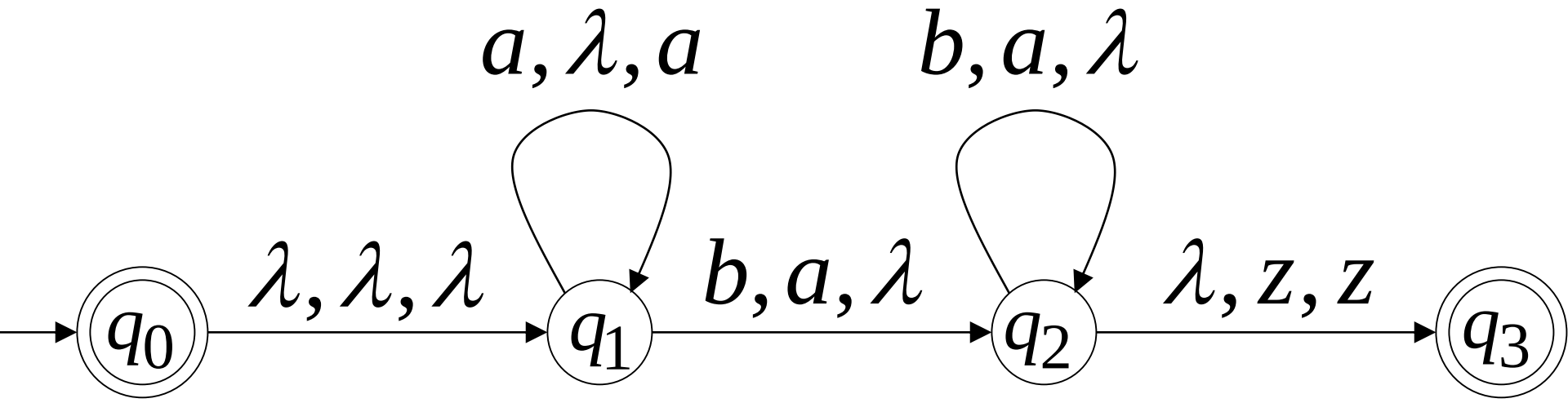
All the input is consumed

AND

The last state is a final state

At the end of the computation, we do not care about the stack contents

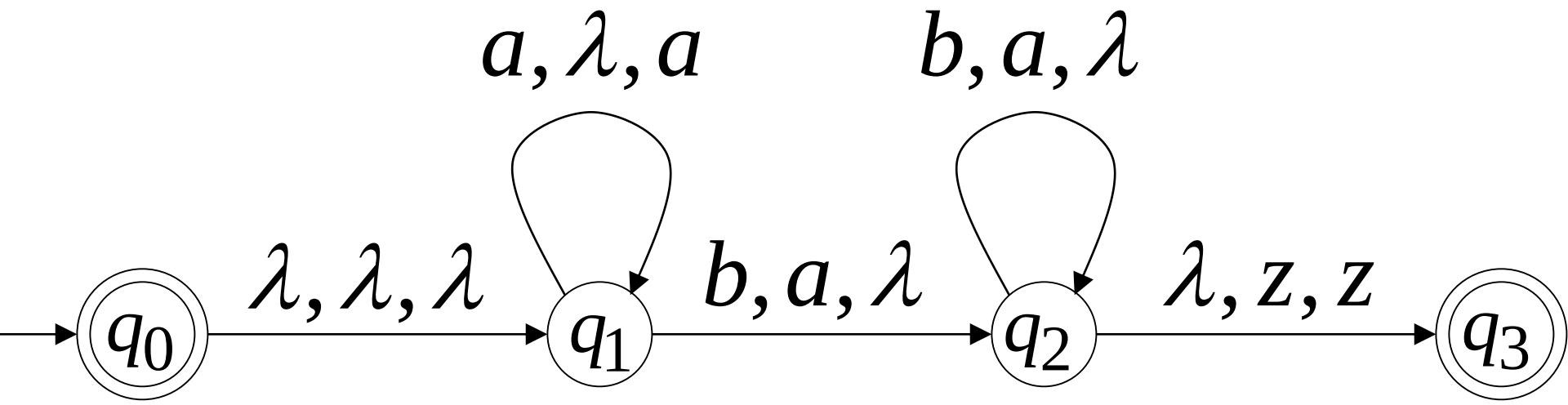
The input string *aaabbb*
is accepted by the NPDA:



In general,

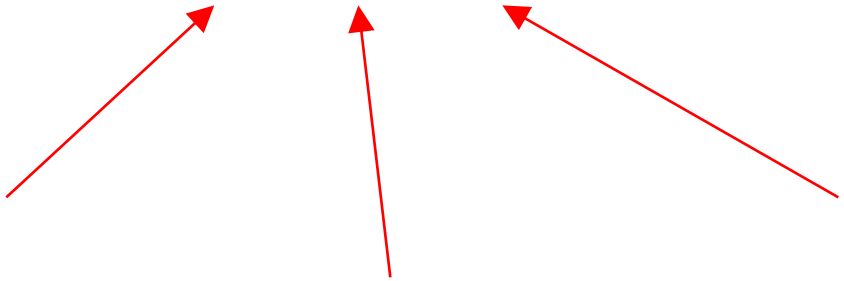
$$L = \{a^n b^n : n \geq 0\}$$

is the language accepted by the NPDA:



Instantaneous Description (ID)

$$(q,\ u,\ s)$$



Current state

Remaining input

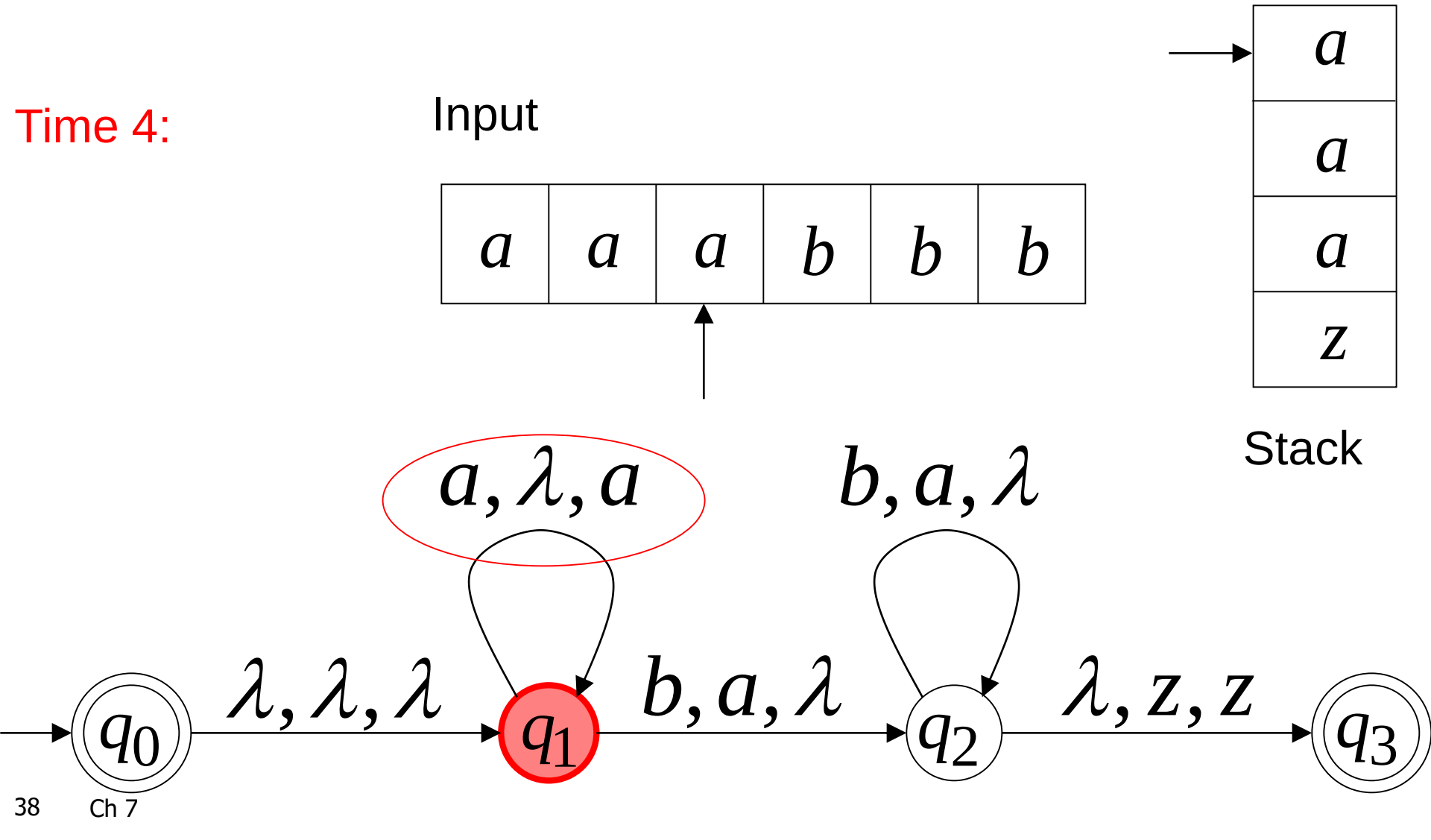
Current stack contents

Example:

Instantaneous Description

$$(q_1, bbb, aaaz)$$

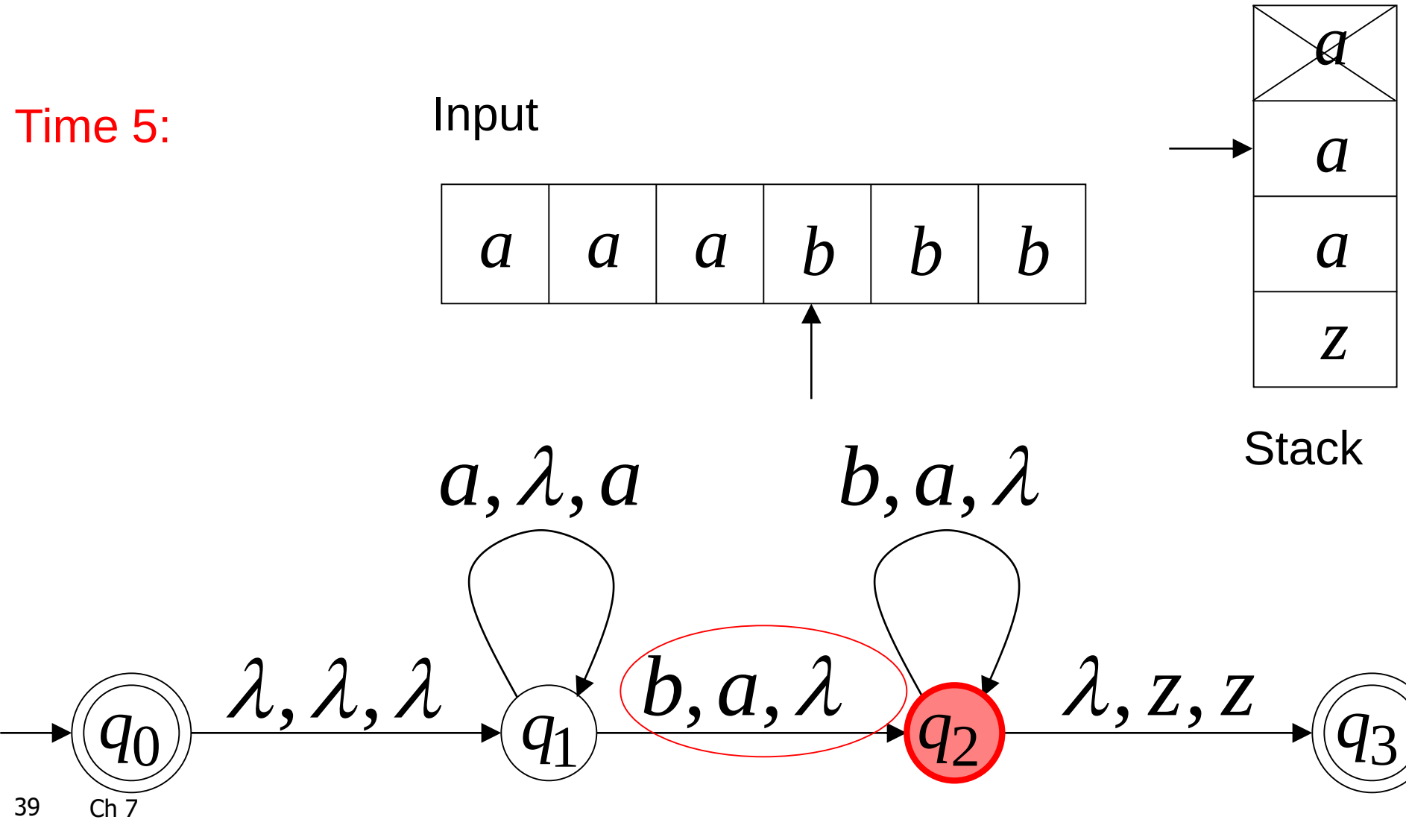
Time 4:



Example: **Instantaneous Description**

$$(q_2, bb, aaz)$$

Time 5:



We write:

$$(q_1, bbb, aaaz) \vdash (q_2, bb, aaz)$$

Time 4

Time 5

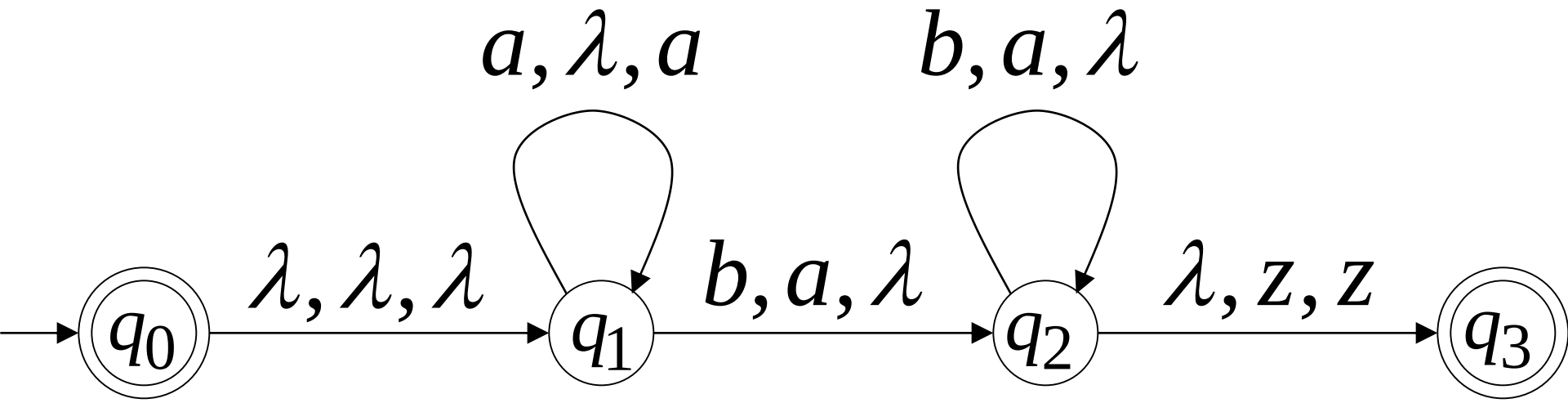
The symbol denotes a move from one ID to another

A computation:

$$(q_0, aaabbb, z) \vdash (q_1, aaabbb, z) \vdash$$

$$(q_1, aabbb, az) \vdash (q_1, abbb, aaz) \vdash (q_1, bbb, aaaz) \vdash$$

$$(q_2, bb, aaz) \vdash (q_2, b, az) \vdash (q_2, \lambda, z) \vdash (q_3, \lambda, z)$$



$$(q_0, aaabbb, z) \vdash (q_1, aaabbb, z) \vdash$$

$$(q_1, aabbb, az) \vdash (q_1, abbb, aaz) \vdash (q_1, bbb, aaaz) \vdash$$

$$(q_2, bb, aaz) \vdash (q_2, b, az) \vdash (q_2, \lambda, z) \vdash (q_3, \lambda, z)$$

For convenience we write:

$$(q_0, aaabbb, z) \overset{*}{\vdash} (q_3, \lambda, z)$$

Formal Definition

Language $L(M)$ of NPDA M:

$$L(M) = \{w: \ (q_0, w, z) \overset{*}{\vdash} (q_f, \lambda, s)\}$$

Initial state

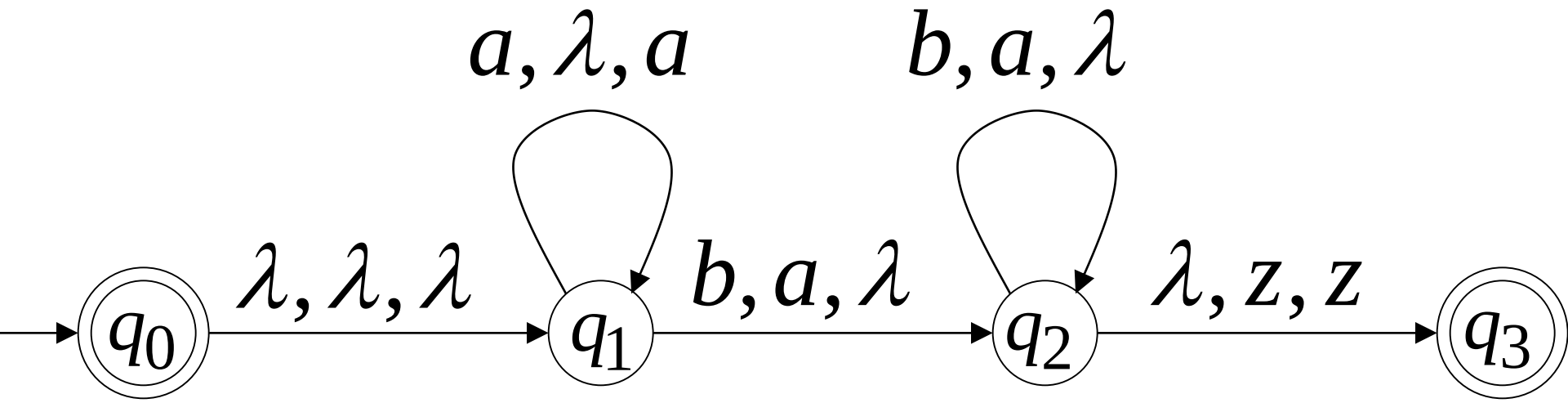
Final state

Example:

$$(q_0, aaabbb, z) \overset{*}{\vdash} (q_3, \lambda, z)$$

$$aaabbb \in L(M)$$

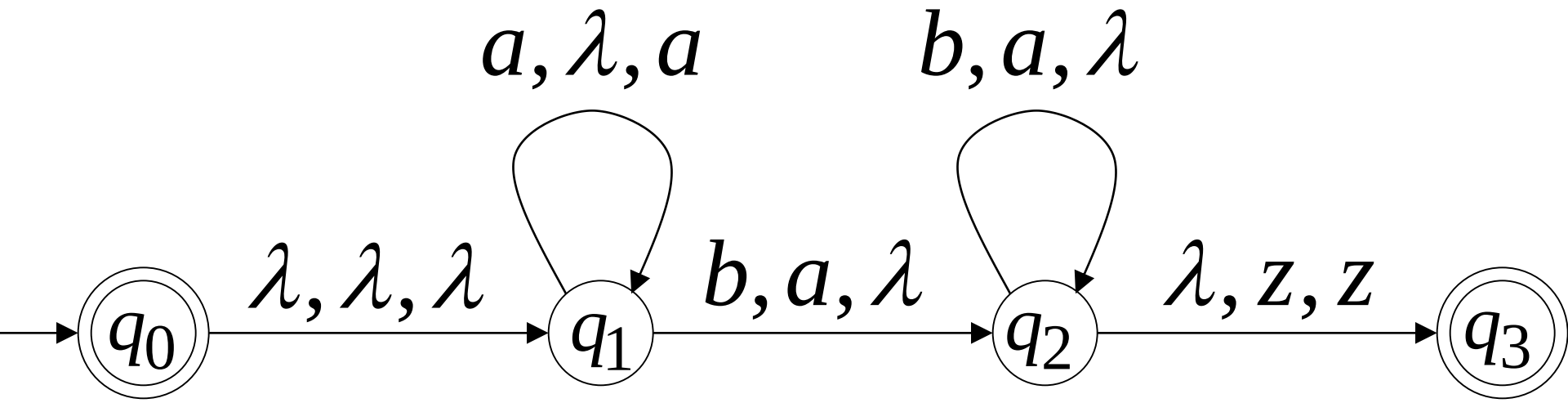
NPDA M :



$$(q_0, a^n b^n, z) \overset{*}{\vdash} (q_3, \lambda, z)$$

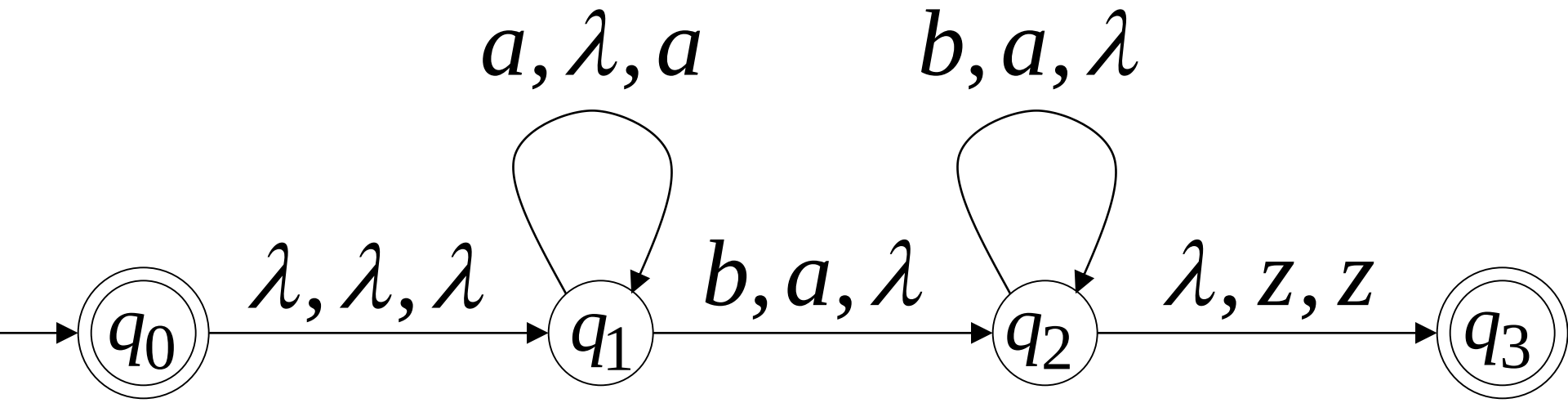
$$a^n b^n \in L(M)$$

NPDA M :



Therefore: $L(M) = \{a^n b^n : n \geq 0\}$

NPDA M :



Example 7.4

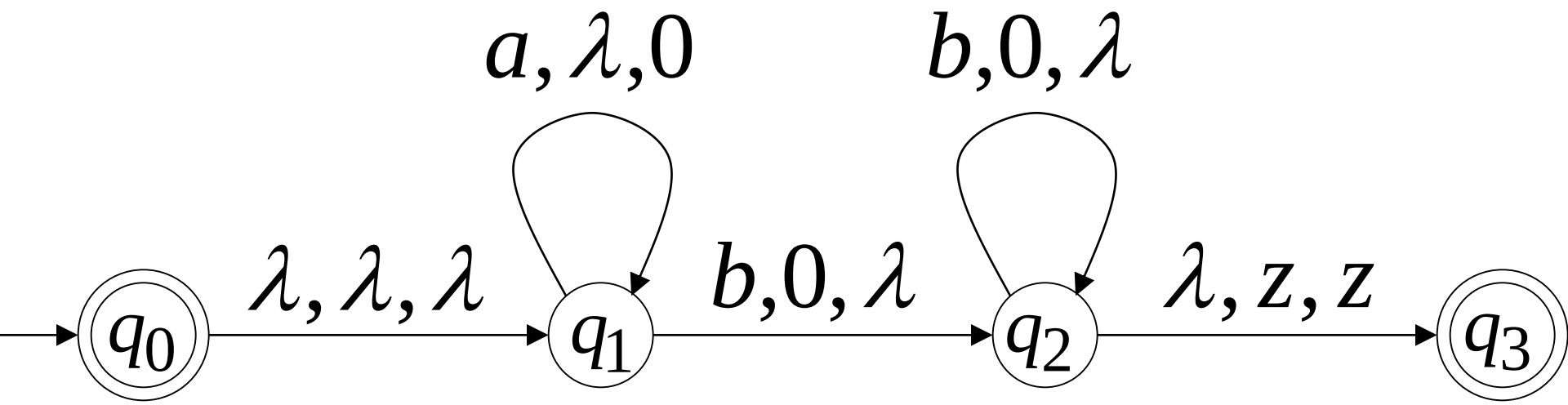
NPDA M

$$L(M) = \{w: \quad n_a = n_b\}$$

Stack

Order of *a* and *b* is don't care

How about more *b*'s than *a*'s in the prefix of *w*?



Example 7.4

NPDA M

$$L(M) = \{w: \quad n_a = n_b\}$$

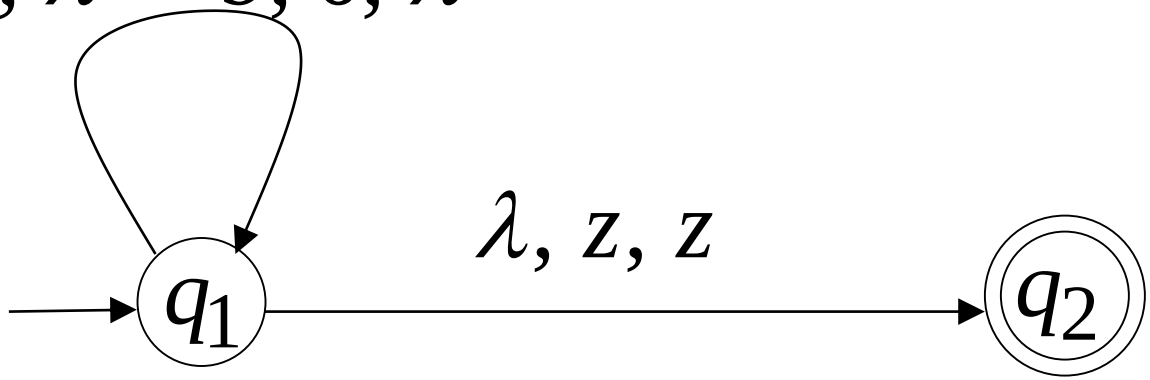
$a, z, 0z \quad b, z, 1z$

$a, 0, 00 \quad b, 1, 11$

$a, 1, \lambda \quad b, 0, \lambda$

Single loop

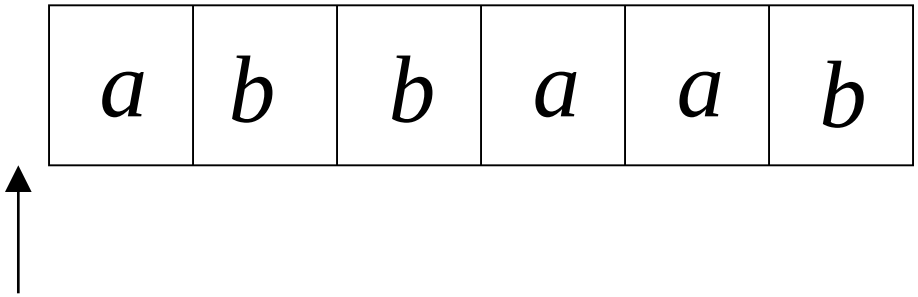
Use negative counter symbol (1)

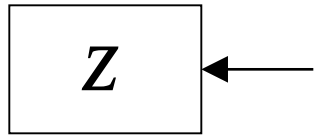


Execution Example:

Time 0

Input





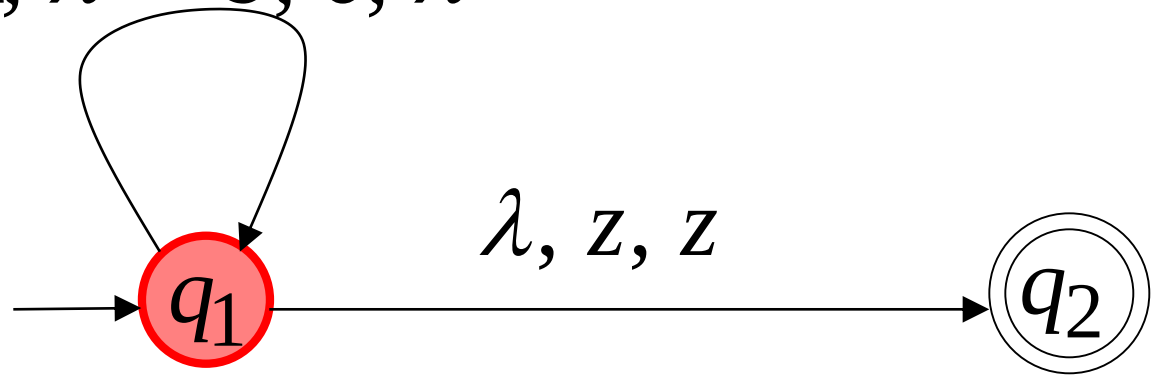
Stack

$a, z, 0z \quad b, z, 1z$

$a, 0, 00 \quad b, 1, 11$

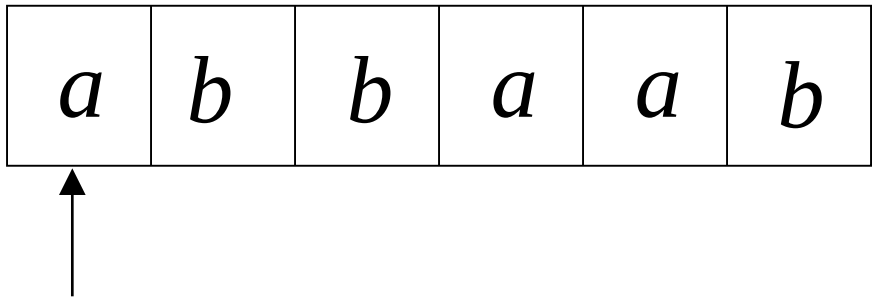
$a, 1, \lambda \quad b, 0, \lambda$

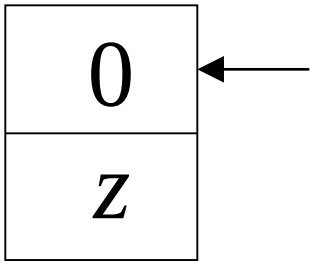
current state



Time 1

Input



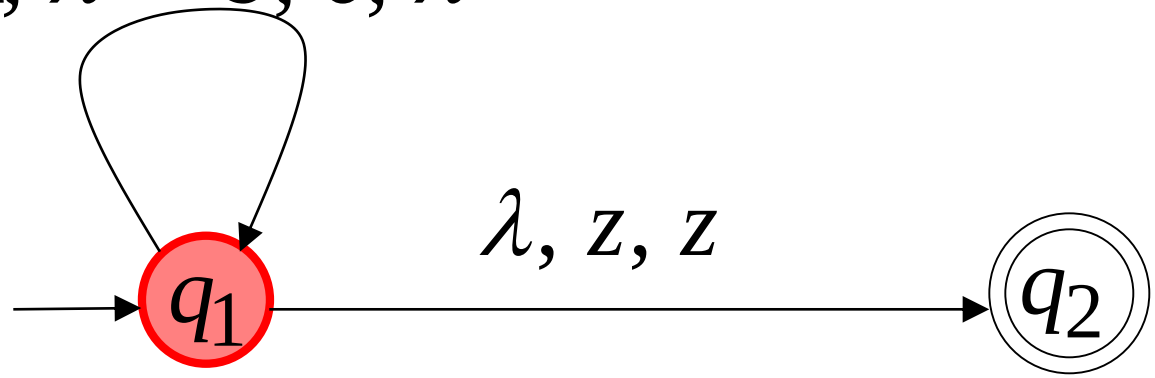


Stack

$a, z, 0z$ $b, z, 1z$

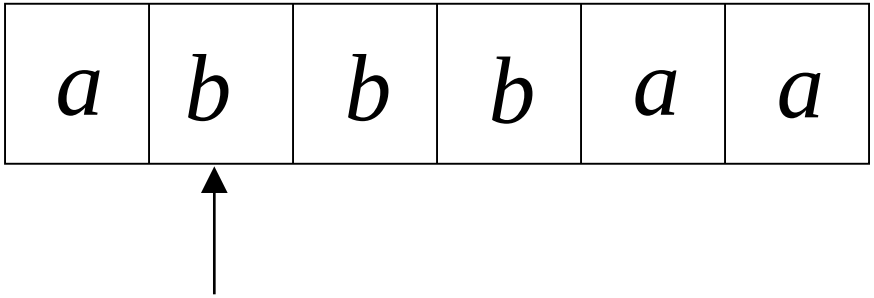
$a, 0, 00$ $b, 1, 11$

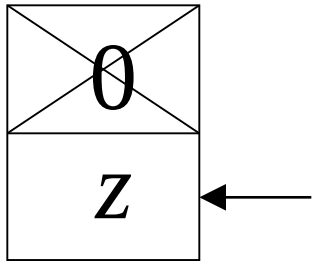
$a, 1, \lambda$ $b, 0, \lambda$



Time 2

Input



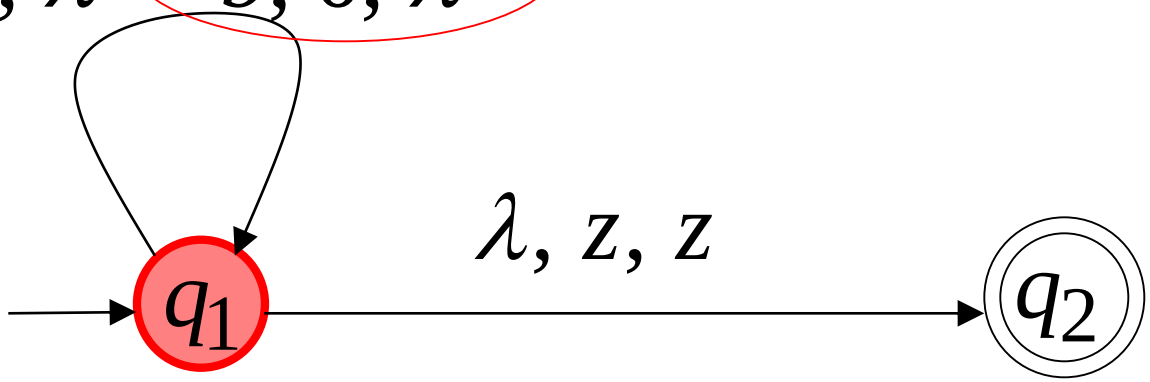


Stack

$a, z, 0z \quad b, z, 1z$

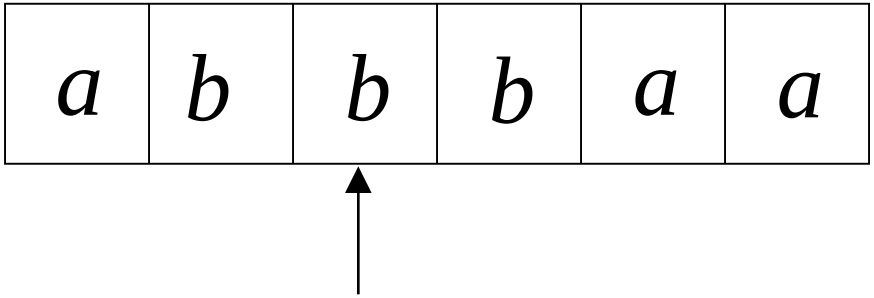
$a, 0, 00 \quad b, 1, 11$

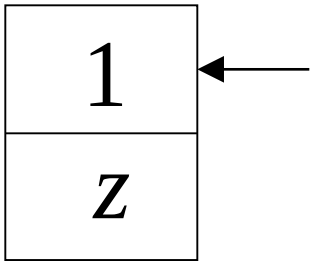
$a, 1, \lambda \quad b, 0, \lambda$



Time 3

Input



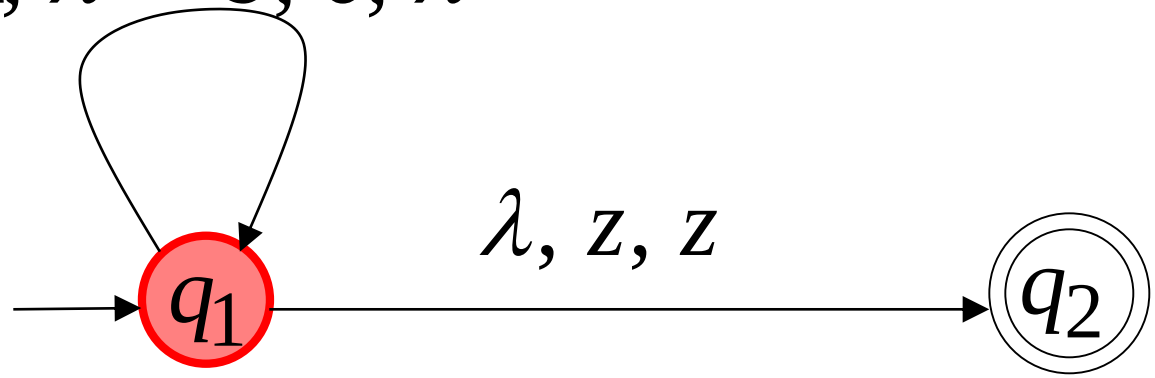


Stack

$a, z, 0z$ $b, z, 1z$

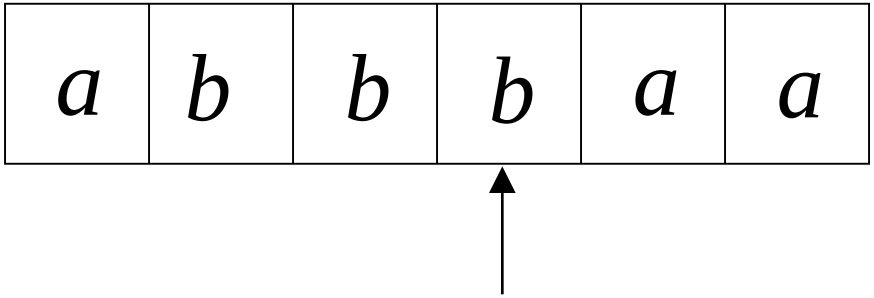
$a, 0, 00$ $b, 1, 11$

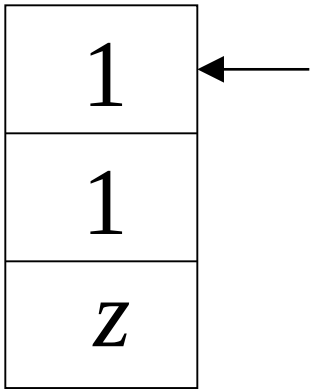
$a, 1, \lambda$ $b, 0, \lambda$



Time 4

Input



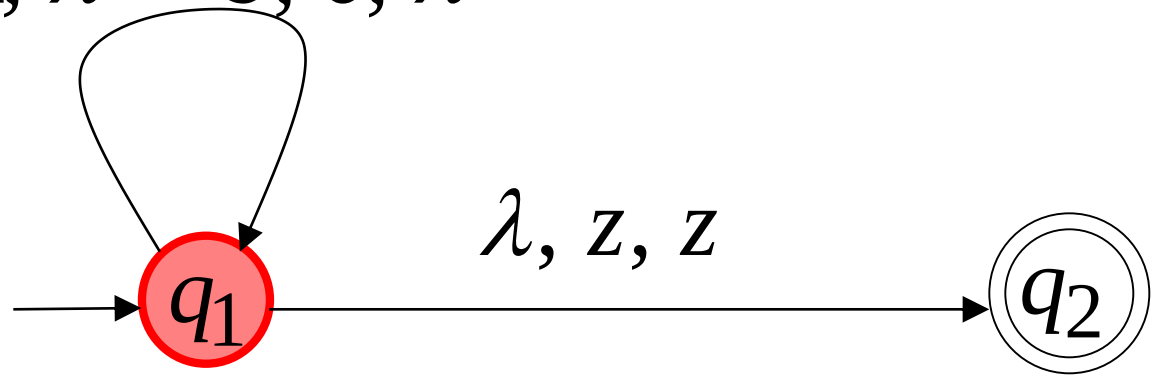


Stack

$a, z, 0z \quad b, z, 1z$

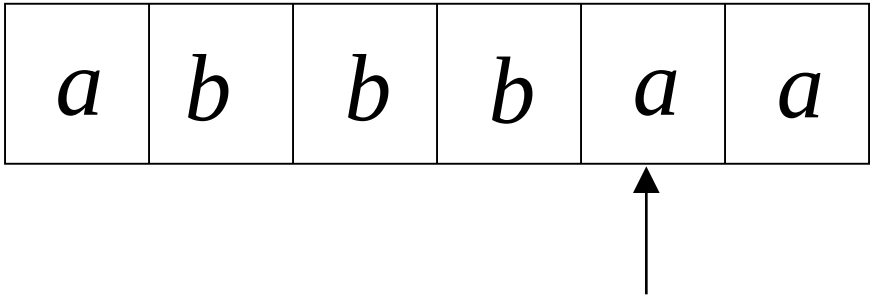
$a, 0, 00 \quad b, 1, 11$

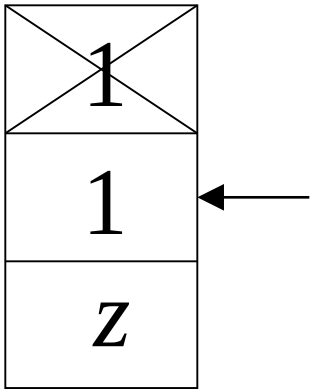
$a, 1, \lambda \quad b, 0, \lambda$



Time 5

Input



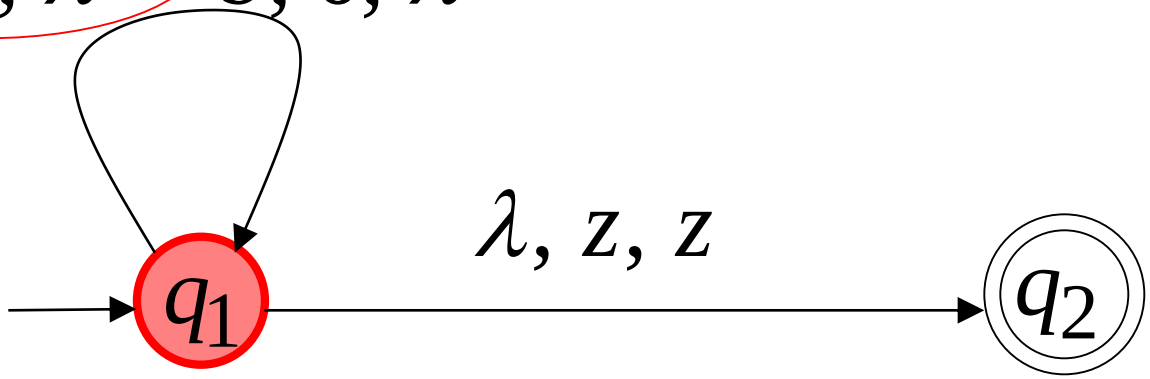


Stack

$a, z, 0z \quad b, z, 1z$

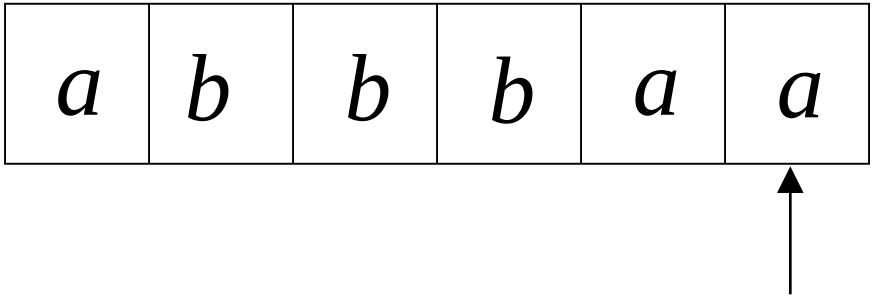
$a, 0, 00 \quad b, 1, 11$

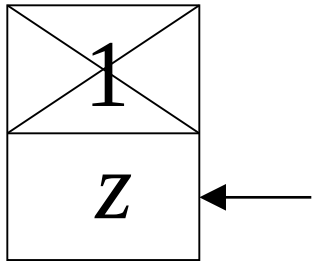
$a, 1, \lambda \quad b, 0, \lambda$



Time 6

Input



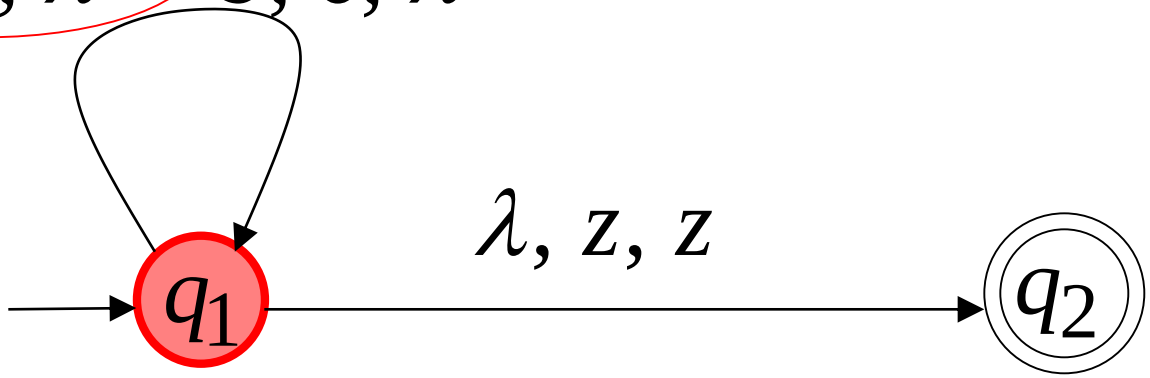


Stack

$a, z, 0z \quad b, z, 1z$

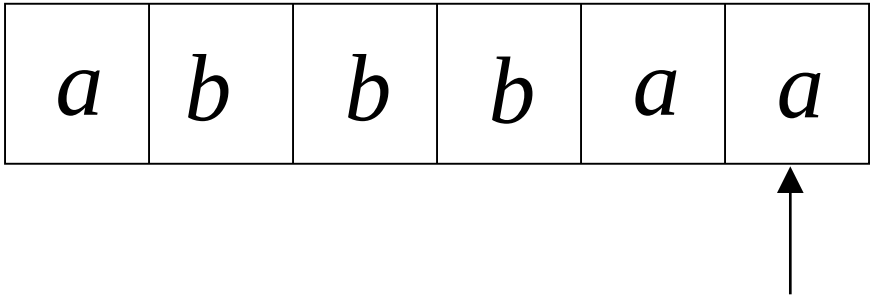
$a, 0, 00 \quad b, 1, 11$

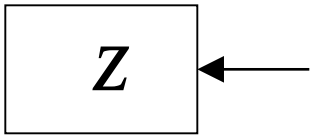
$a, 1, \lambda \quad b, 0, \lambda$



Time 7

Input



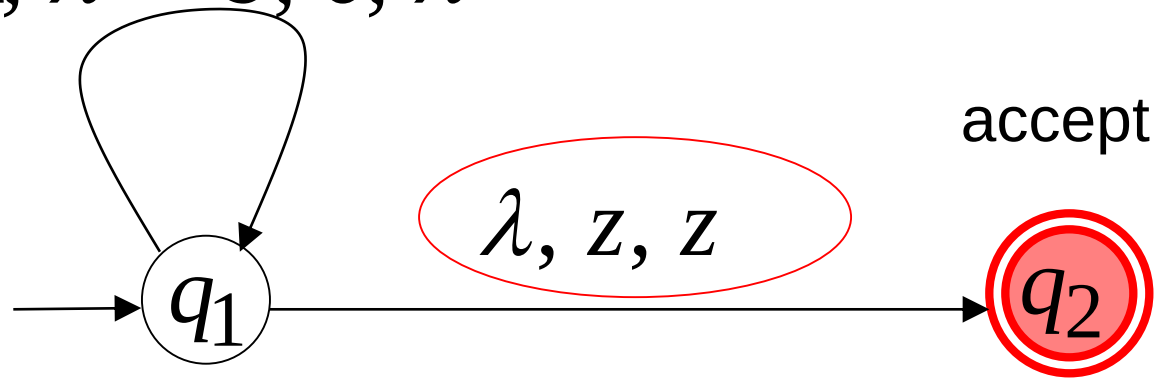


Stack

$a, z, 0z \quad b, z, 1z$

$a, 0, 00 \quad b, 1, 11$

$a, 1, \lambda \quad b, 0, \lambda$

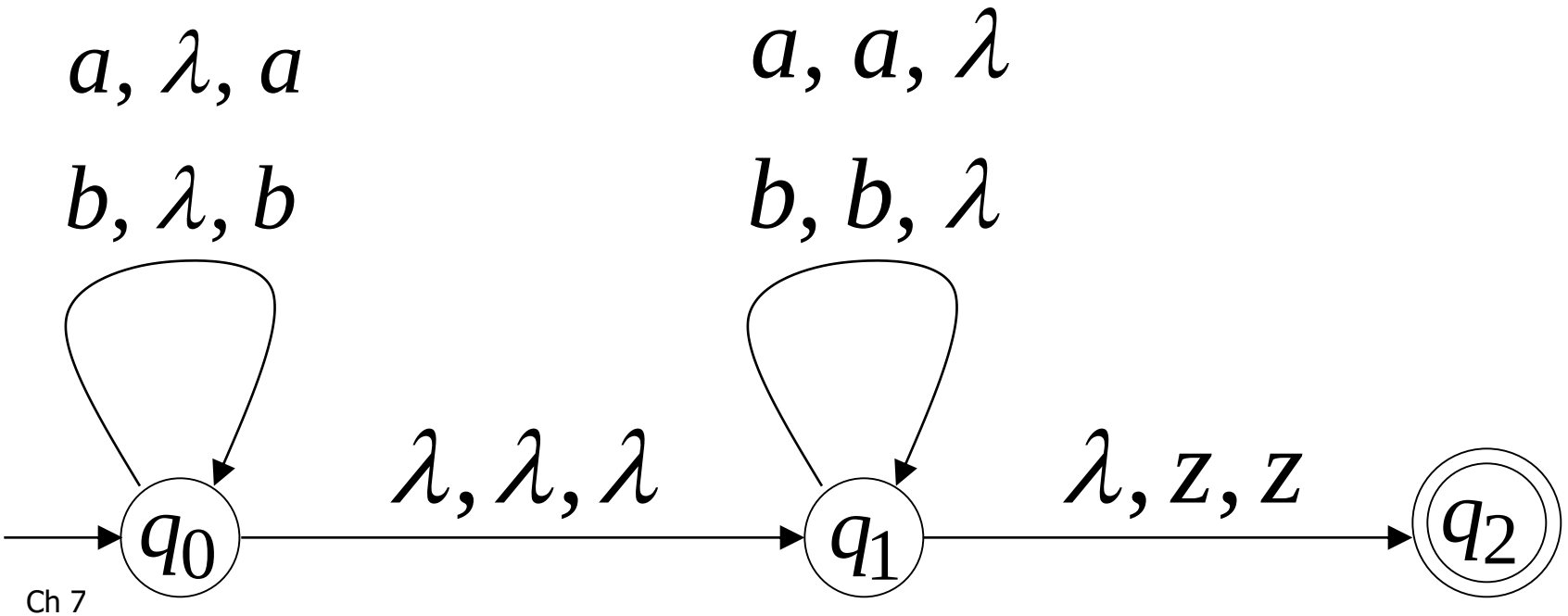


Example 7.5

NPDA M

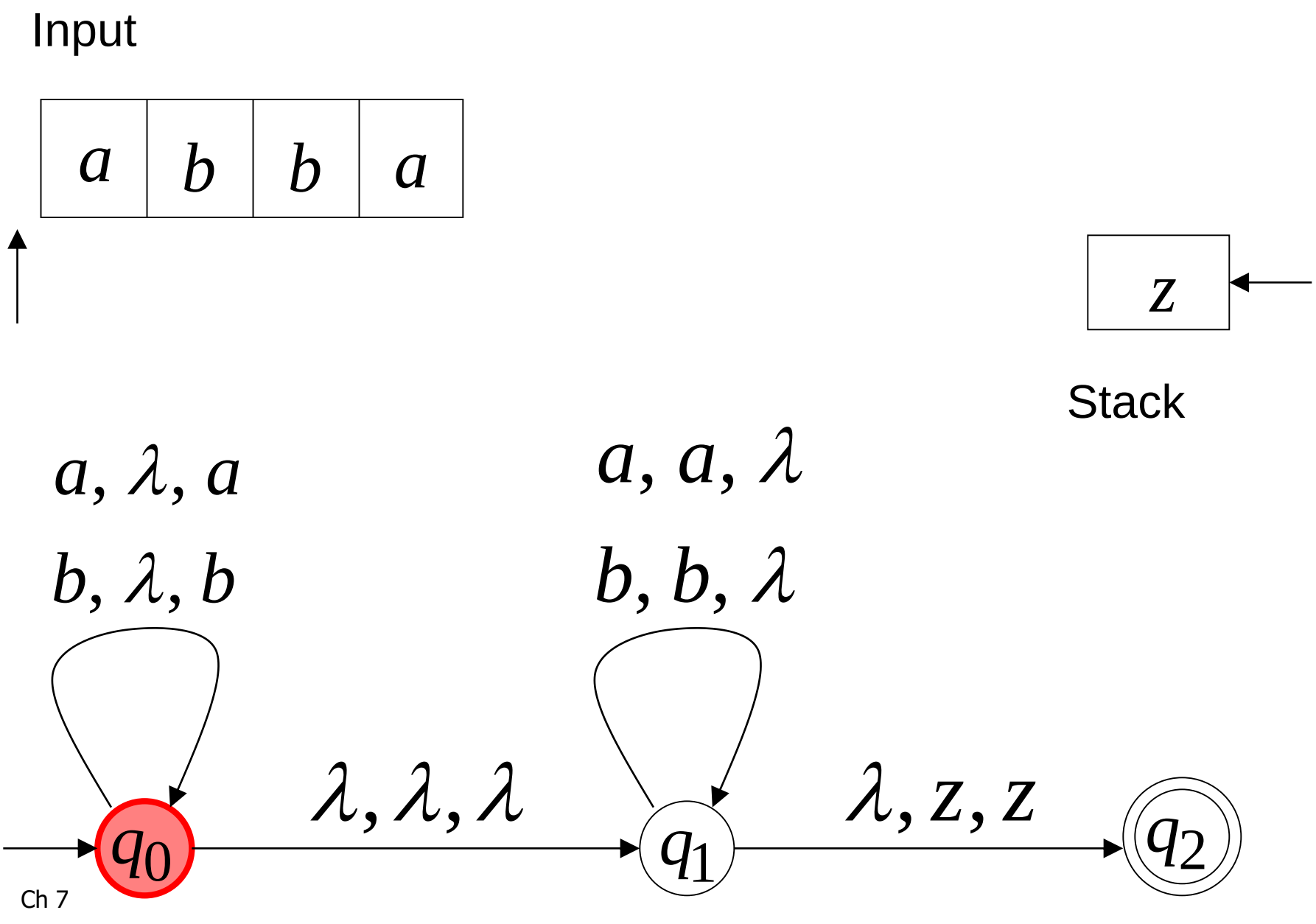
$$L(M) = \{ww^R\}$$

How do we know the middle of the string?



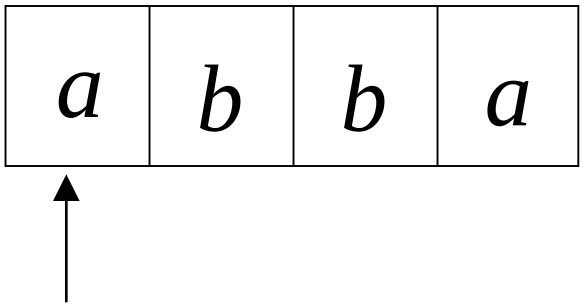
Execution Example:

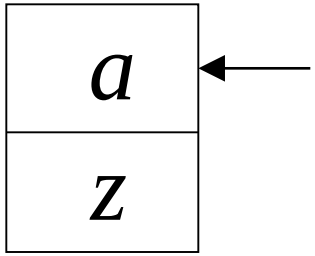
Time 0



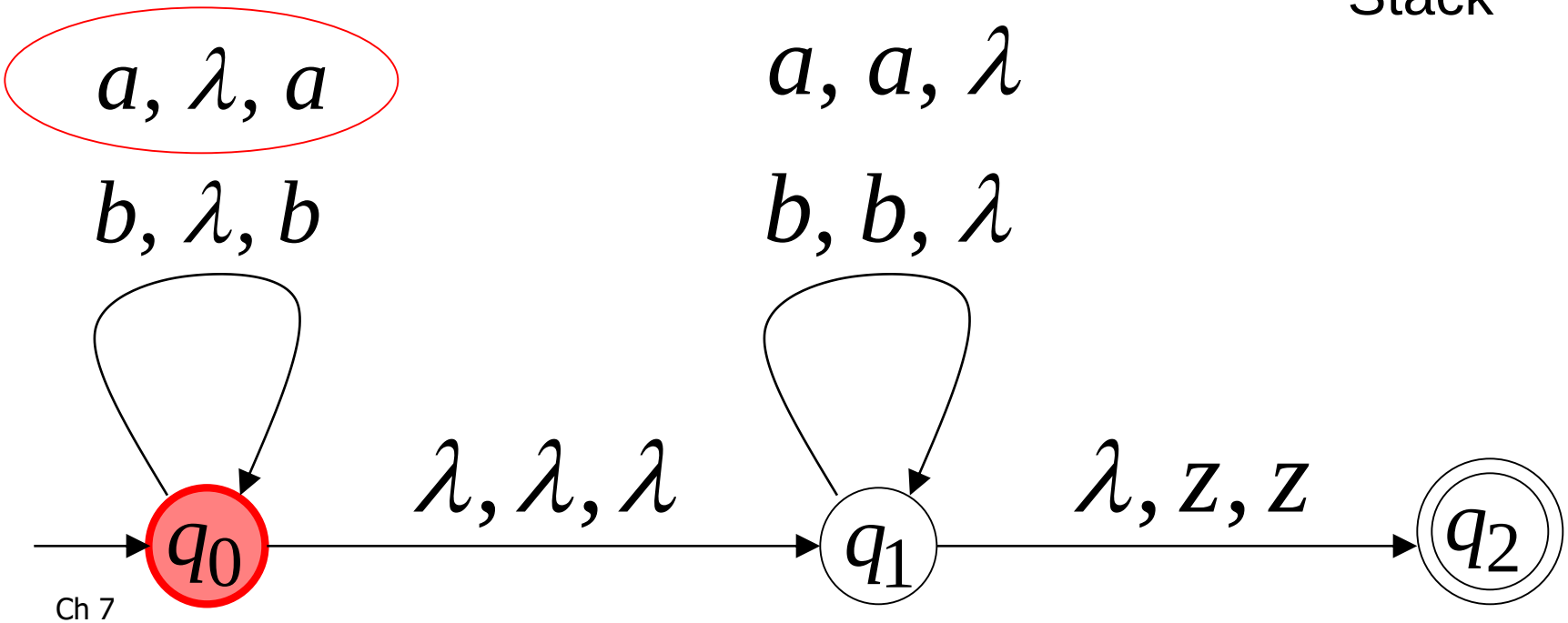
Time 1

Input

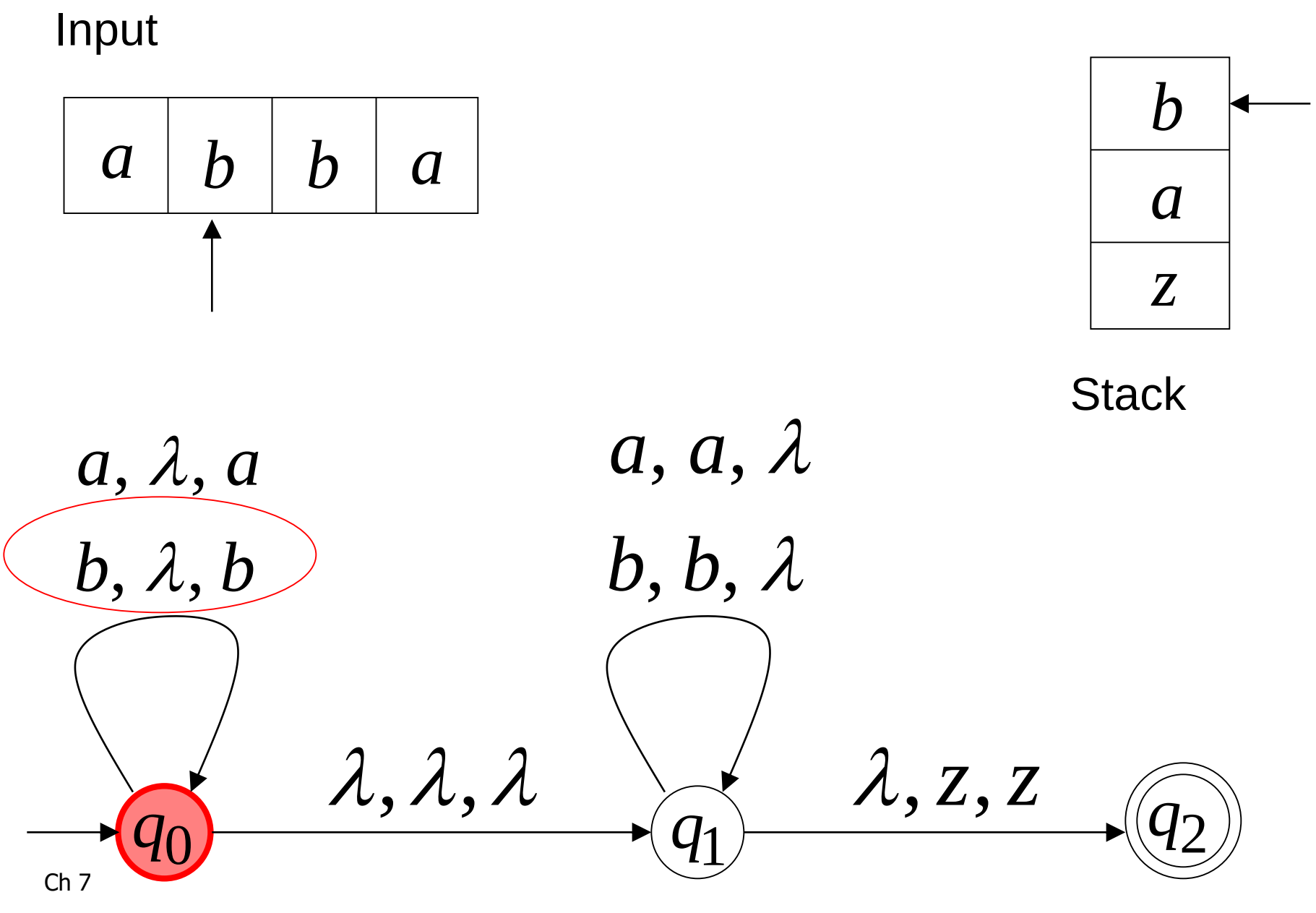




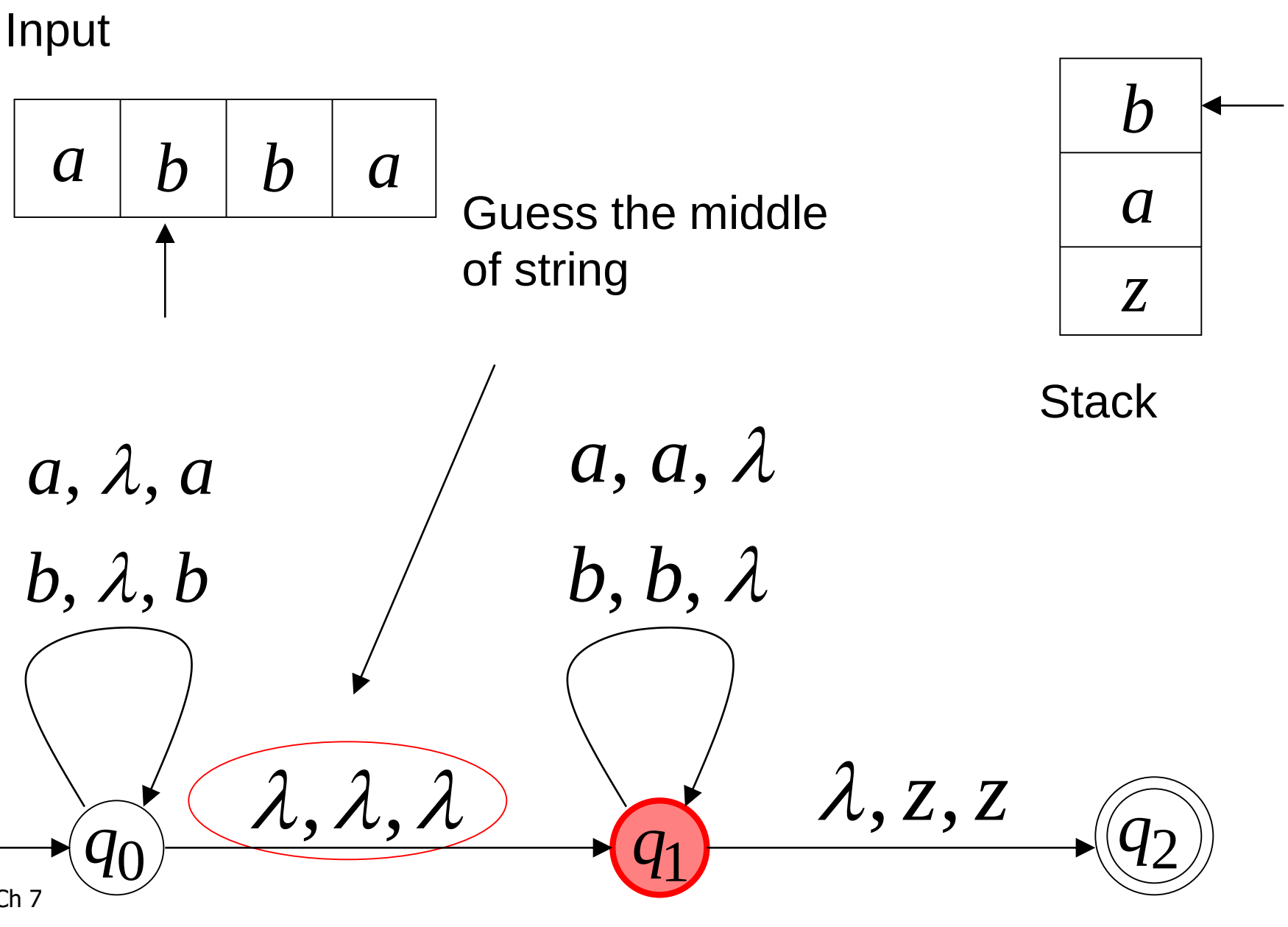
Stack



Time 2

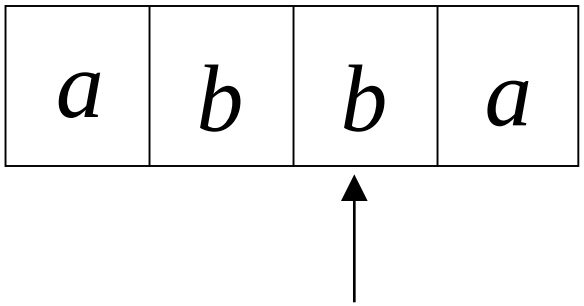


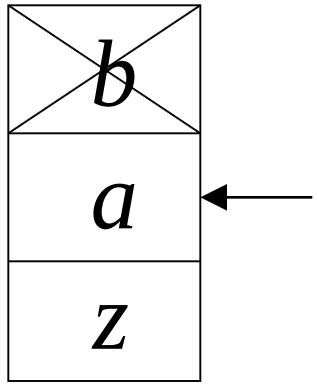
Time 3


Input
a b b a
Guess the middle of string
b
a
z
Stack
a, λ, a
b, λ, b
a, a, λ
b, b, λ
λ, λ, λ
λ, z, z
q0
q1
q2

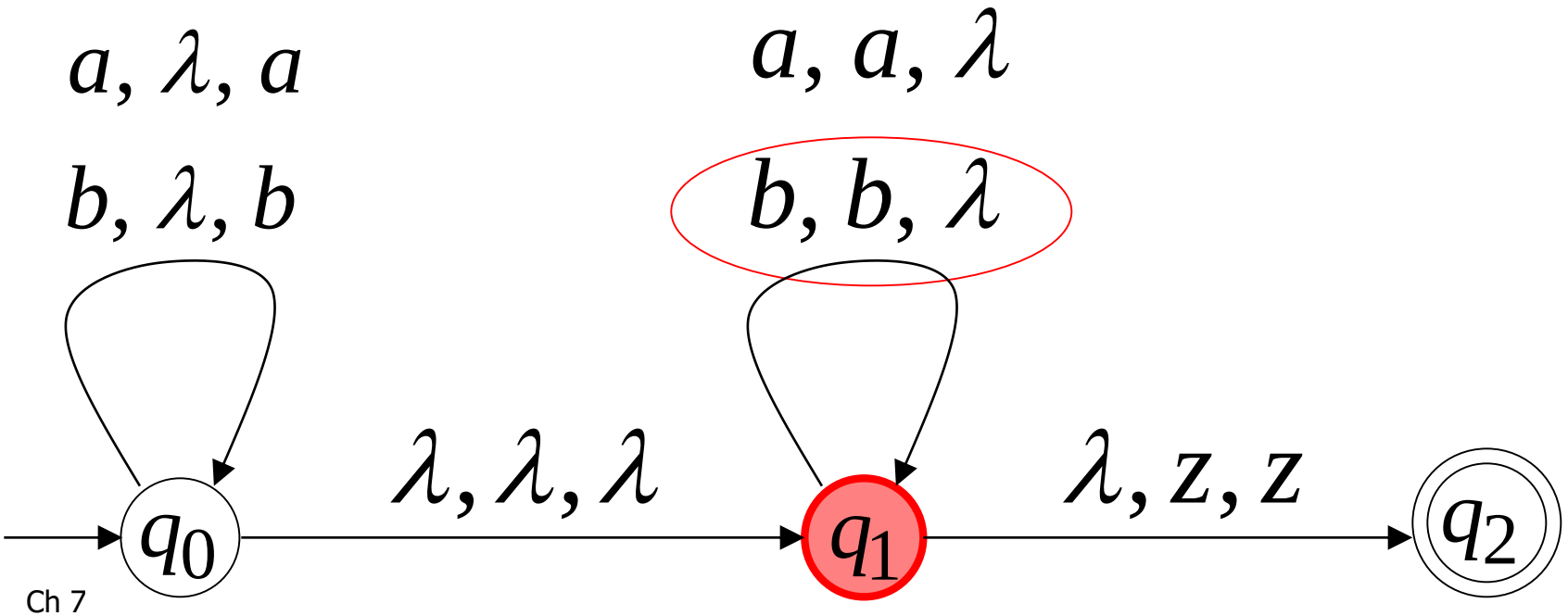
Time 4

Input



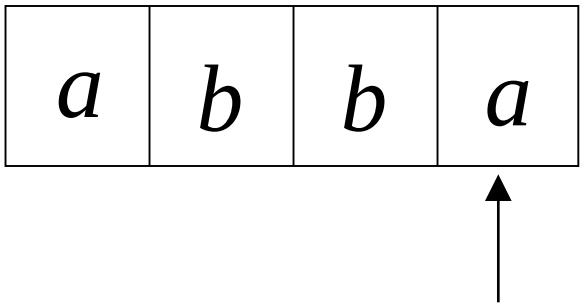


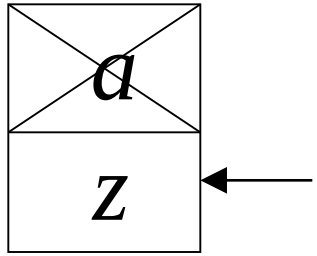
Stack



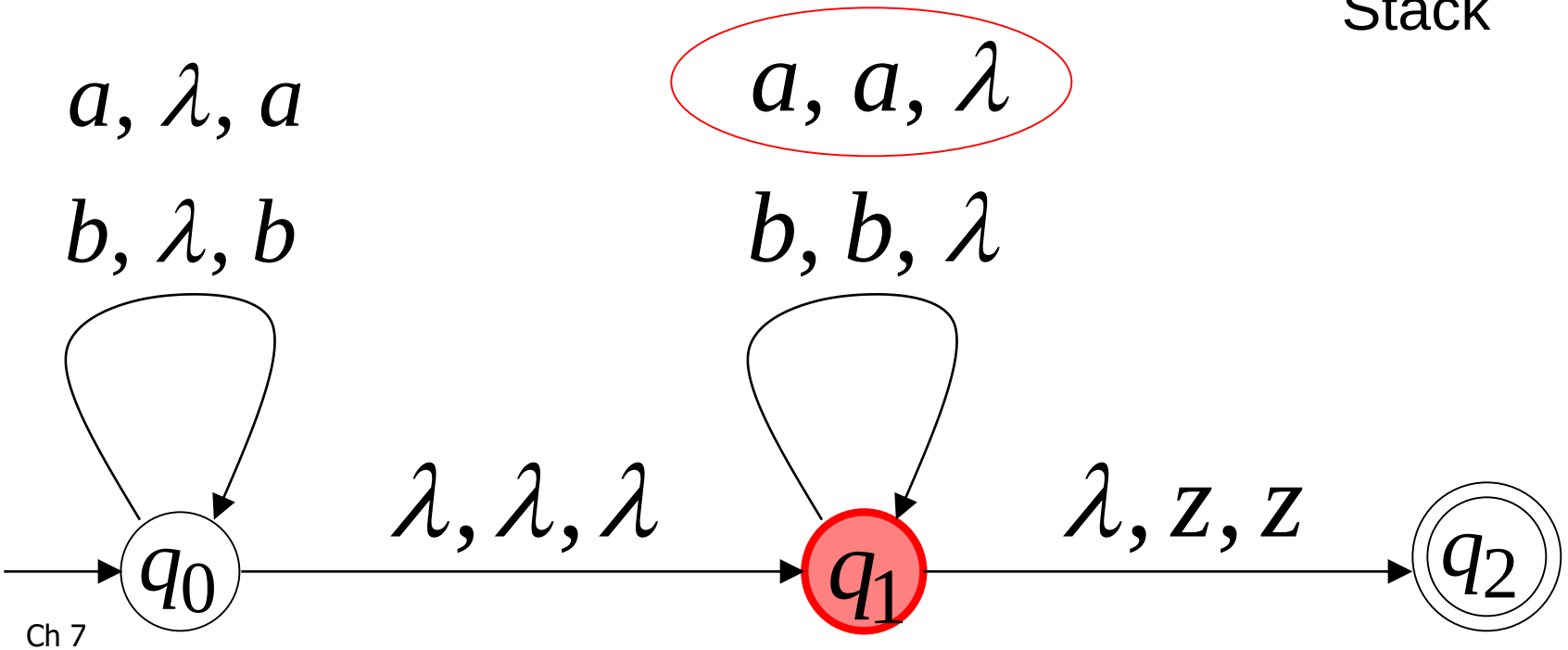
Time 5

Input

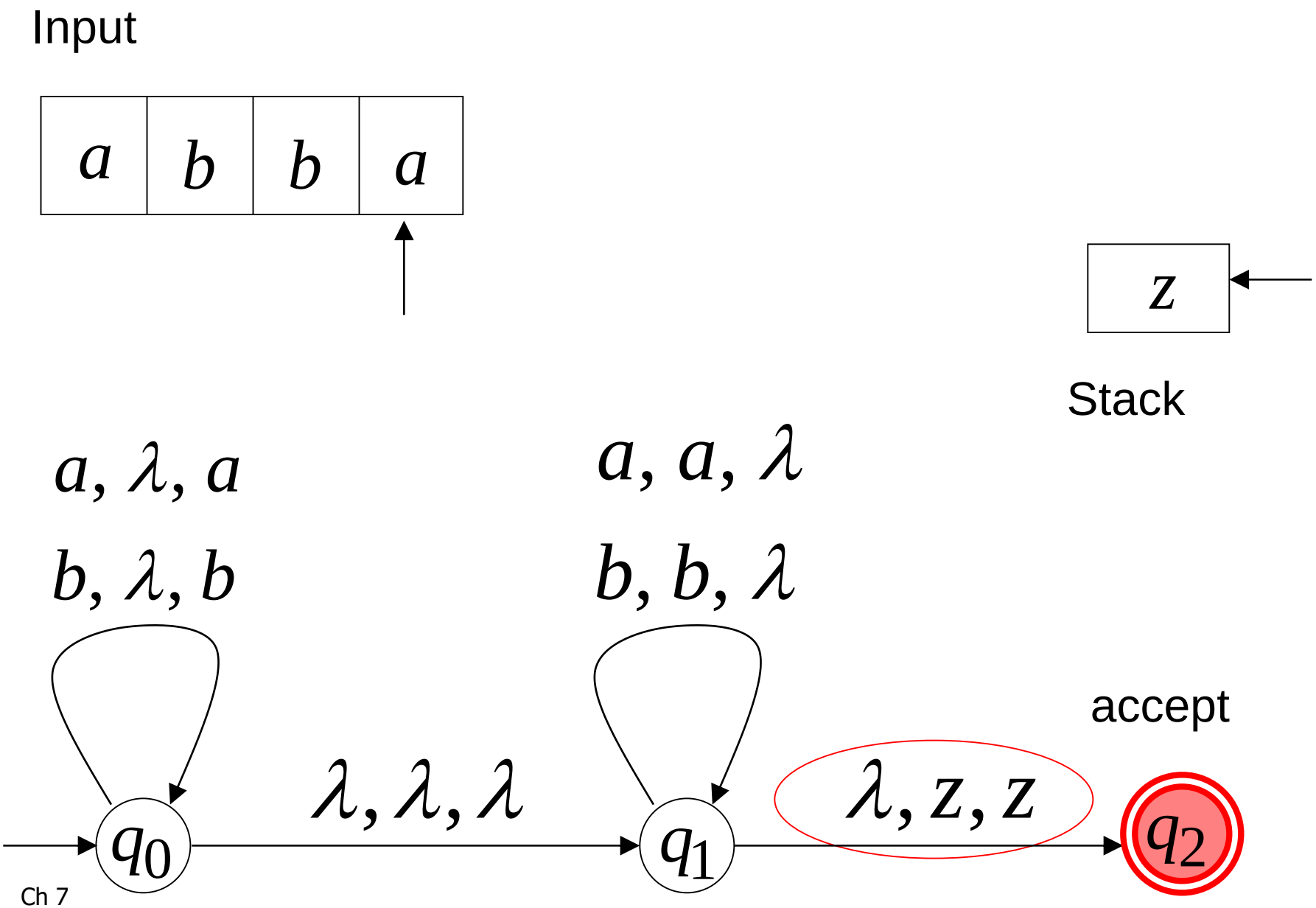




Stack

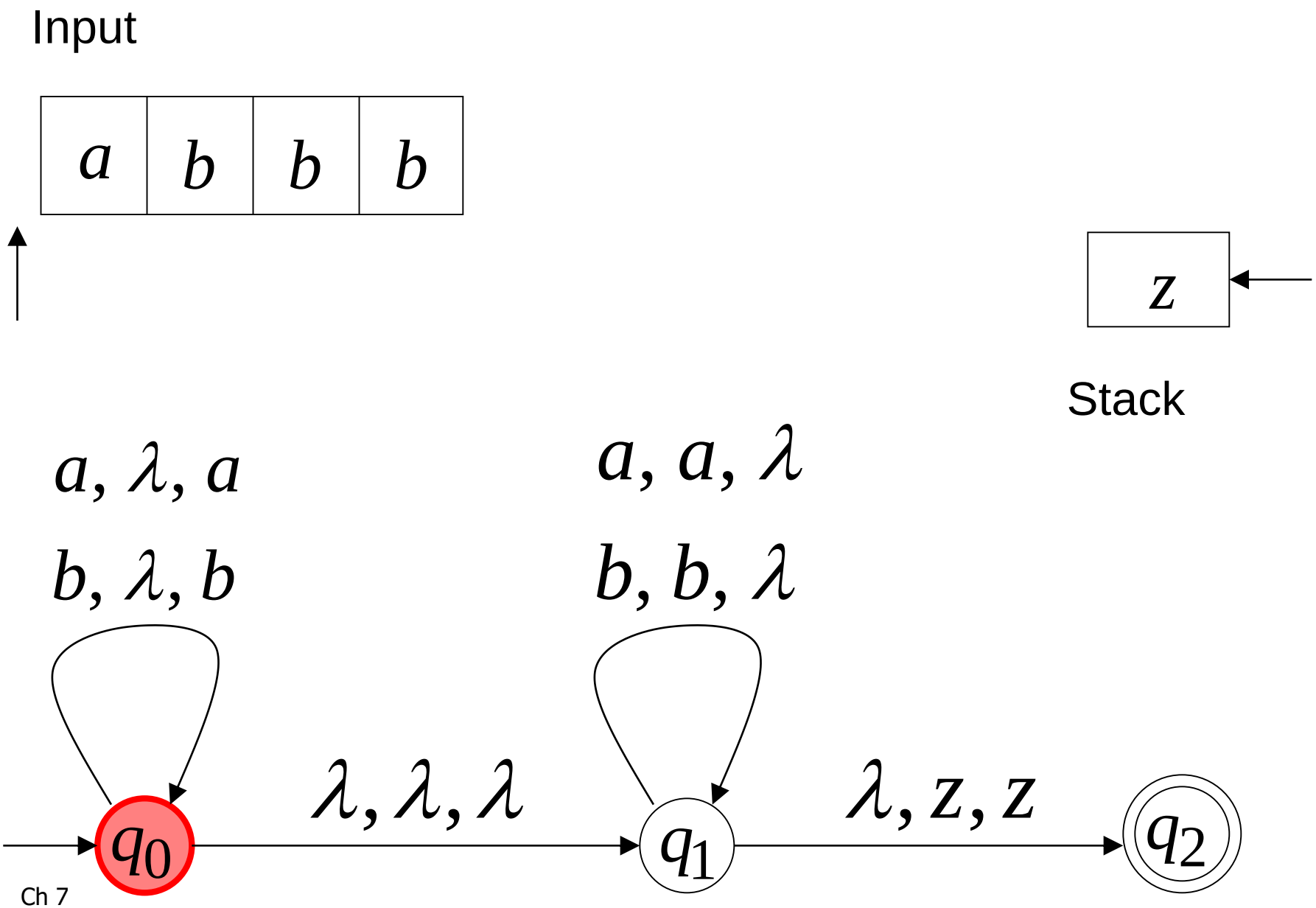


Time 6



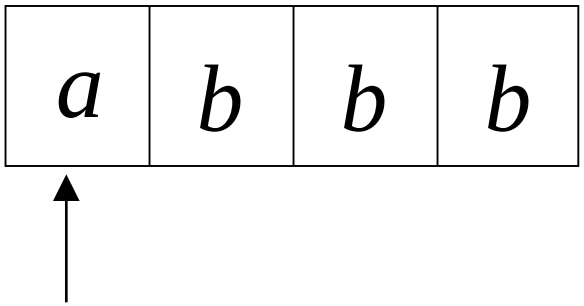
Rejection Example:

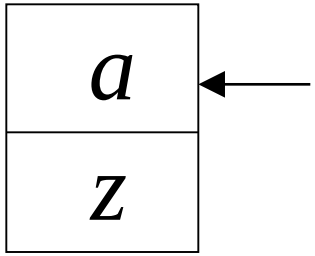
Time 0



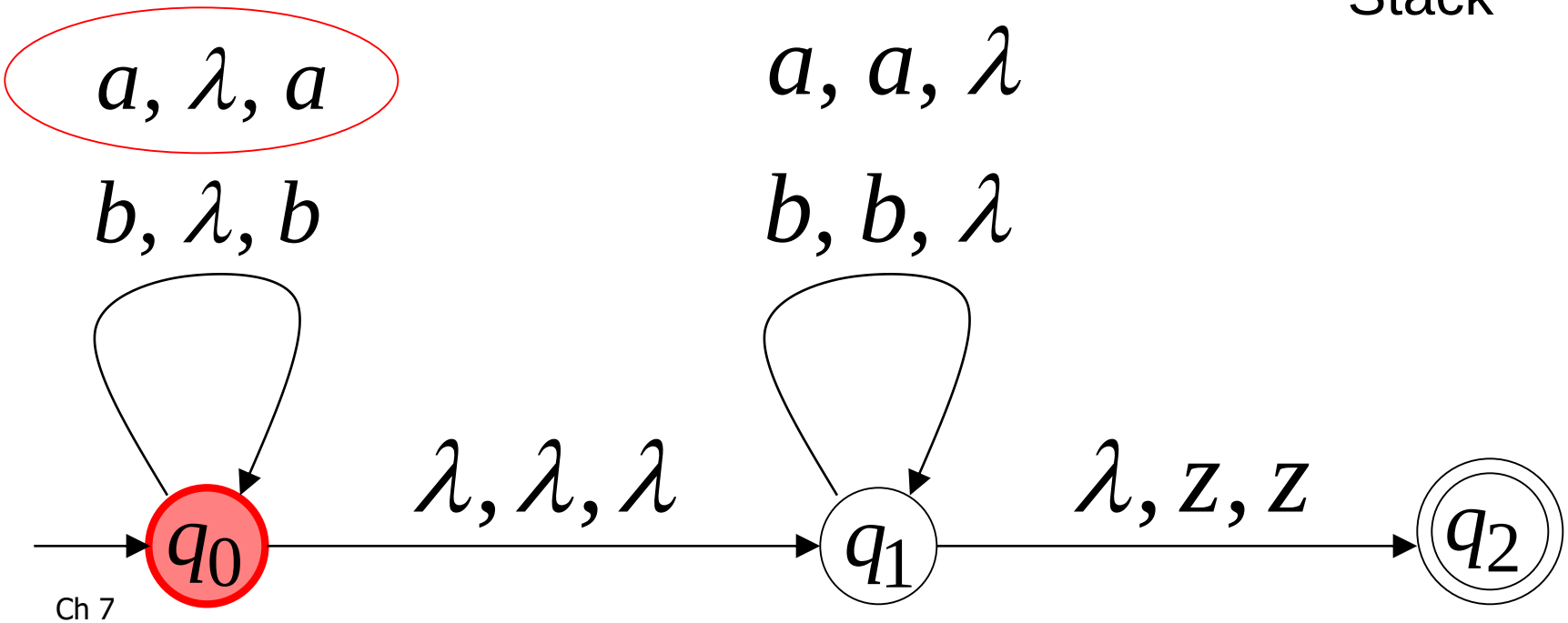
Time 1

Input

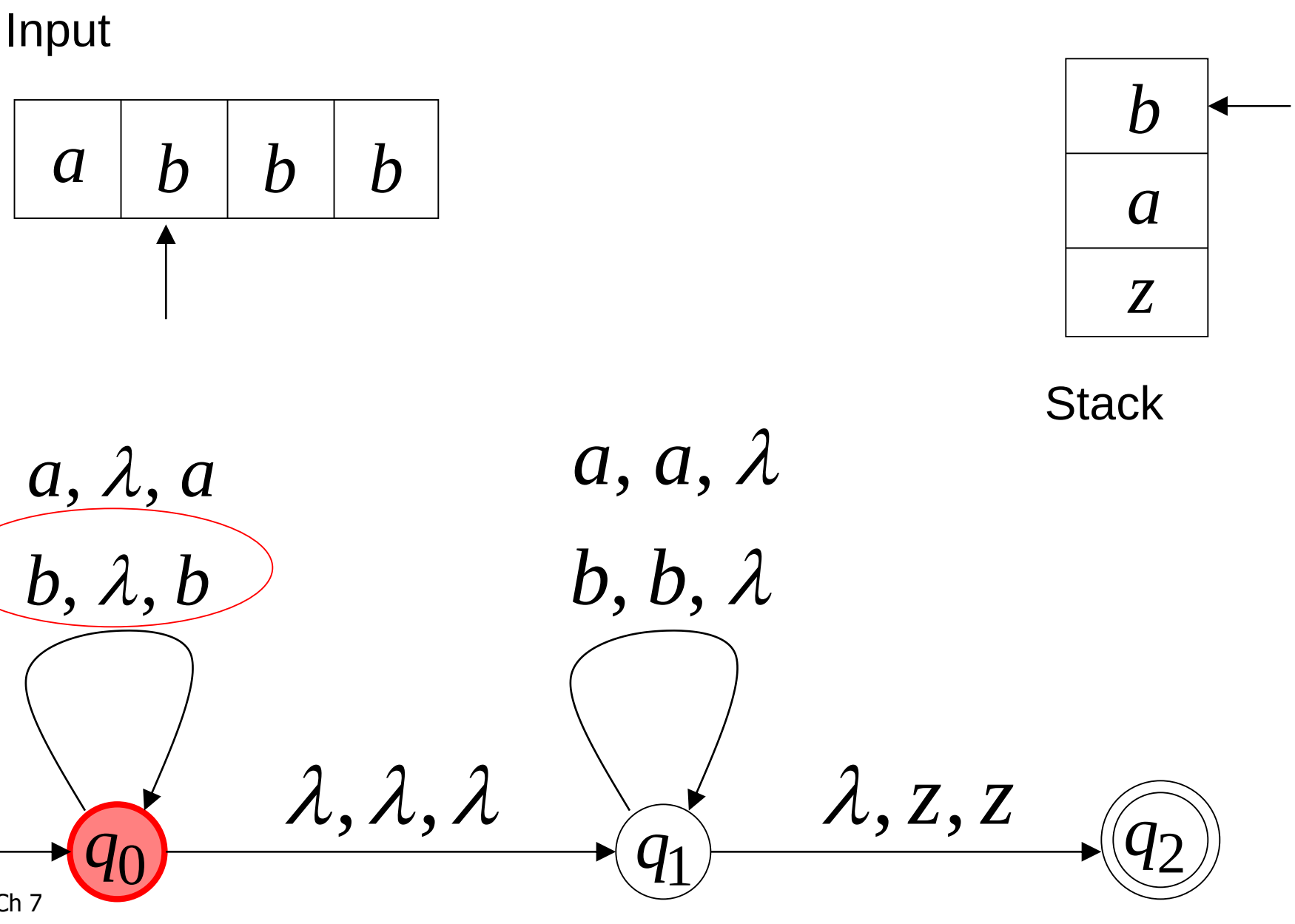




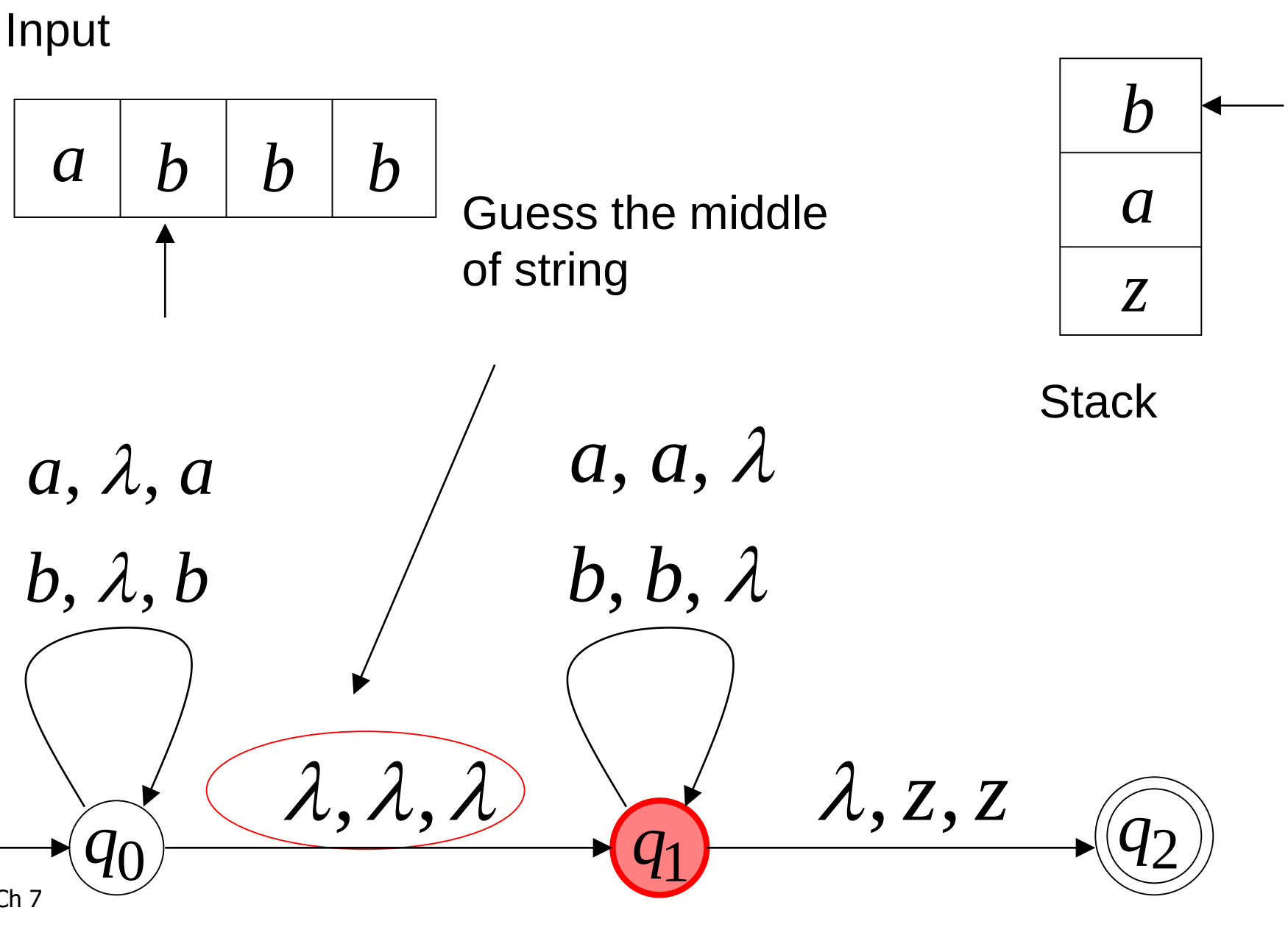
Stack



Time 2

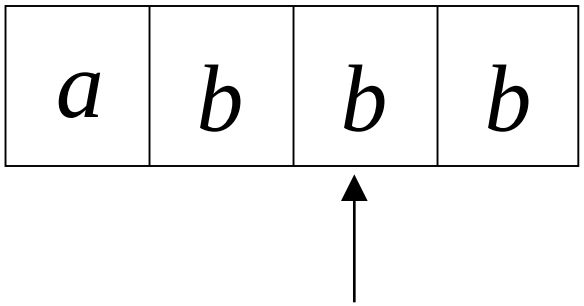


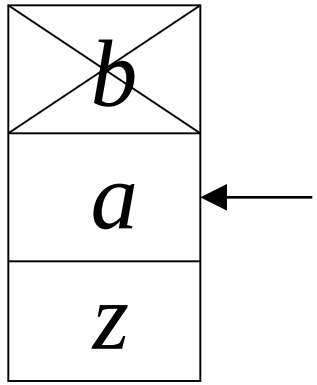
Time 3


Input
a
b
b
b
Guess the middle of string
b
a
z
Stack
a, λ, a
b, λ, b
a, a, λ
b, b, λ
λ, λ, λ
λ, z, z
q0
q1
q2

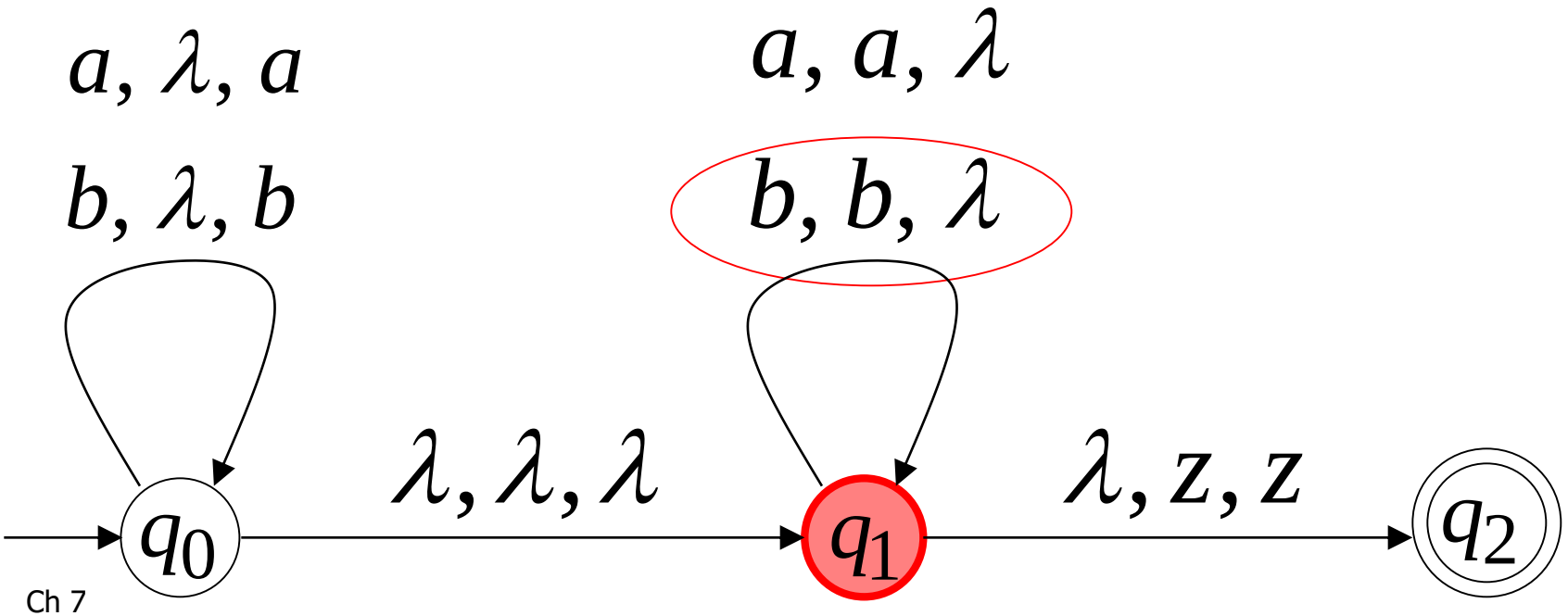
Time 4

Input





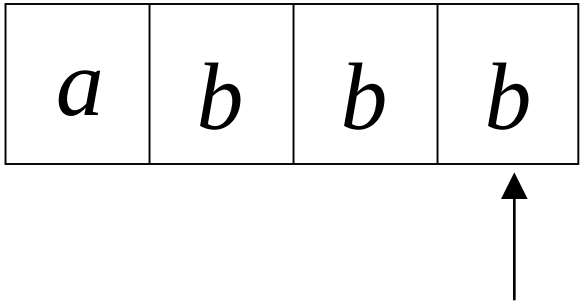
Stack



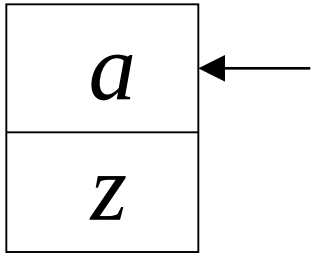
Time 5

There is no possible transition.

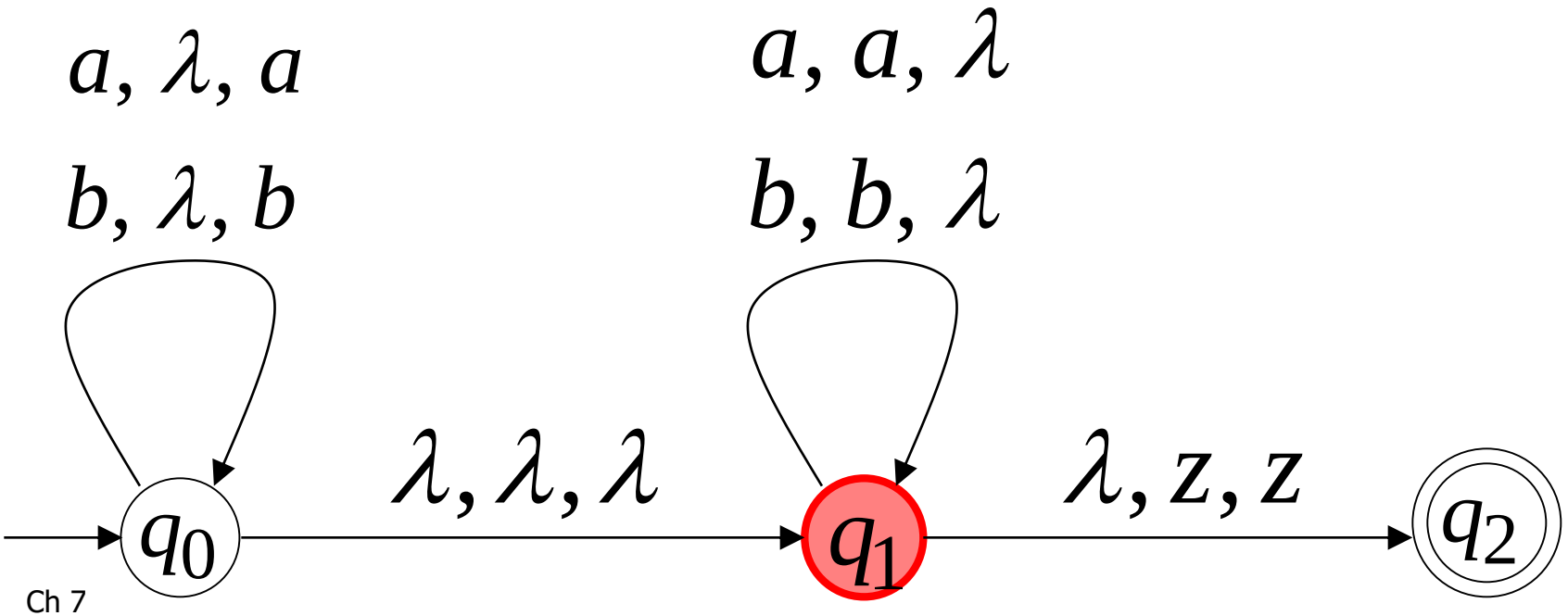
Input



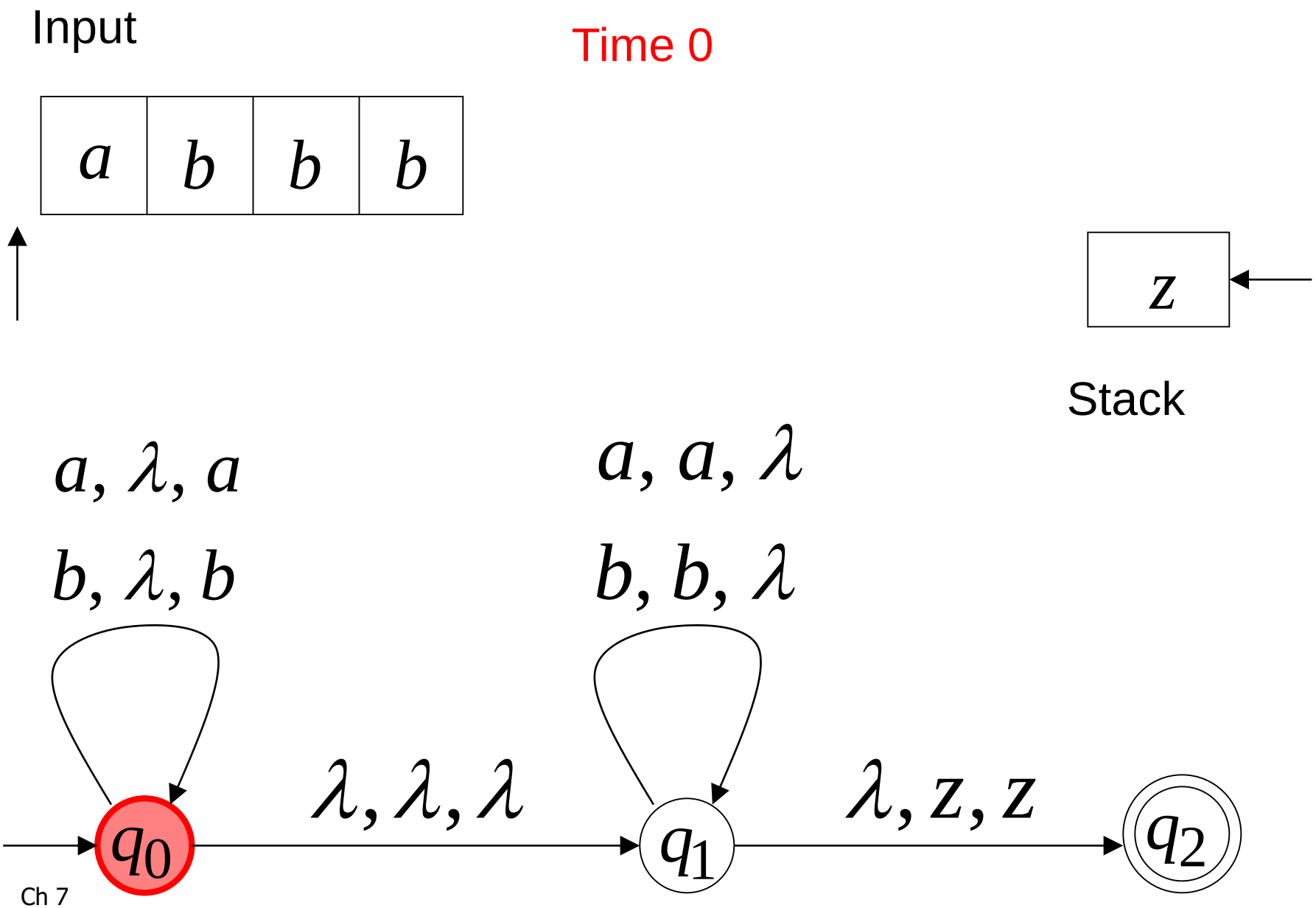
Input is not consumed



Stack

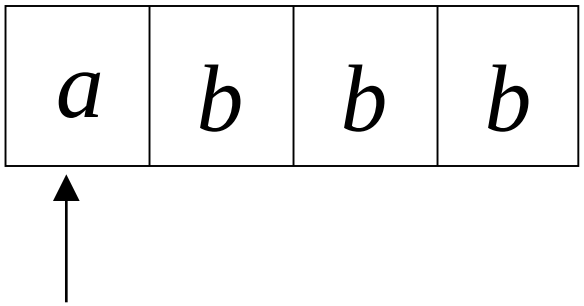


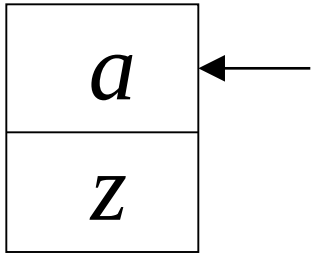
Another computation on same string:



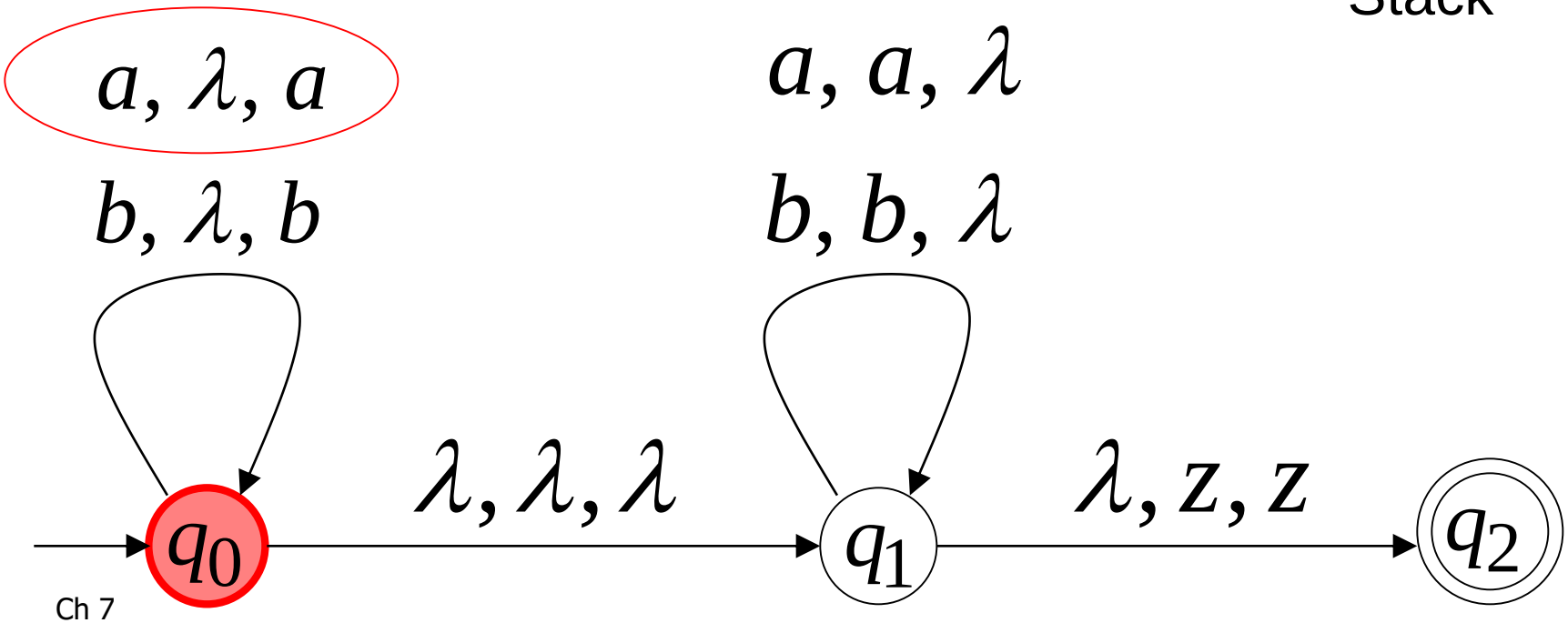
Time 1

Input

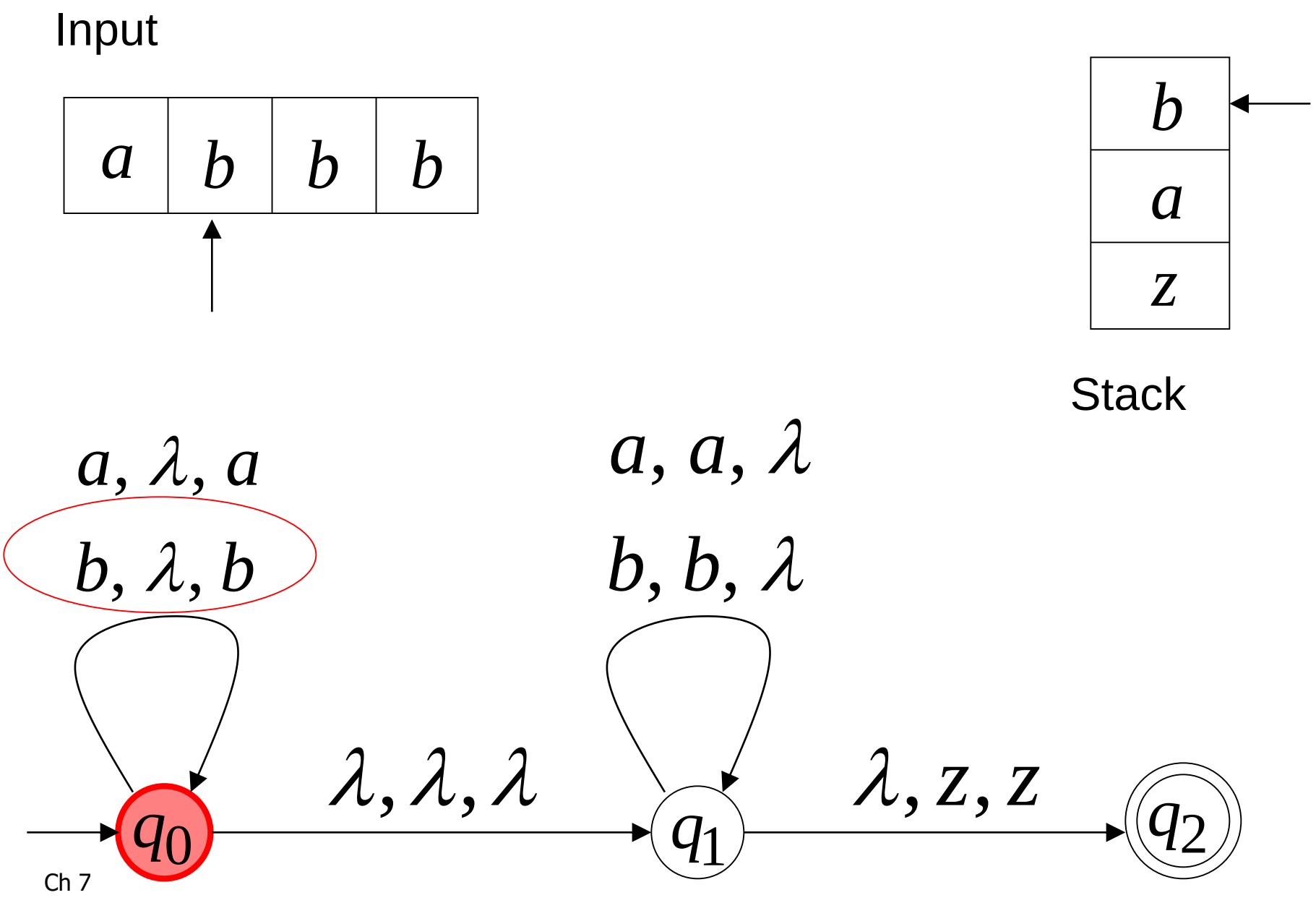




Stack

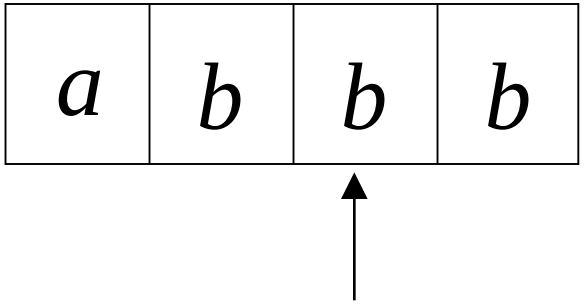


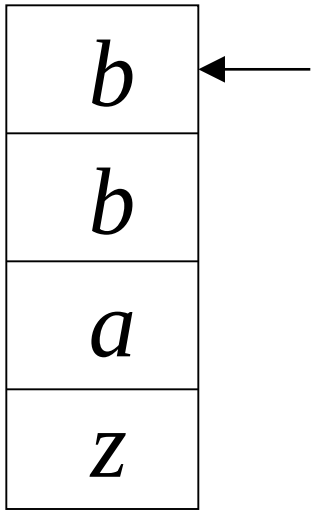
Time 2



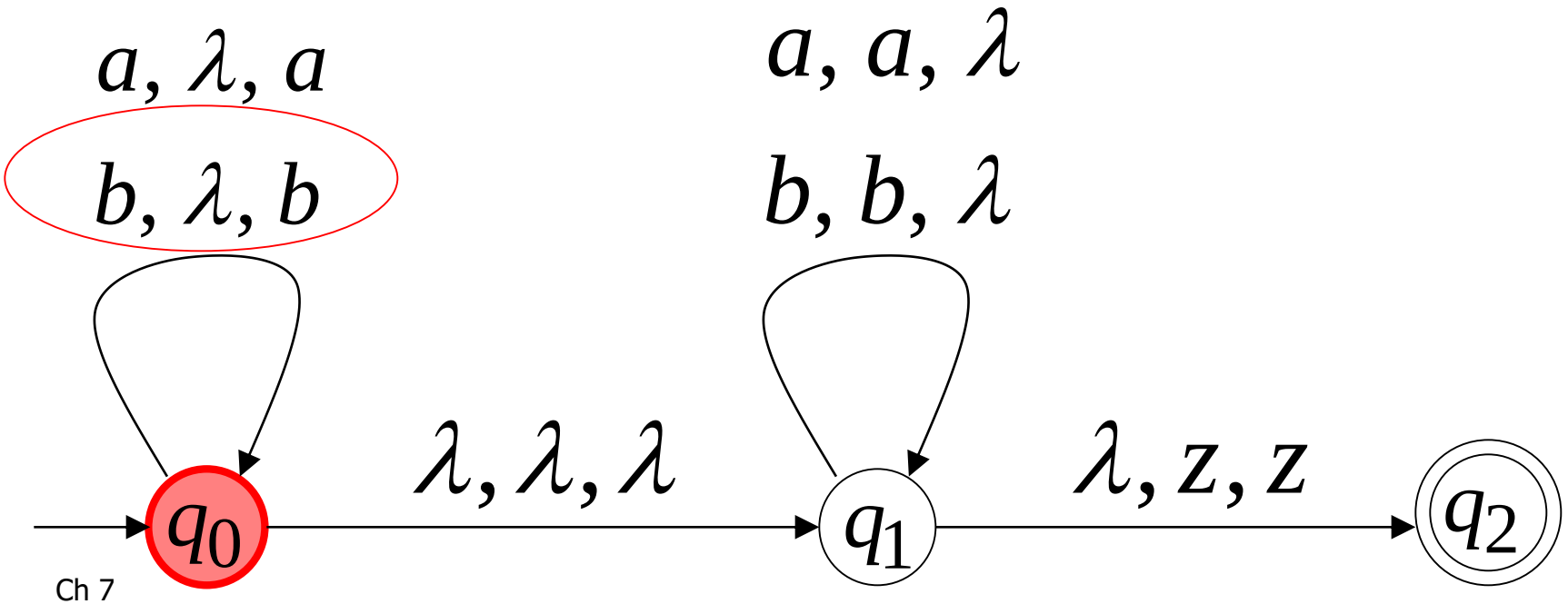
Time 3

Input



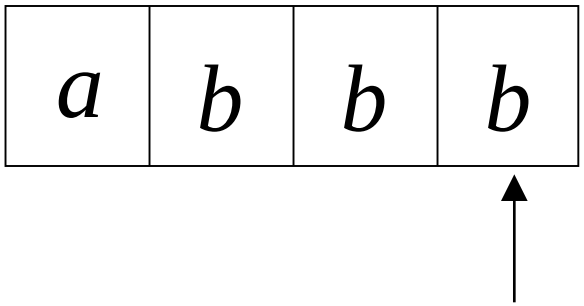


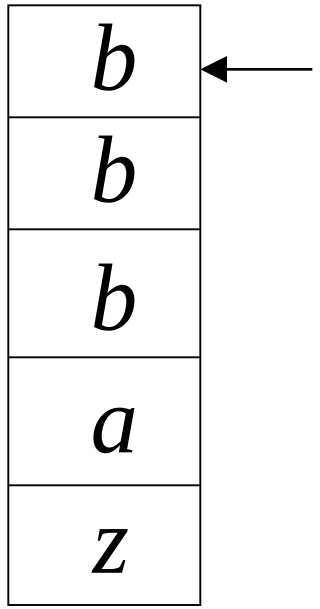
Stack



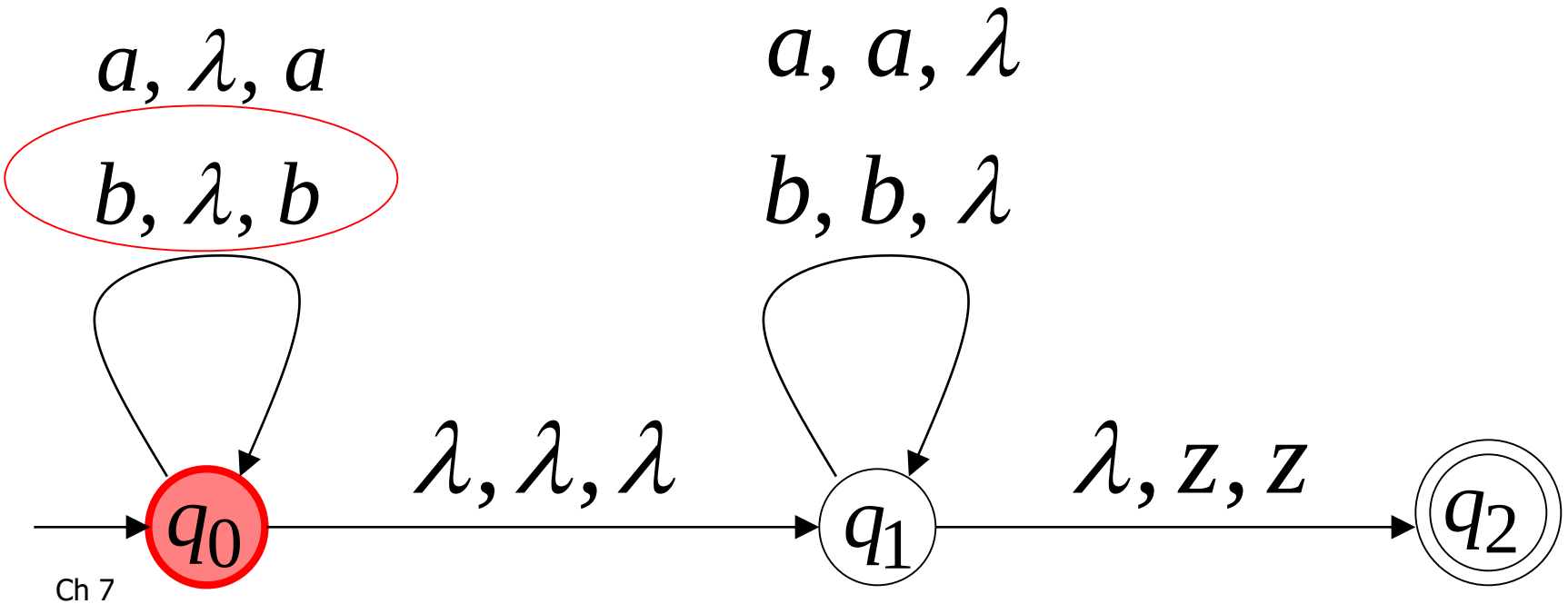
Time 4

Input



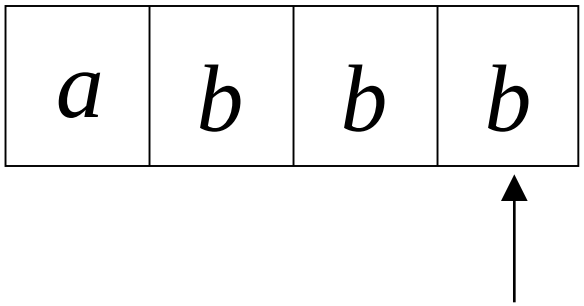


Stack

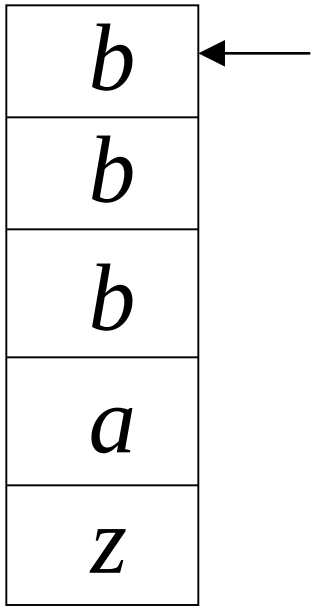


Time 5

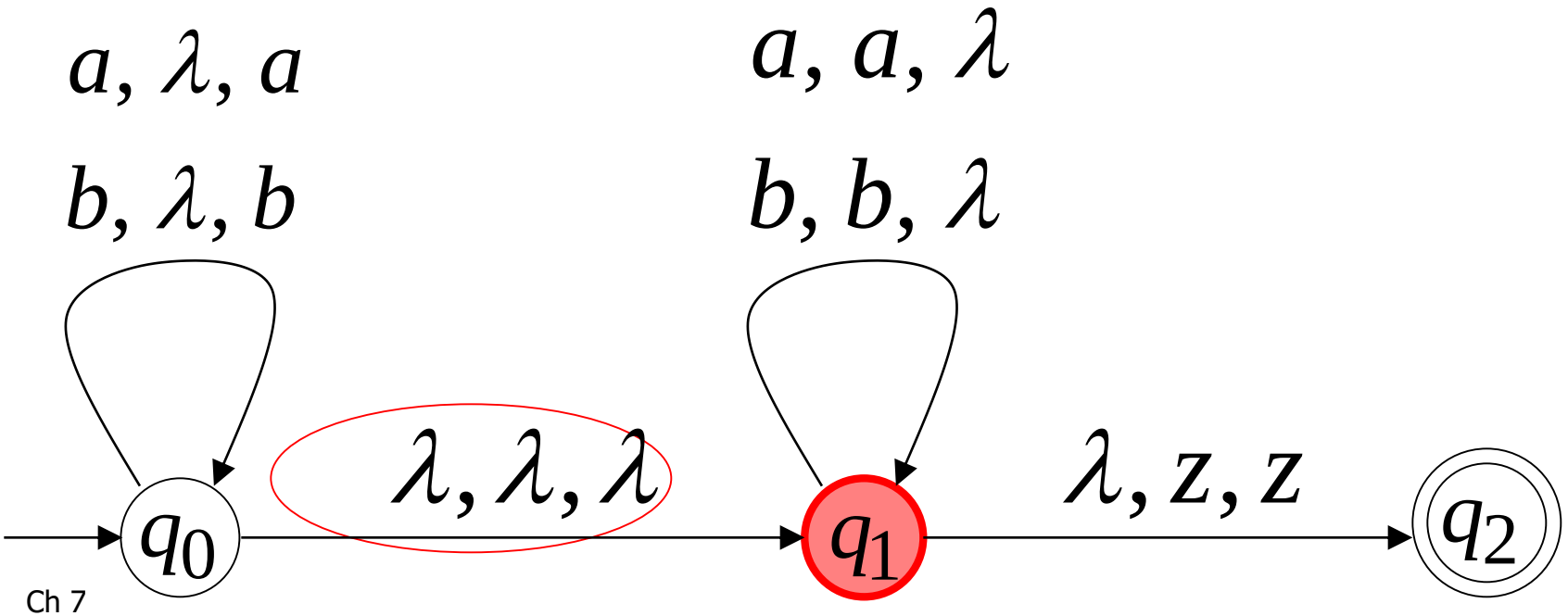
Input



No final state
is reached

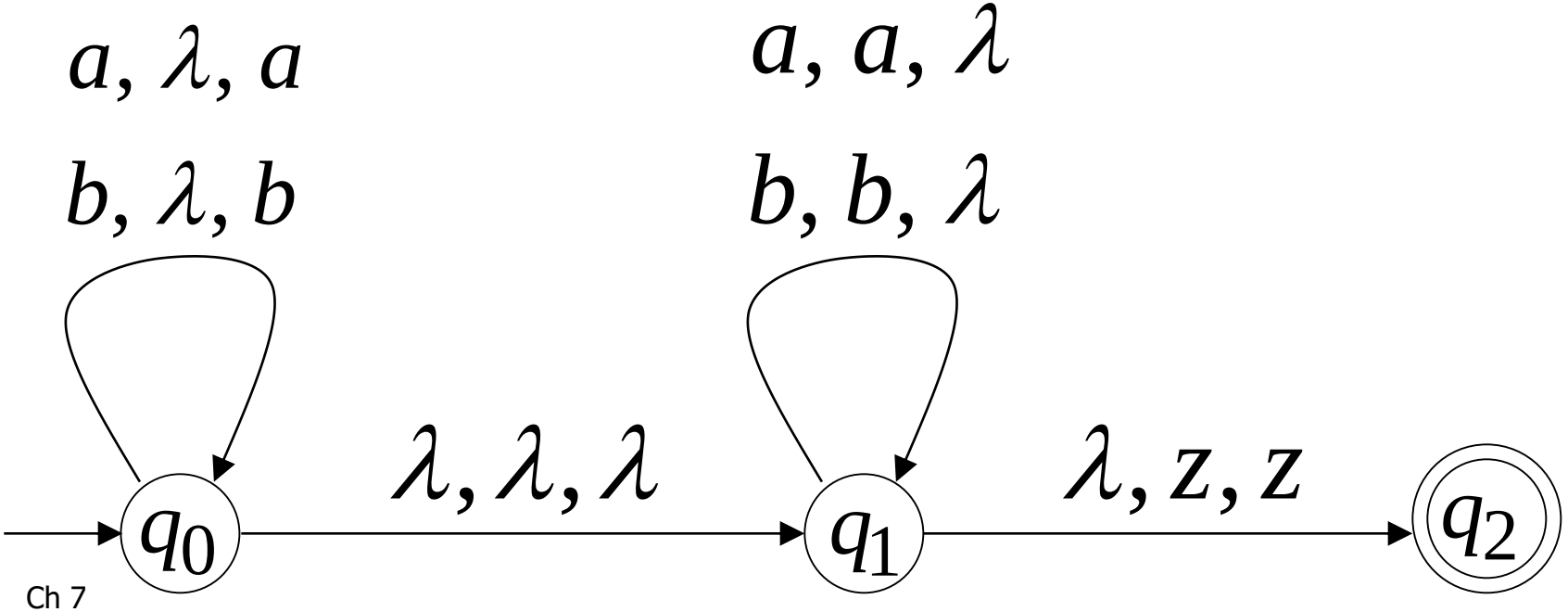


Stack



There is no computation that accepts string $abbb$

$$abbb \notin L(M)$$



A string is rejected if there is
NO computation such that:

All the input is consumed

AND

The last state is a final state

At the end of the computation,
we do not care about the stack contents

In other words, a string is rejected if in every computation with this string:

The input cannot be consumed

OR

The input is consumed and the last state is not a final state

OR

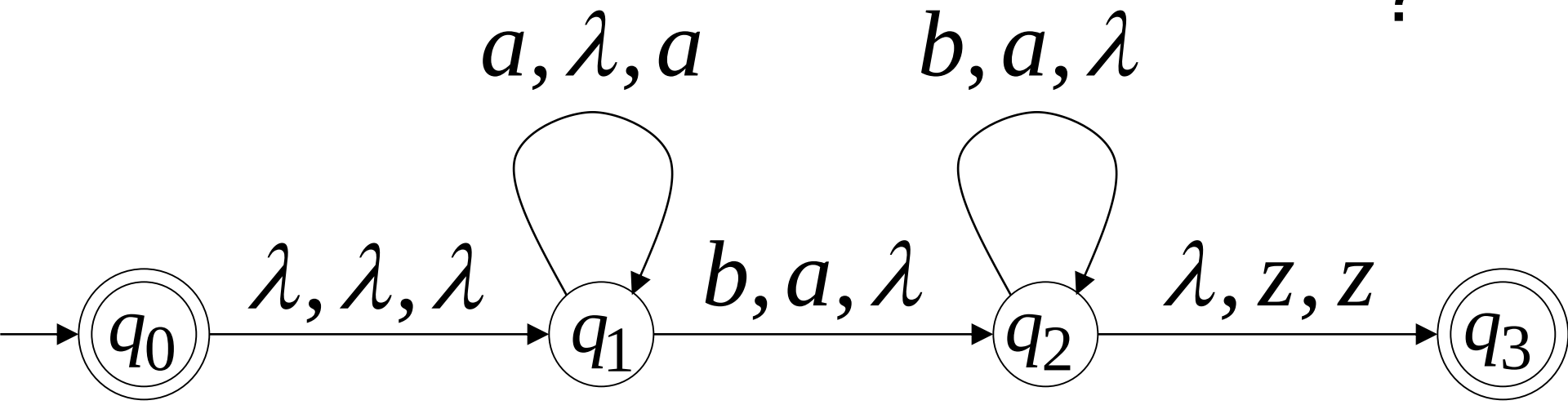
The stack head moves below the bottom of the stack

Another NPDA example

NPDA M

$$L(M) = \{a^n b^m \ : \ n \geq m-1\}$$

?



Another NPDA example

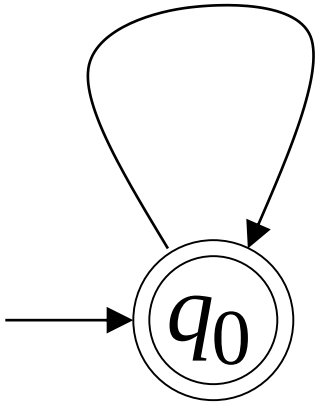
NPDA M

$$L(M) = \{a^n b^m \ : \ n_a \geq n_b - 1\}$$

a, λ, a

b, a, λ

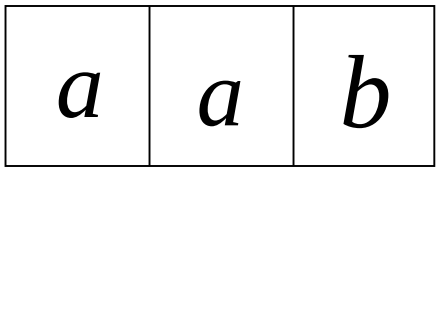
b, z, λ



Execution Example:

Time 0

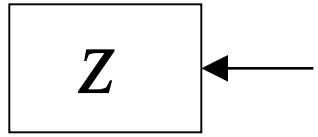
Input



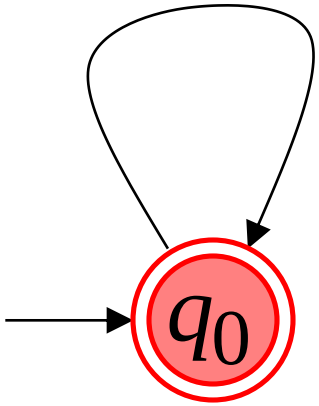
a, λ, a

b, a, λ

b, z, λ

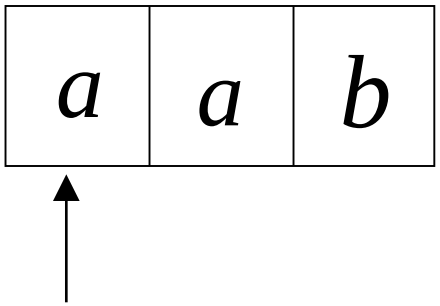


Stack



Time 1

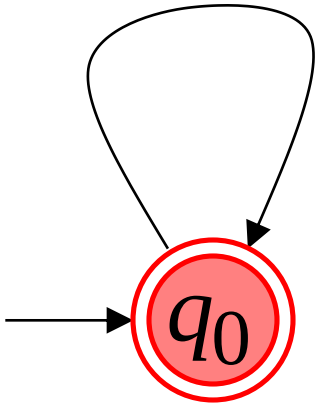
Input

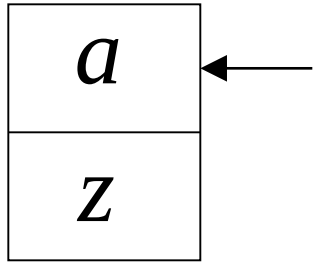


a, λ, a

b, a, λ

b, z, λ

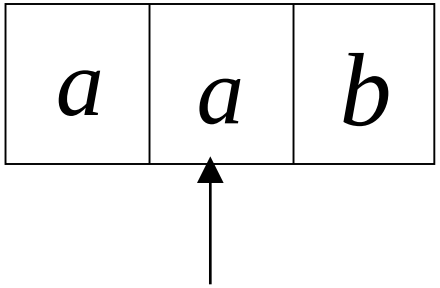




Stack

Time 2

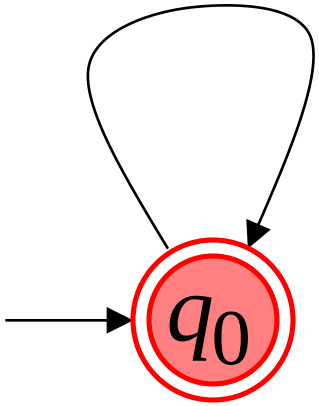
Input

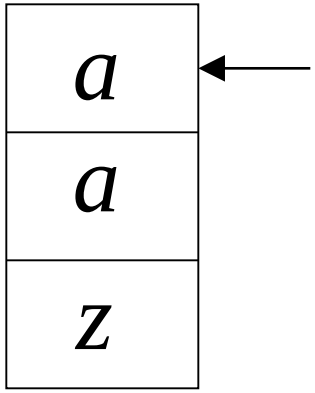


a, λ, a

b, a, λ

b, z, λ

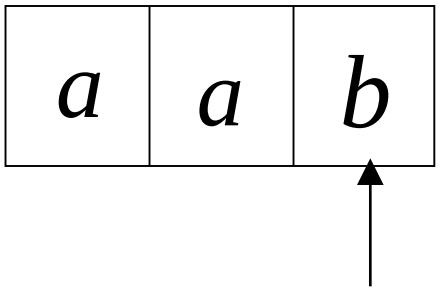




Stack

Time 3

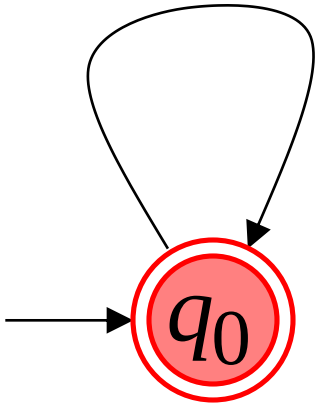
Input

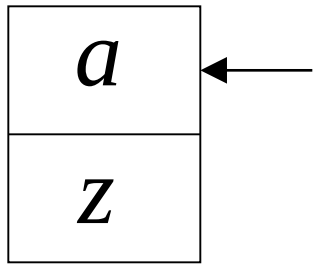


a, λ, a

b, a, λ

b, z, λ





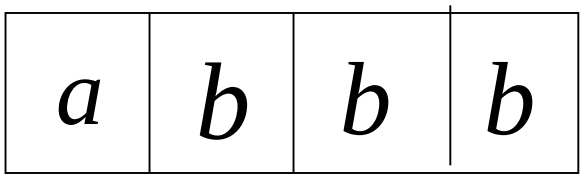
Stack

accept

Rejection example:

Time 0

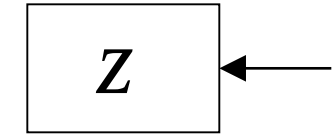
Input



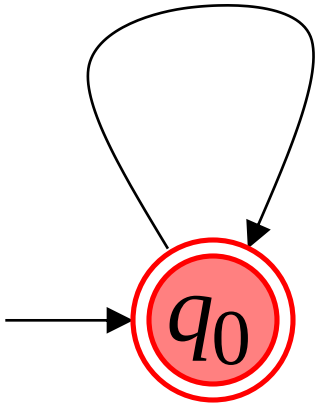
a, λ, a

b, a, λ

b, z, λ

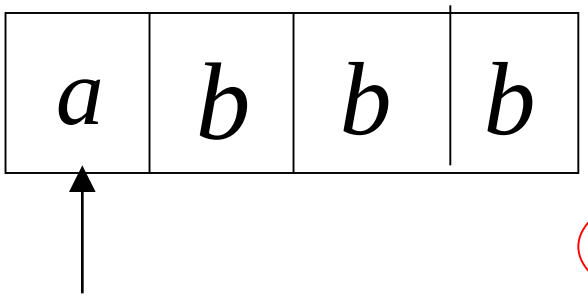


Stack



Time 1

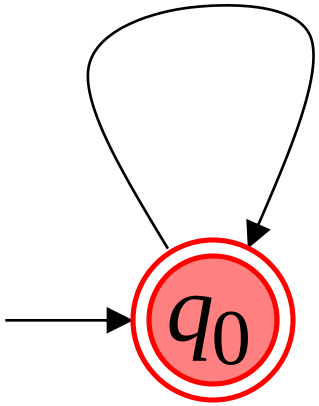
Input

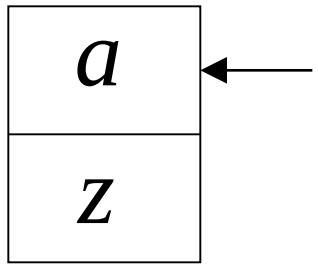


a, λ, a

b, a, λ

b, z, λ

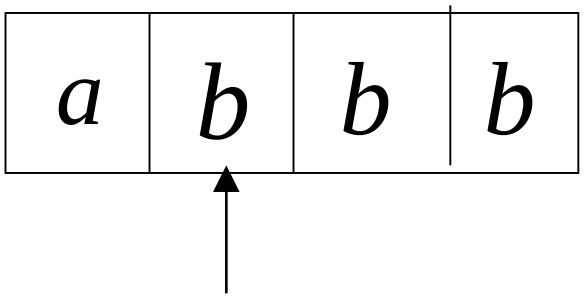




Stack

Time 2

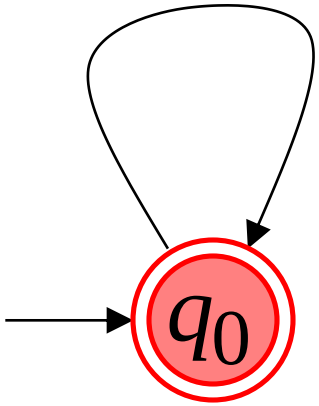
Input

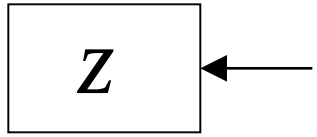


a, λ, a

b, a, λ

b, z, λ

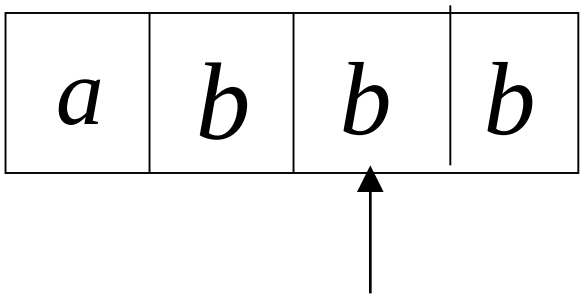




Stack

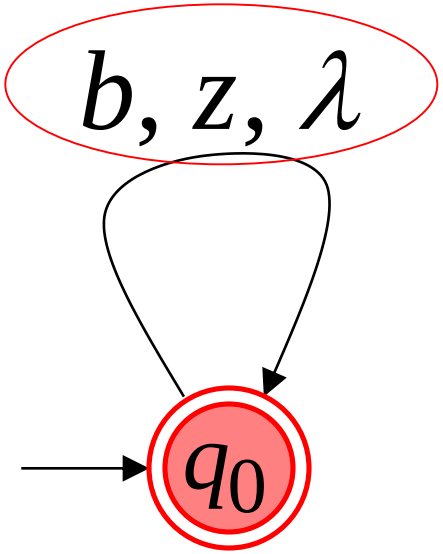
Time 3

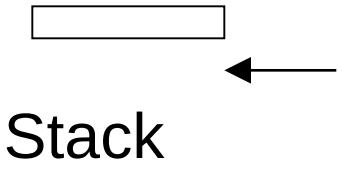
Input



a, λ, a

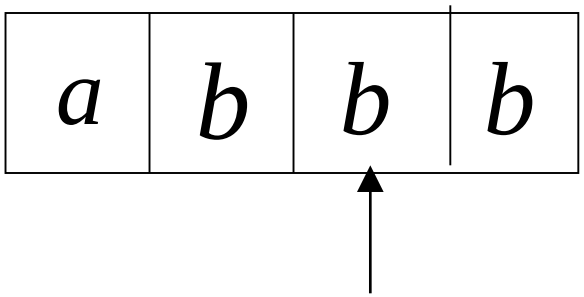
b, a, λ





Time 4

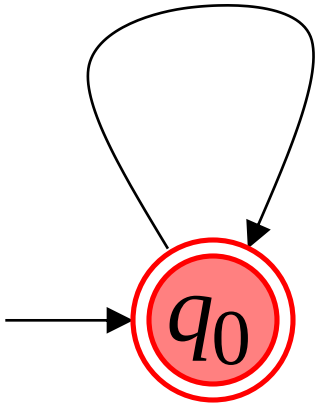
Input

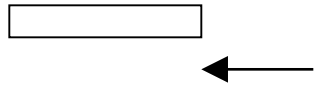


a, λ, a

b, a, λ

b, z, λ





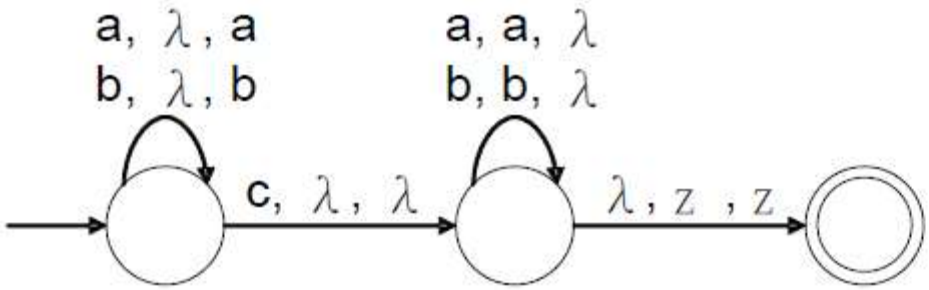
Stack

Halt and Reject

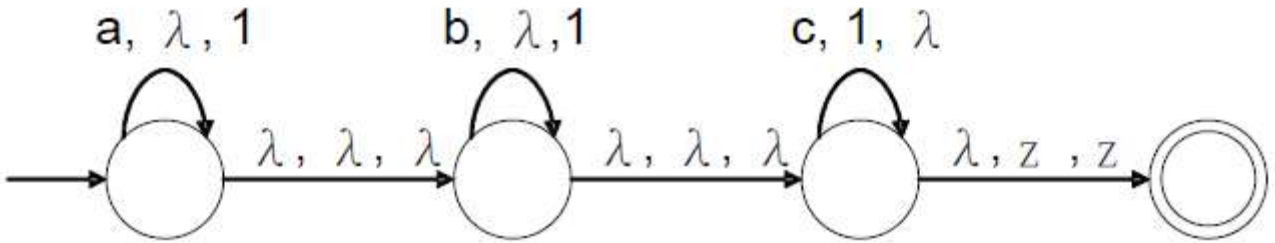
Quiz

- $L=\{wcw^R : w \in \{a,b\}^*\}$
- $L=\{a^n b^m c^{n+m} : n \geq 0, m \geq 0\}$

(b)



(c)



Outline



Nondeterministic Pushdown Automata



Pushdown Automata and Context-Free Languages



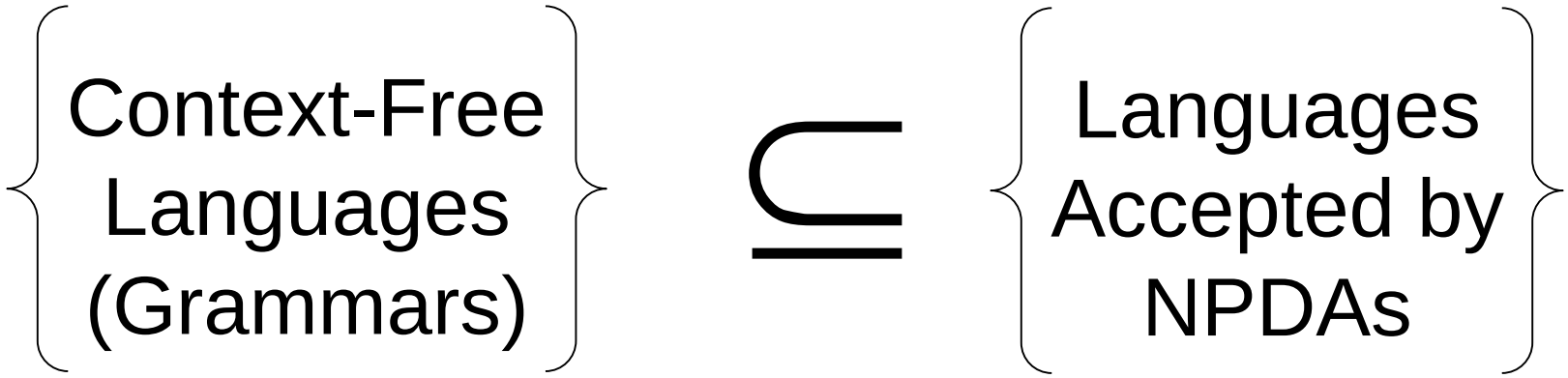
Deterministic Pushdown Automata and Deterministic CFLs

NPDAs Accept Context-Free Languages

Theorem:

$$\left\{\begin{array}{c}\text{Context-Free}\\ \text{Languages}\\ \text{(Grammars)}\end{array}\right\} = \left\{\begin{array}{c}\text{Languages}\\ \text{Accepted by}\\ \text{NPDAs}\end{array}\right\}$$

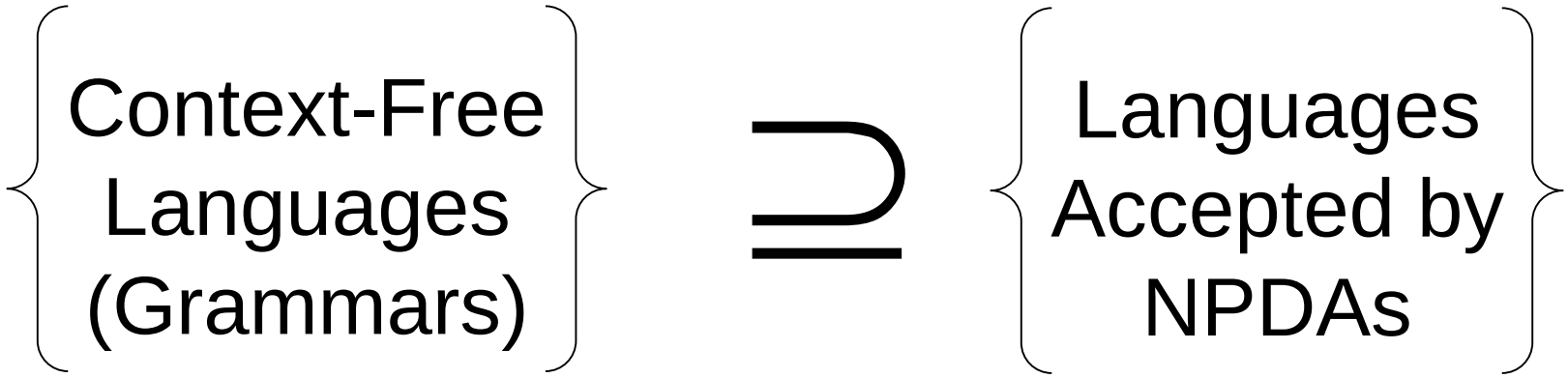
Proof - Step 1:



Theorem 7.1

Convert any context-free grammar G to an NPDA M with: $L(G) = L(M)$

Proof - Step 2:



Theorem 7.2

Convert any NPDA M to a context-free grammar G with: $L(G) = L(M)$

Proof - step 1

Converting Context-Free Grammars to NPDAs

We will convert any context-free grammar G

to an NPDA automaton M

Such that:

M simulates leftmost derivations of G

Assume G in Greibach normal form

$$A \rightarrow a\, V_1 V_2 \cdots V_k$$

Top of stack

Input

New Variable in stack

Leftmost derivation

G :

$$S \Rightarrow \cdots\cdots \Rightarrow \sigma_1\sigma_2\cdots\sigma_k V_1 V_2 \cdots V_m \Rightarrow \cdots$$

Input processed

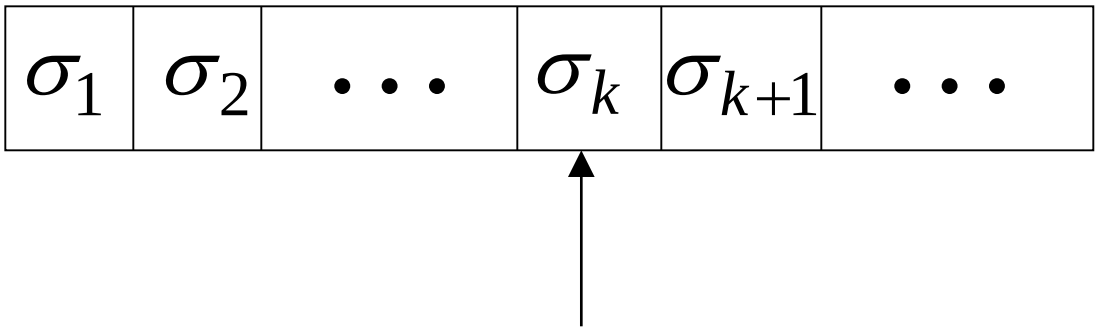
Stack contents

leftmost variable

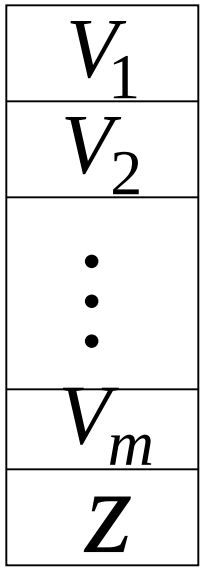
M :

Simulation of derivation

Input



Stack



Leftmost derivation

G:

$$S \Rightarrow \cdots\cdots\cdots \Rightarrow \sigma_1\sigma_2\cdots\sigma_n$$

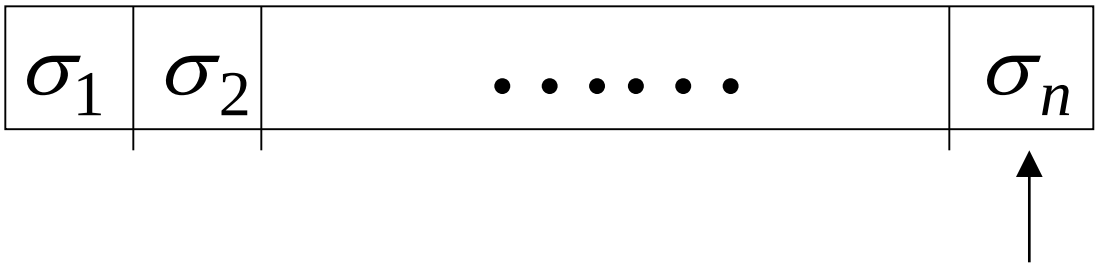
string of terminals

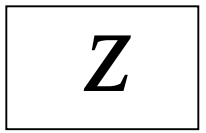
M:

Simulation of derivation

Stack

Input





end of input is reached

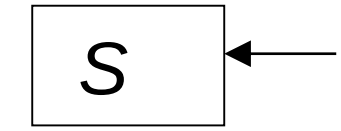
Example 7.6

$S \rightarrow aSbb \mid a$ Greibach Normal Form

$S \rightarrow aSA \mid a$

$A \rightarrow bB$

$B \rightarrow b$



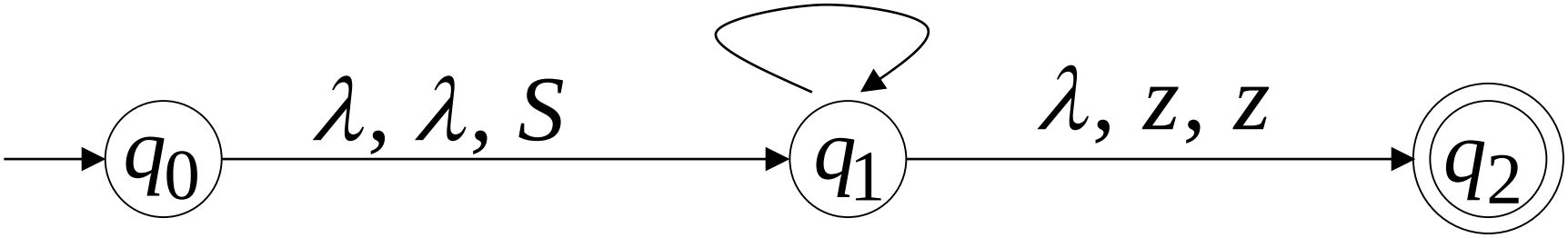
Stack

a, S, SA

a, S, λ

b, A, B

b, B, λ



Example 7.7

$S \rightarrow aA$

$A \rightarrow aABC \mid bB \mid a$

$B \rightarrow b$

$C \rightarrow c$

a, S, A

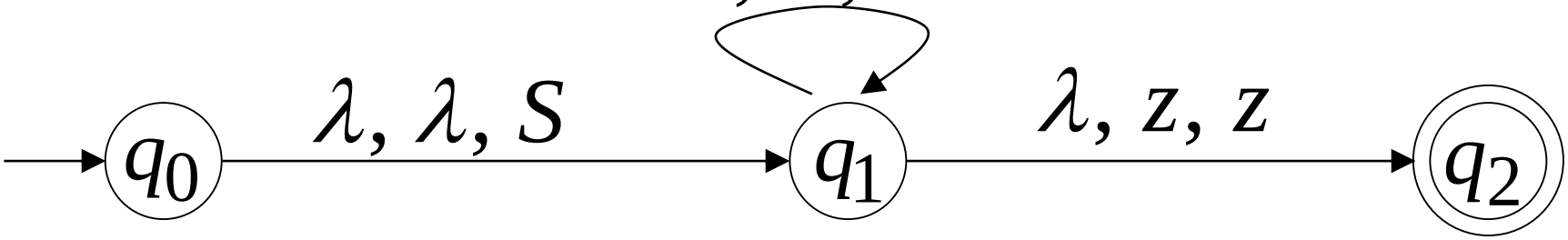
a, A, ABC

a, A, λ

b, A, B

b, B, λ

c, C, λ



Yet another approach…

An example grammar:

$$S \rightarrow aSTb$$

$$S \rightarrow b$$

$$T \rightarrow Ta$$

$$T \rightarrow \lambda$$

What is the equivalent NPDA?

Grammar:

$$S \to aSTb$$

$$S \to b$$

$$T \to Ta$$

$$T \to \lambda$$

A leftmost derivation:

$$S \Rightarrow aSTb \Rightarrow abTb \Rightarrow abTab \Rightarrow abab$$

$$S \Rightarrow aSTb \Rightarrow abTb \Rightarrow abTab \Rightarrow abab$$

Grammar:

$S \rightarrow aSTb$

$S \rightarrow b$

$T \rightarrow Ta$

$T \rightarrow \lambda$

NPDA:

$\lambda, S, aSTb$

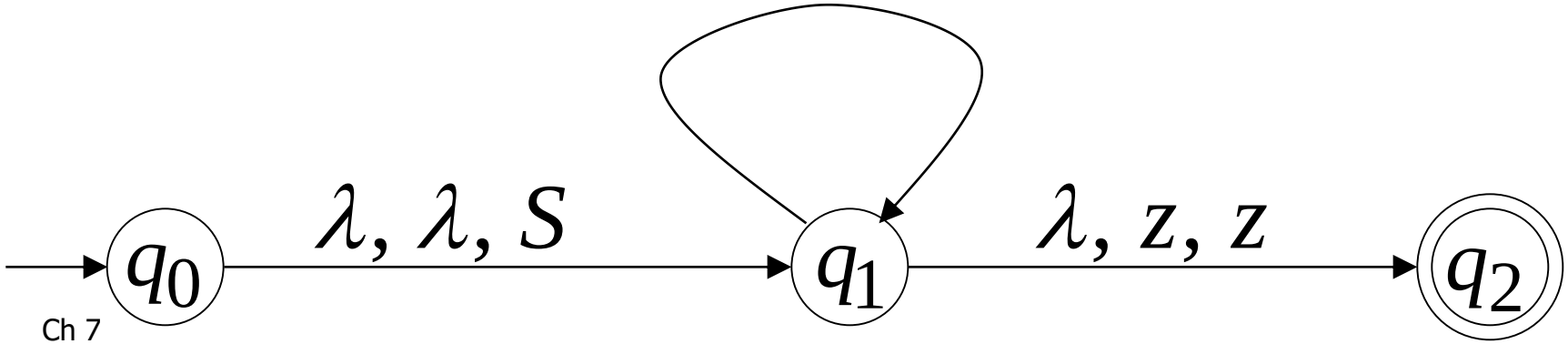
λ, S, b

λ, T, Ta

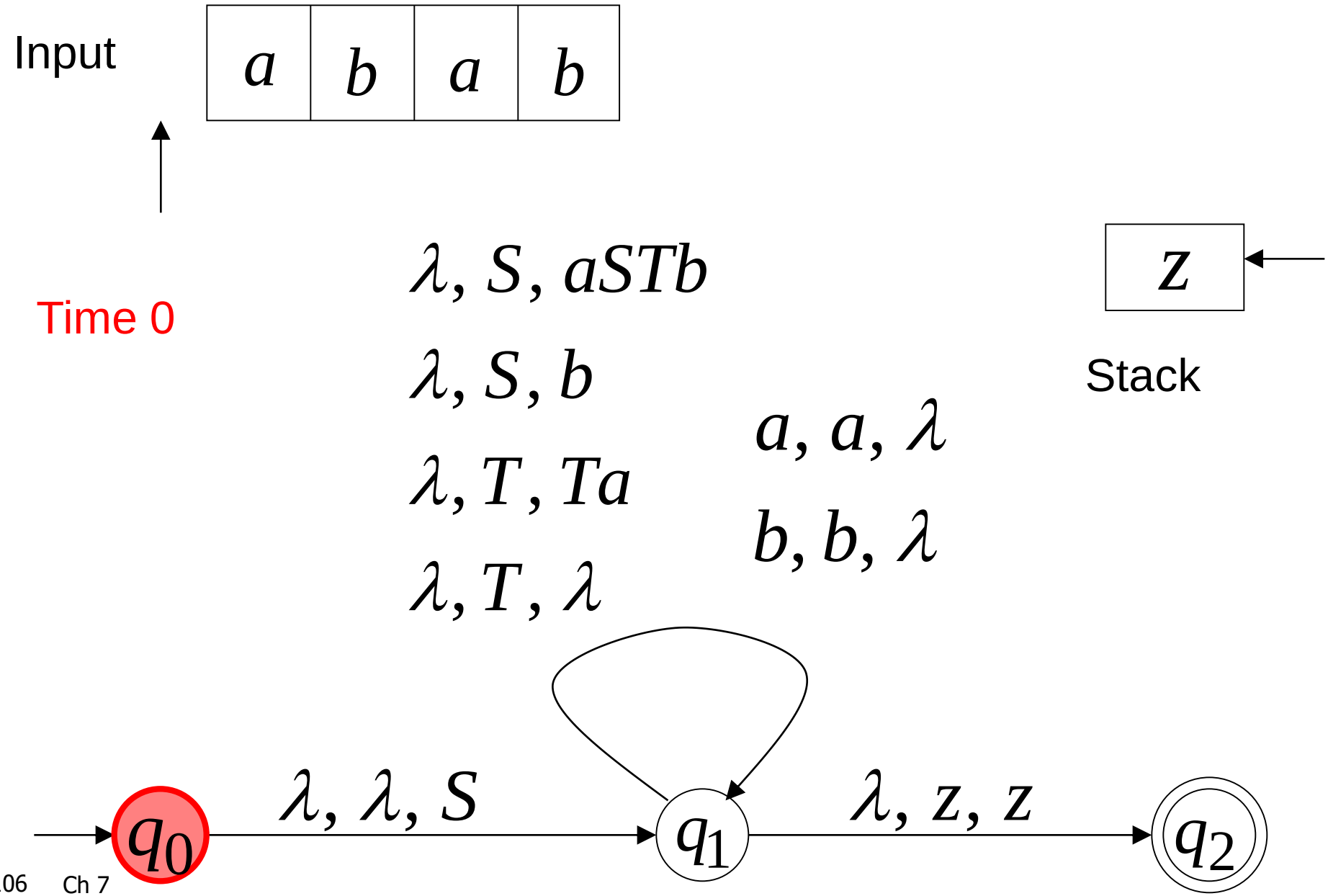
λ, T, λ

a, a, λ

b, b, λ



Derivation:


Input
a
b
a
b
Time 0
λ, S, aSTb
λ, S, b
λ, T, Ta
λ, T, λ
a, a, λ
b, b, λ
Z
Stack
q0
λ, λ, S
q1
λ, z, z
q2

Derivation: S

Input

a	b	a	b

Time 0

S
z

Stack

$\lambda, S, aSTb$

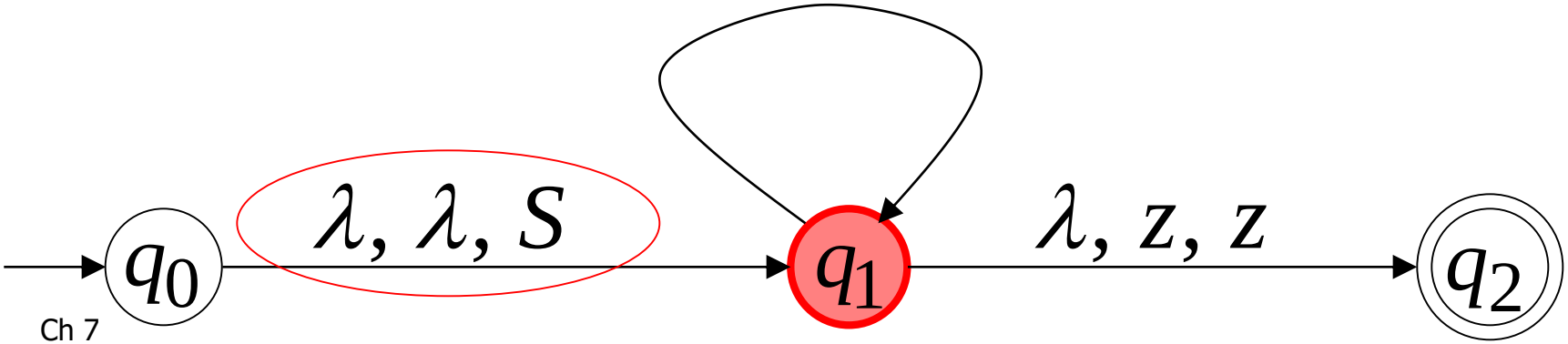
λ, S, b

λ, T, Ta

λ, T, λ

a, a, λ

b, b, λ



Derivation: $S \Rightarrow aSTb$

Input

a	b	a	b

a
S
T
b
z

Stack

Time 1

$\lambda, S, aSTb$

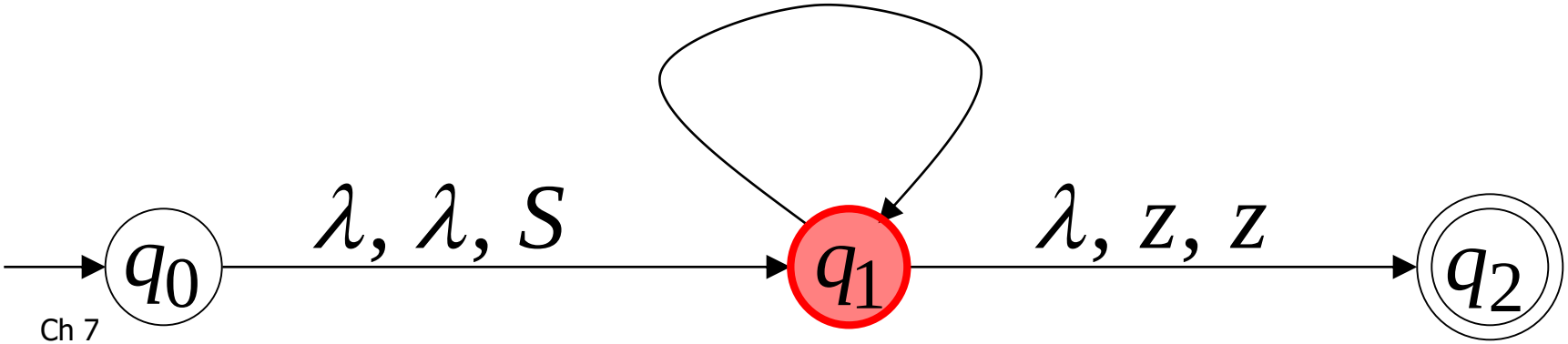
λ, S, b

λ, T, Ta

λ, T, λ

a, a, λ

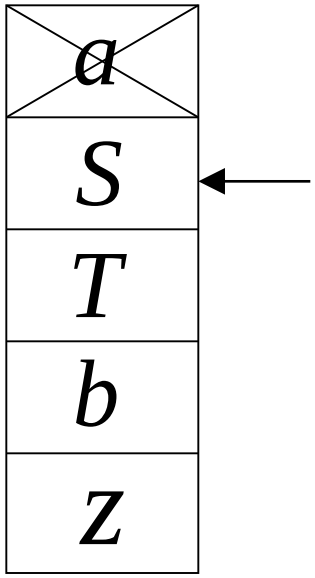
b, b, λ



Derivation: $S \Rightarrow aSTb$

Input

a	b	a	b



Stack

Time 2

$\lambda, S, aSTb$

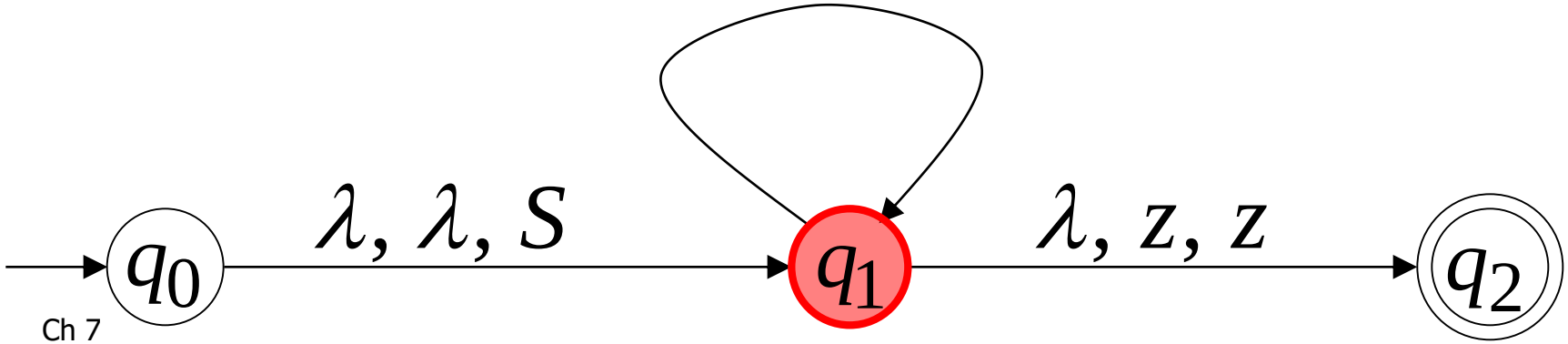
λ, S, b

λ, T, Ta

λ, T, λ

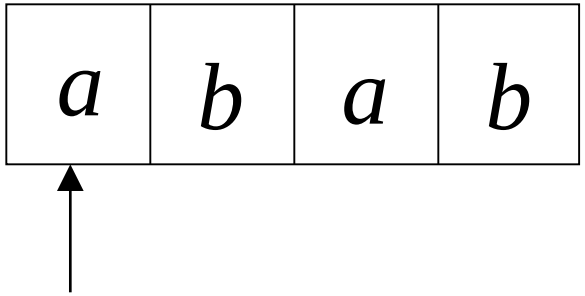
a, a, λ

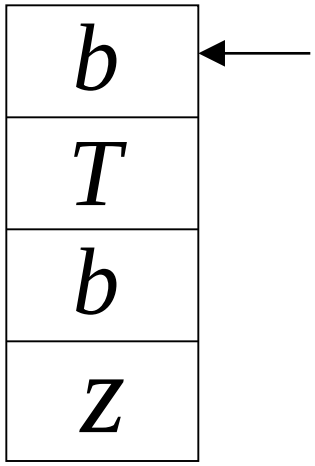
b, b, λ



Derivation: $S \Rightarrow aSTb \Rightarrow abTb$

Input





Stack

Time 3

$\lambda, S, aSTb$

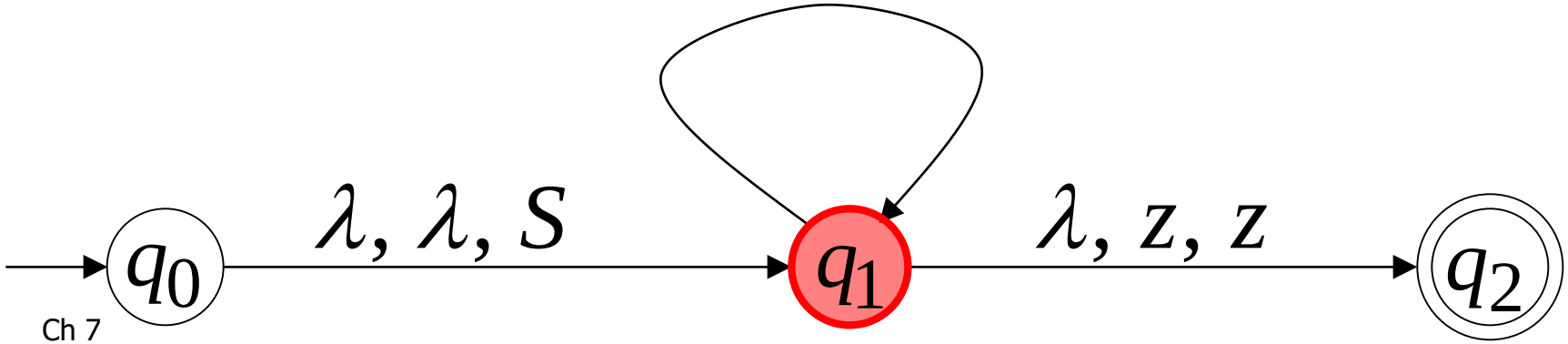
λ, S, b

λ, T, Ta

λ, T, λ

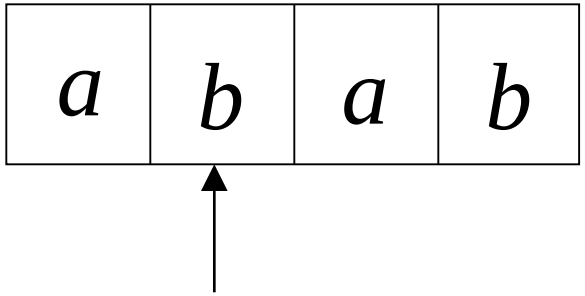
a, a, λ

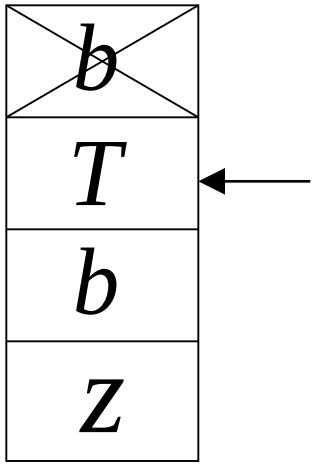
b, b, λ



Derivation: $S \Rightarrow aSTb \Rightarrow abTb$

Input





Stack

Time 4

$\lambda, S, aSTb$

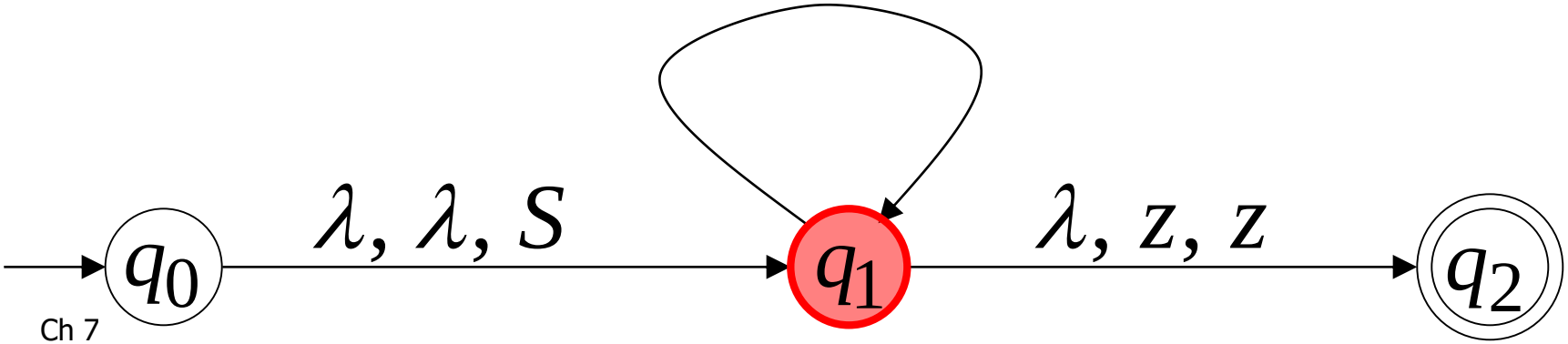
λ, S, b

λ, T, Ta

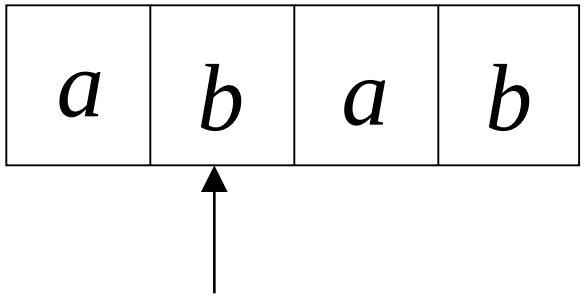
λ, T, λ

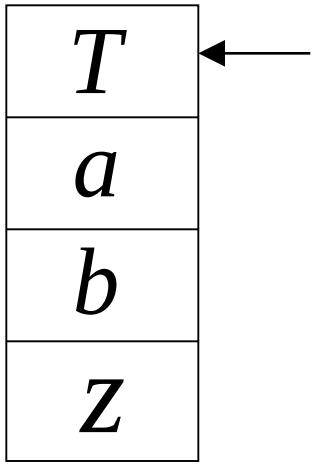
a, a, λ

b, b, λ



Derivation: $S \Rightarrow aSTb \Rightarrow abTb \Rightarrow abTab$





Time 5

$\lambda, S, aSTb$

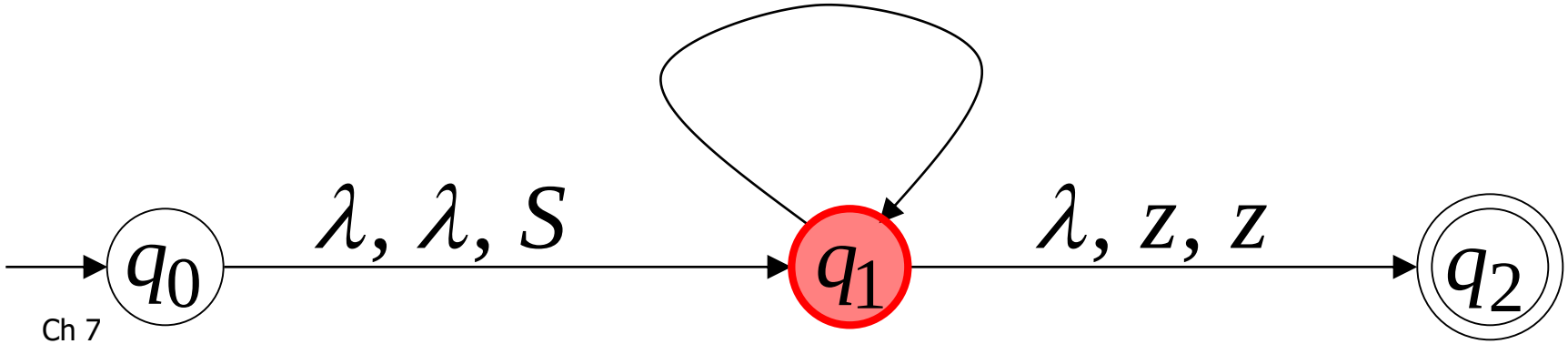
λ, S, b

λ, T, Ta

λ, T, λ

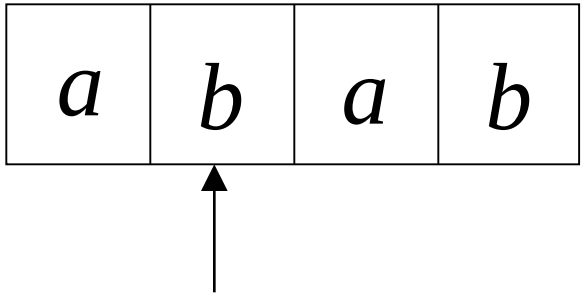
a, a, λ

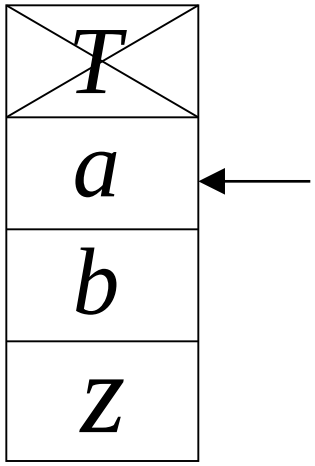
b, b, λ



Derivation: $S \Rightarrow aSTb \Rightarrow abTb \Rightarrow abTab \Rightarrow abab$

Input





Stack

Time 6

$\lambda, S, aSTb$

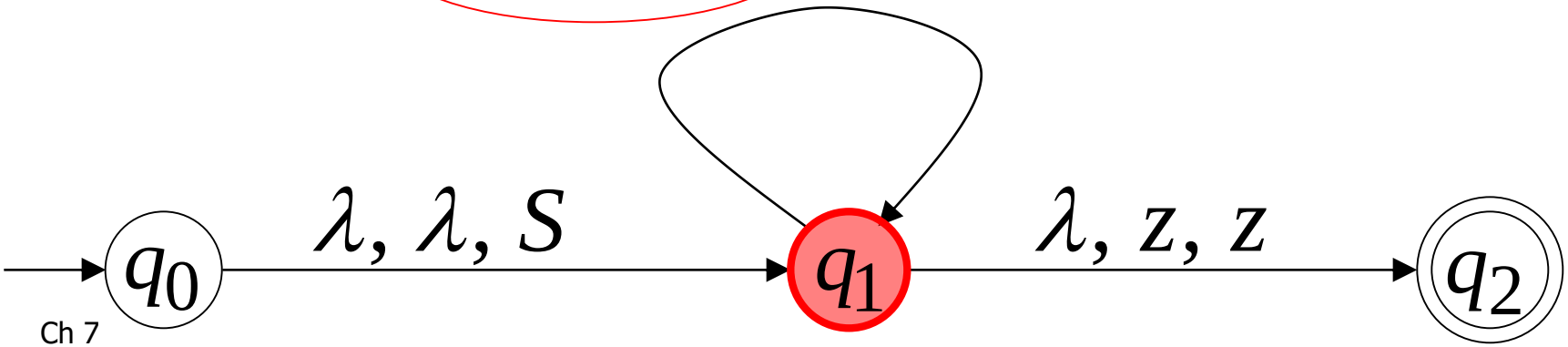
λ, S, b

λ, T, Ta

λ, T, λ

a, a, λ

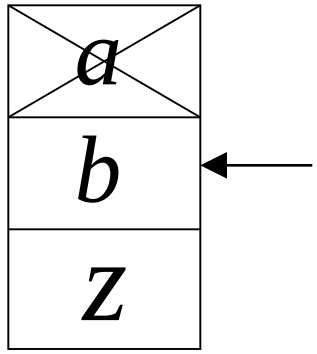
b, b, λ



Derivation: $S \Rightarrow aSTb \Rightarrow abTb \Rightarrow abTab \Rightarrow abab$

Input

a	b	a	b



Stack

Time 7

$\lambda, S, aSTb$

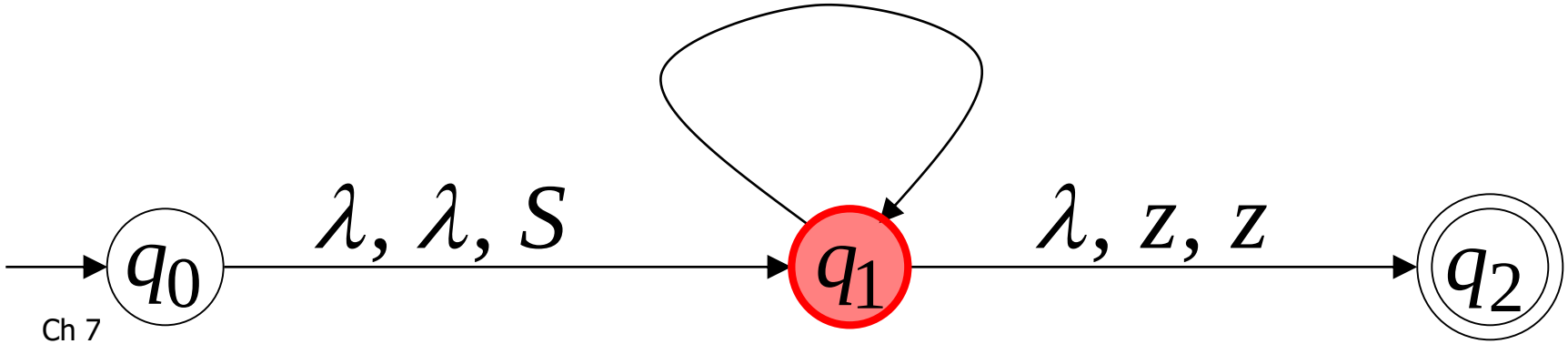
λ, S, b

λ, T, Ta

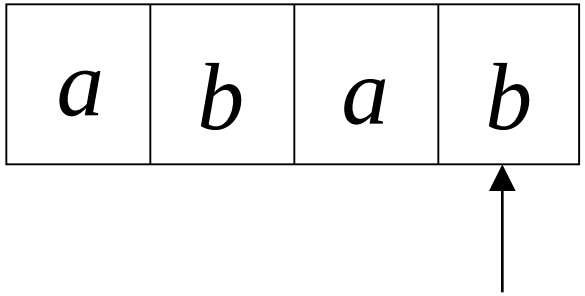
λ, T, λ

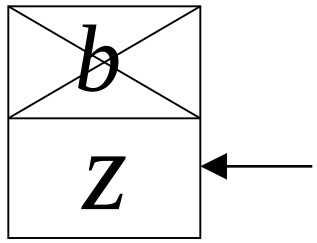
a, a, λ

b, b, λ



Derivation:





Stack

Time 8

$\lambda, S, aSTb$

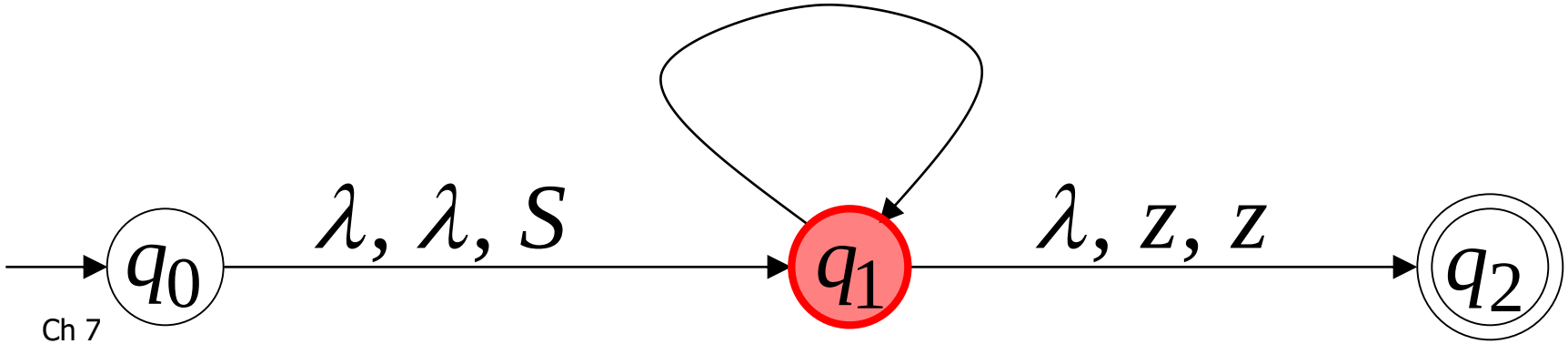
λ, S, b

λ, T, Ta

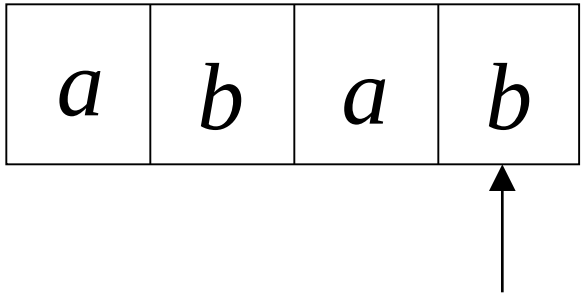
λ, T, λ

a, a, λ

b, b, λ



Derivation:



Time 9

$\lambda, S, aSTb$

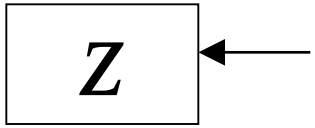
λ, S, b

λ, T, Ta

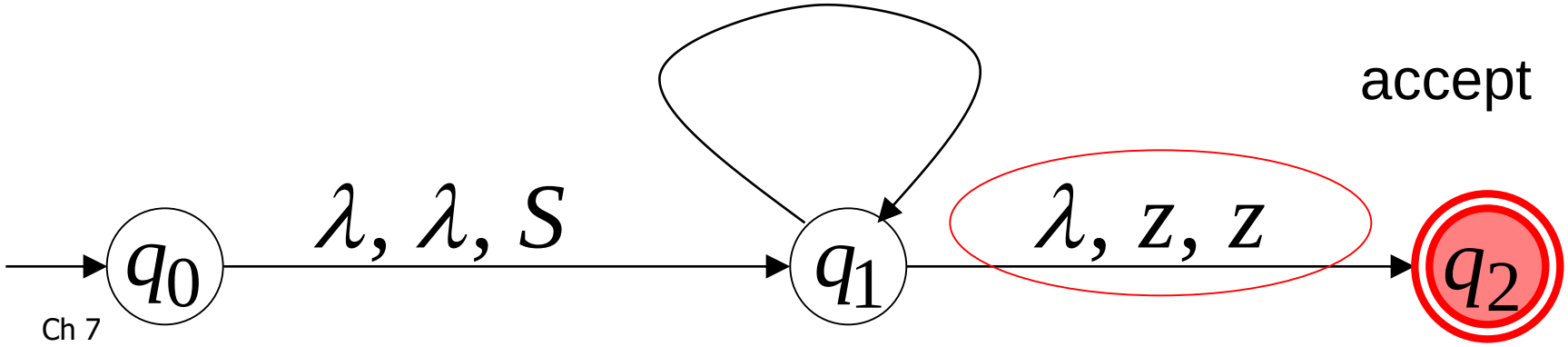
λ, T, λ

a, a, λ

b, b, λ



Stack



Constructing NPDA M from grammar G :

For any production

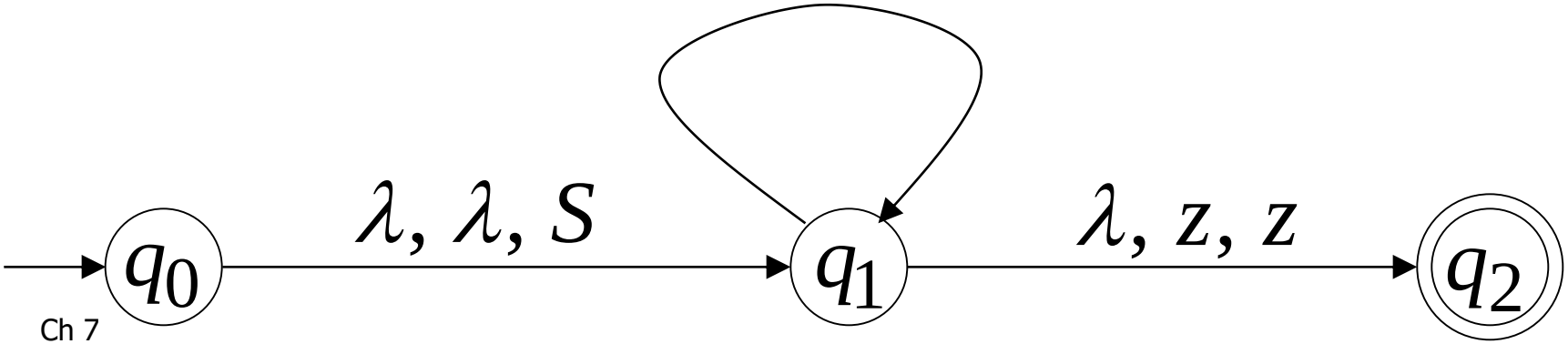
$A \to w$

λ, A, w

For any terminal

a

a, a, λ



In general:

Given any CFG G

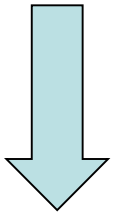
We can construct an NPDA M

With $L(G) = L(M)$

CFG G generates string w

if and only if

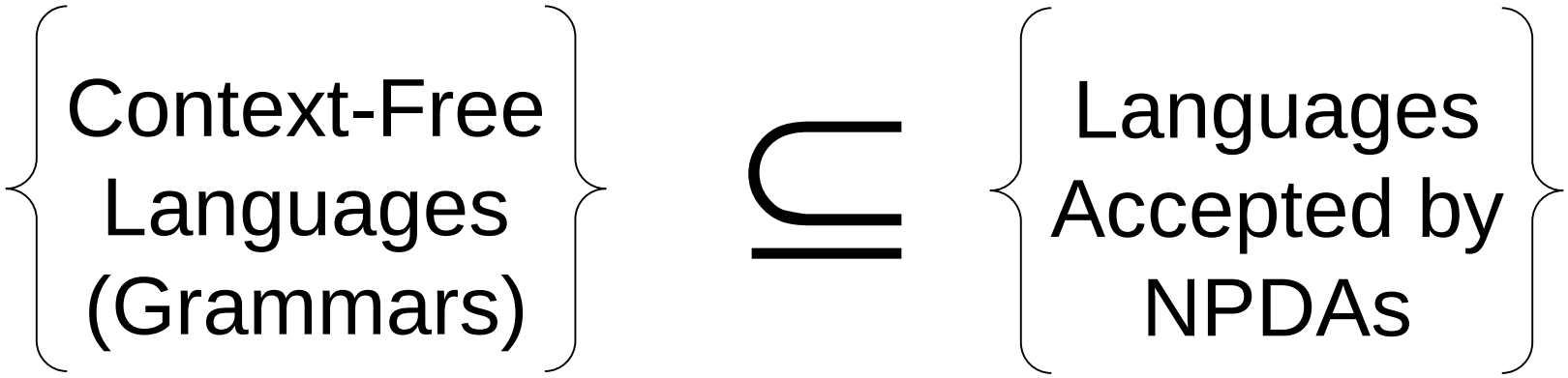
NPDA M accepts w



$$L(G) = L(M)$$

Therefore:

For any context-free language
there is an NPDA
that accepts the same language



Proof - step 2

Converting NPDAs to Context-Free Grammars

For any NPDA M

we will construct

a context-free grammar G with

$$L(M) = L(G)$$

Intuition: The grammar simulates the machine

A derivation in Grammar G :

terminals variables

$$S \Rightarrow \cdots \Rightarrow abc\ldots ABC\ldots \Rightarrow \ldots \Rightarrow abc\ldots$$



Input processed Stack contents

Current configuration in NPDA M

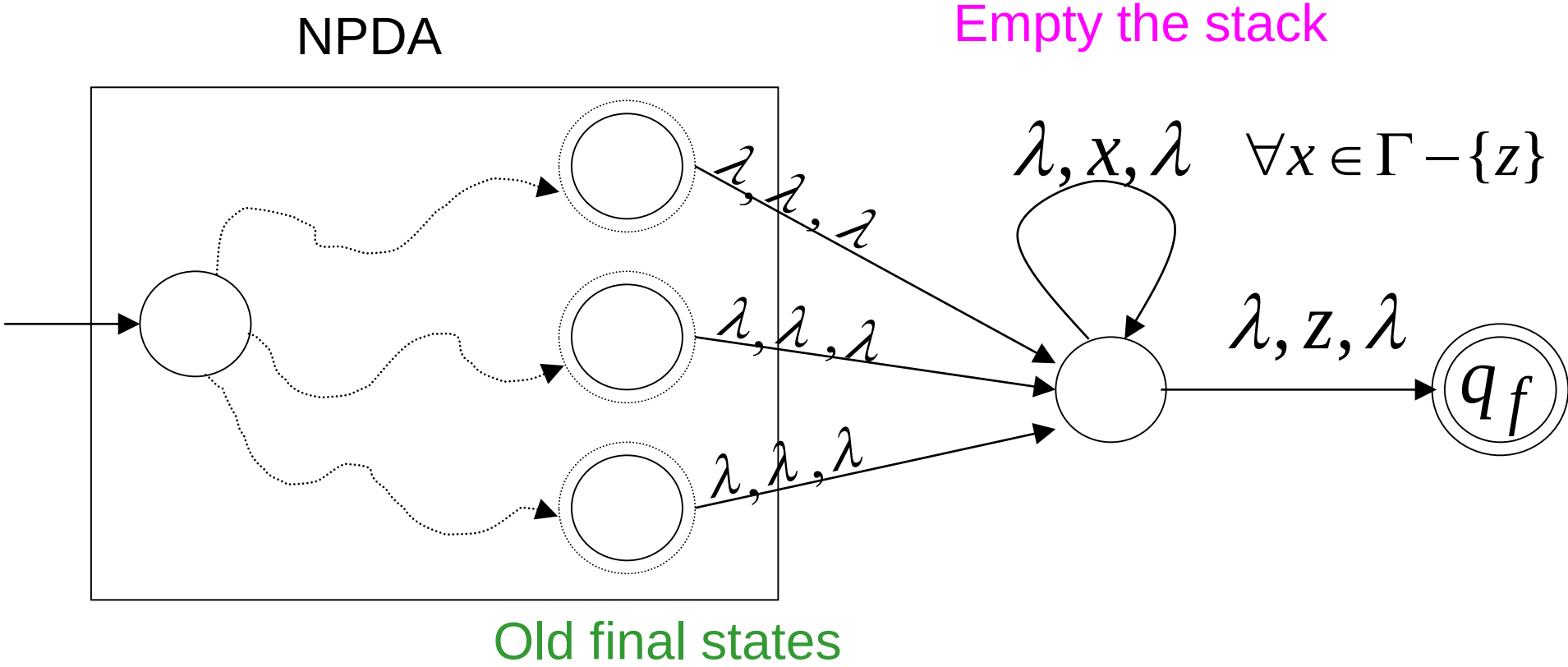
Some Necessary Modifications

Modify (if necessary) the NPDA so that:

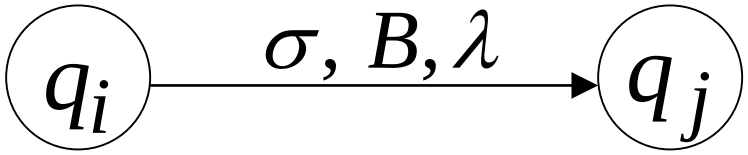
1) It has a single final state
   and empties the stack when it accepts a string

2) Has transitions in a special form

1) Modify the NPDA so that
it empties the stack
and has a unique final state



2) modify the NPDA so that
transitions have the following forms:



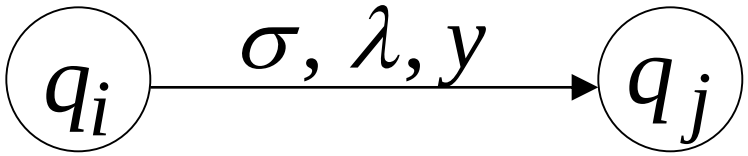
OR

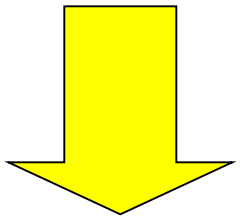
B, C, D : stack symbols

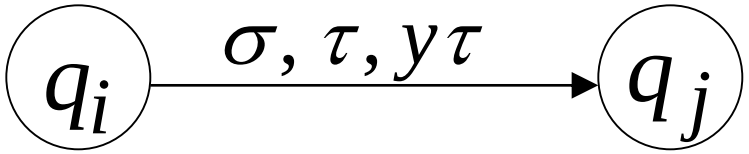


Each move either increases or decreases
the stack content by a single symbol

Convert:

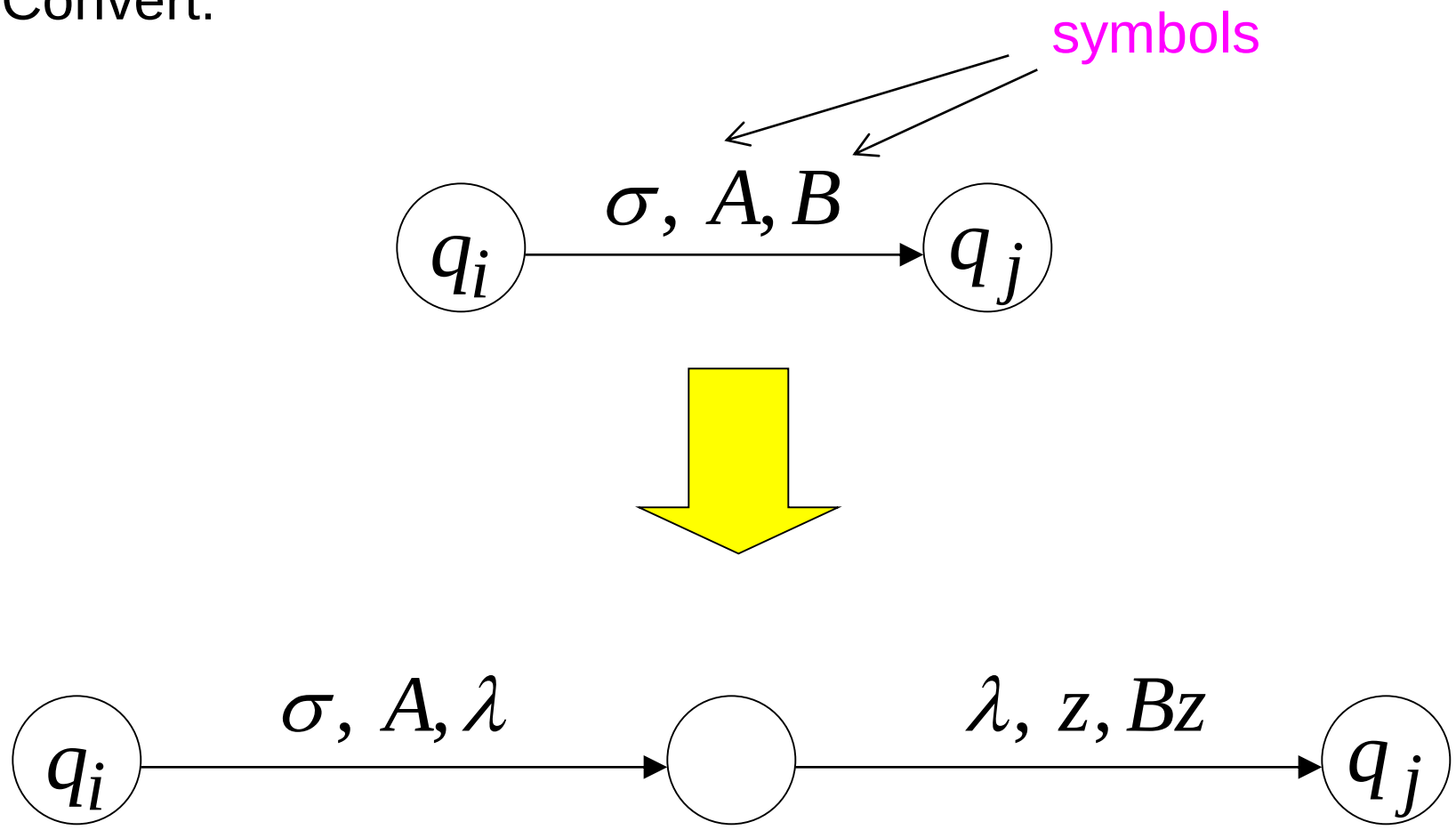




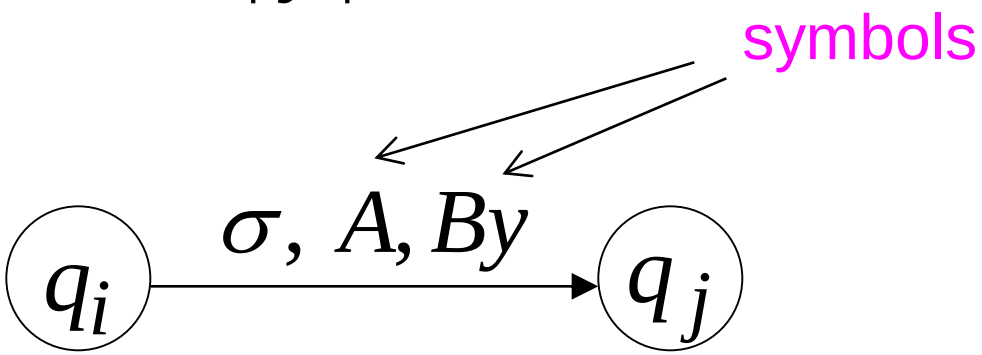


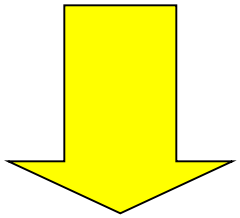
$$\forall \tau \in \Gamma - \{z\}$$

Convert:

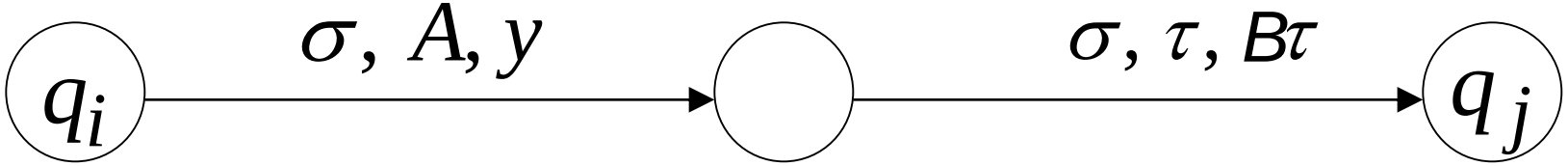


Convert: $|y| \geq 2$





Convert recursively



$\forall \tau \in \Gamma - \{z\}$

Example of an NPDA in correct form:

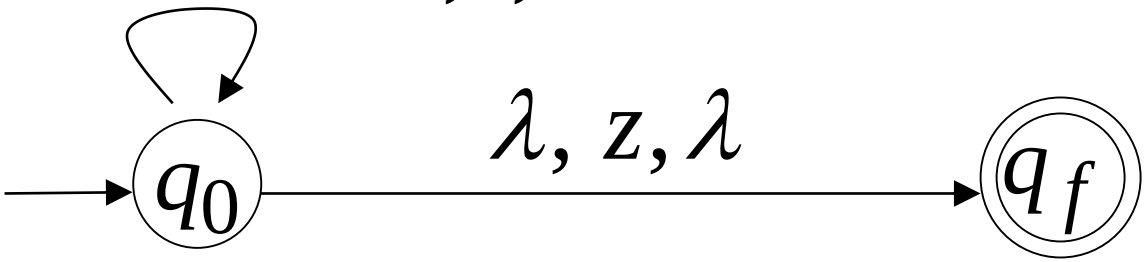
$$L(M) = \{w: \quad n_a = n_b\}$$

z : initial stack symbol

$a,\ z, 0z$ $\qquad$ $b,\ z, 1z$

$a, 0, 00$ $\qquad$ $b, 1, 11$

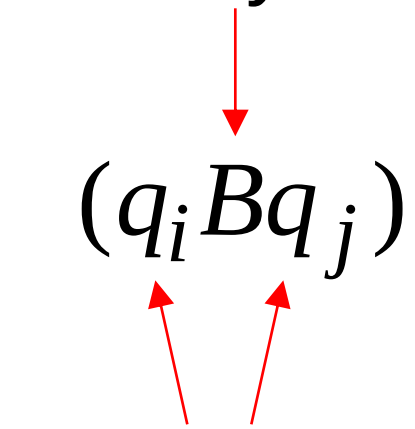
$a, 1, \lambda$ $\qquad$ $b, 0, \lambda$



The Grammar Construction

In grammar G :

Stack symbol



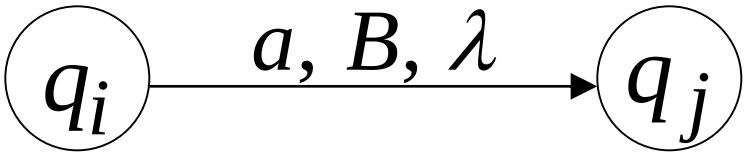
Variables: $(q_i B q_j)$

states

Terminals:

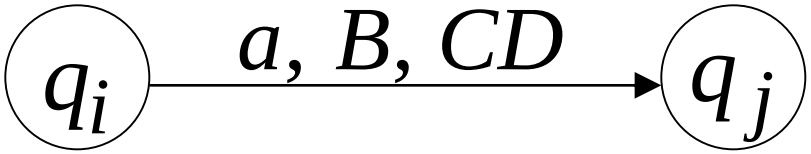
Input symbols of NPDA

For each transition



We add production $(q_i B q_j) \rightarrow a$

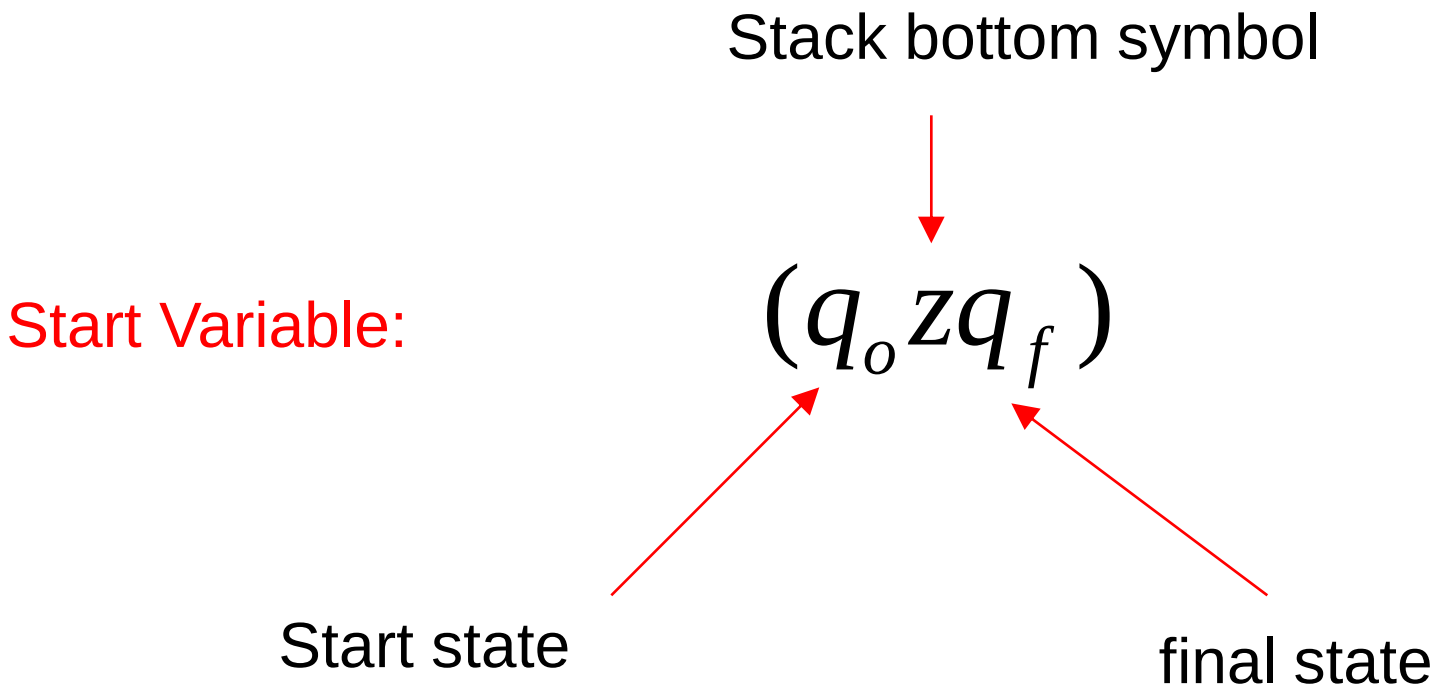
For each transition



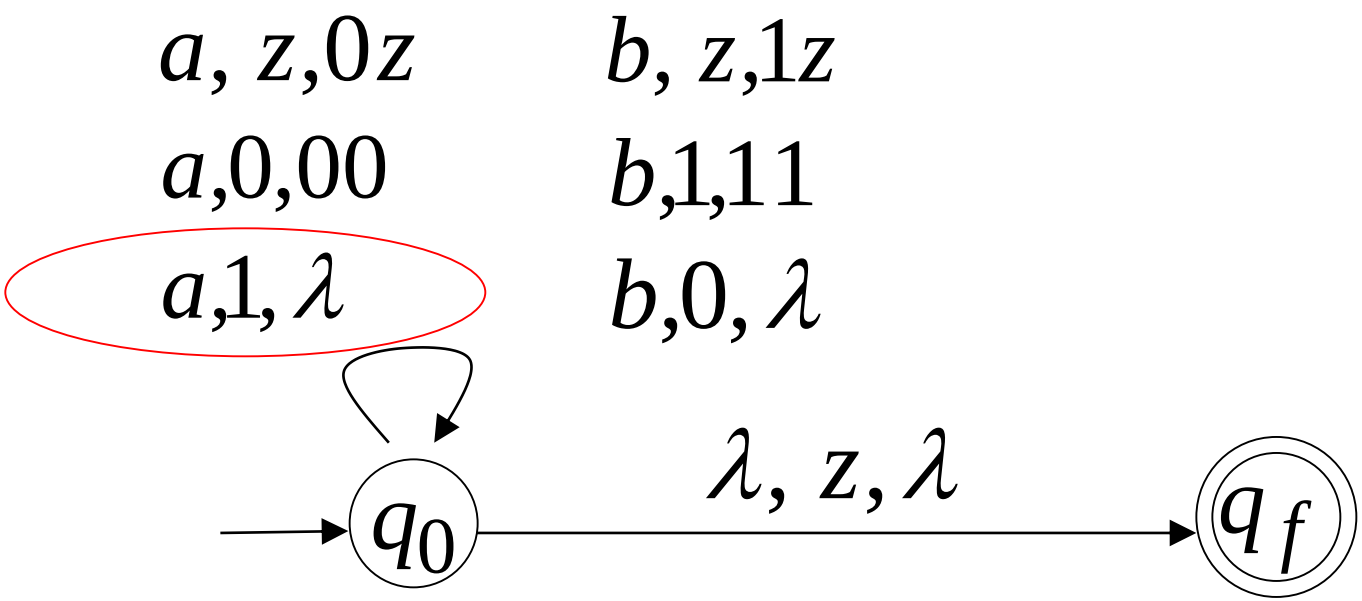
We add productions

$$(q_i B q_k) \rightarrow a(q_j C q_l)(q_l D q_k)$$

For all possible states q_k, q_l
in the automaton


Stack bottom symbol
Start Variable:
$(q_o z q_f)$
Start state
final state

Example:



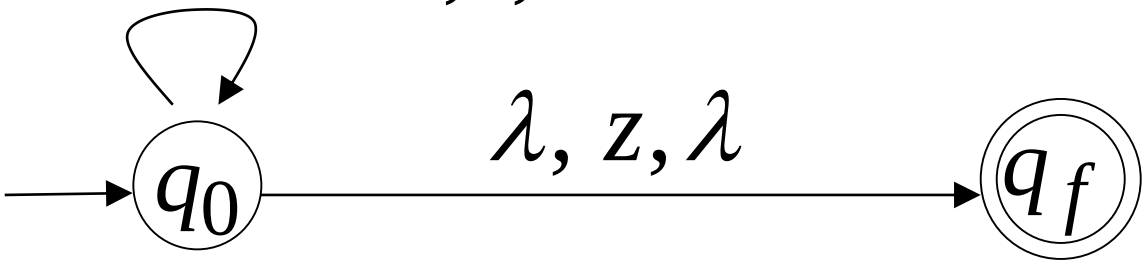
Grammar production: $(q_0 1 q_0) \rightarrow a$

Example:

$a, z, 0z$ $\quad$ $b, z, 1z$

$a, 0, 00$ $\quad$ $b, 1, 11$

$a, 1, \lambda$ $\quad$ $b, 0, \lambda$



Grammar productions:

$$(q_0 z q_0) \rightarrow b(q_0 1 q_0)(q_0 z q_0) \mid b(q_0 1 q_f)(q_f z q_0)$$

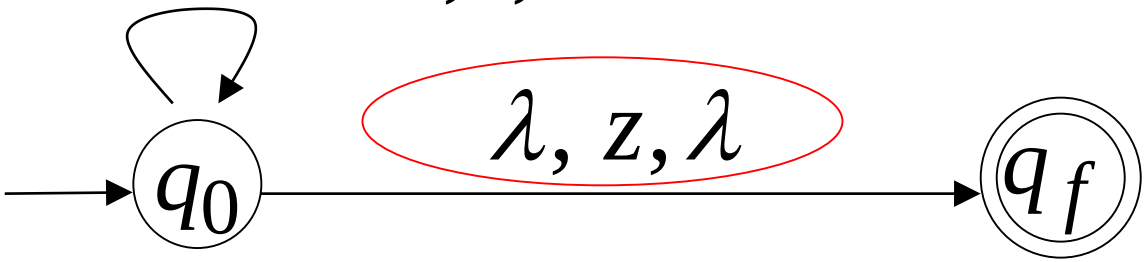
$$(q_0 z q_f) \rightarrow b(q_0 1 q_0)(q_0 z q_f) \mid b(q_0 1 q_f)(q_f z q_f)$$

Example:

$a, z, 0z$ $\quad$ $b, z, 1z$

$a, 0, 00$ $\quad$ $b, 1, 11$

$a, 1, \lambda$ $\quad$ $b, 0, \lambda$



Grammar production: $(q_0 z q_f) \rightarrow \lambda$

Resulting Grammar:

$(q_0 z q_f)$: start variable

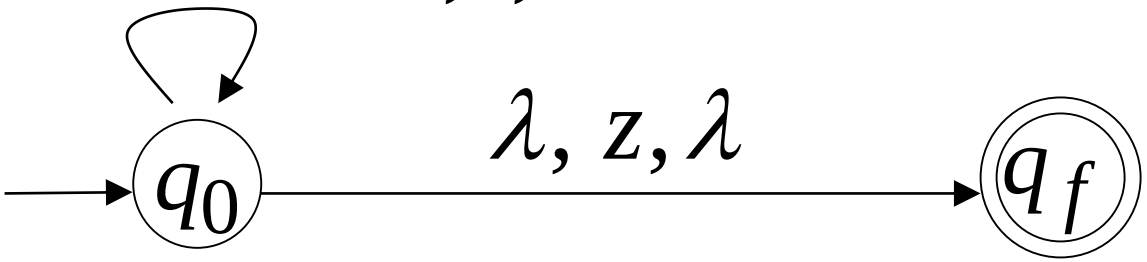
$(q_0 z q_0) \rightarrow b(q_0 1 q_0)(q_0 z q_0) \mid \cancel{b(q_0 1 q_f)(q_f z q_0)}$

$(q_0 z q_f) \rightarrow b(q_0 1 q_0)(q_0 z q_f) \mid \cancel{b(q_0 1 q_f)(q_f z q_f)}$

$(q_0 1 q_0) \rightarrow b(q_0 1 q_0)(q_0 1 q_0) \mid \cancel{b(q_0 1 q_f)(q_f 1 q_0)}$

$(q_0 1 q_f) \rightarrow b(q_0 1 q_0)(q_0 1 q_f) \mid \cancel{b(q_0 1 q_f)(q_f 1 q_f)}$

$a,\ z, 0z$	$b,\ z, 1z$
$a, 0, 00$	$b, 1, 11$
$a, 1, \lambda$	$b, 0, \lambda$



$(q_0 z q_0) \rightarrow a(q_0 0 q_0)(q_0 z q_0) \mid \cancel{a(q_0 0 q_f)(q_f z q_0)}$

$(q_0 z q_f) \rightarrow a(q_0 0 q_0)(q_0 z q_f) \mid \cancel{a(q_0 0 q_f)(q_f z q_f)}$

$(q_0 0 q_0) \rightarrow a(q_0 0 q_0)(q_0 0 q_0) \mid \cancel{a(q_0 0 q_f)(q_f 0 q_0)}$

$(q_0 0 q_f) \rightarrow a(q_0 0 q_0)(q_0 0 q_f) \mid \cancel{a(q_0 0 q_f)(q_f 0 q_f)}$

$(q_0 1 q_0) \rightarrow a$

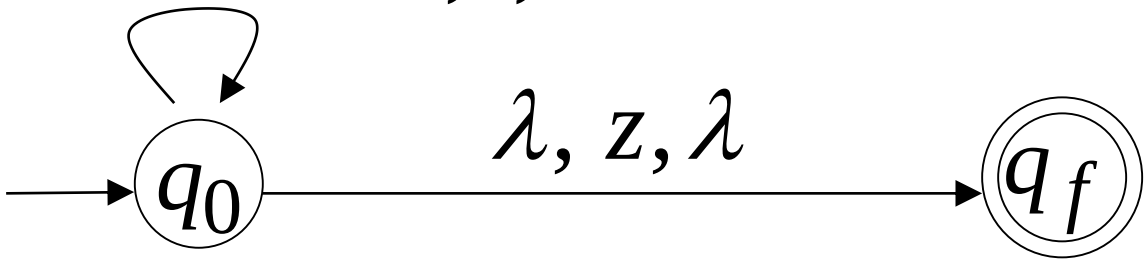
$(q_0 0 q_0) \rightarrow b$

$(q_0 z q_f) \rightarrow \lambda$

$a, z, 0z$ $\quad b, z, 1z$

$a, 0, 00$ $\quad b, 1, 11$

$a, 1, \lambda$ $\quad b, 0, \lambda$



Derivation of string $abba$

$$(q_0 z q_f) \Rightarrow a(q_0 0 q_0)(q_0 z q_f) \Rightarrow$$

$$ab(q_0 z q_f) \Rightarrow$$

$$abb(q_0 1 q_0)(q_0 z q_f) \Rightarrow$$

$$abba(q_0 z q_f) \Rightarrow abba$$

Example 7.8

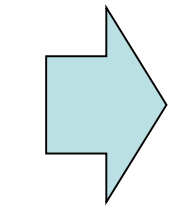
- Consider the NPDA with transitions

$$\delta(q_0, a, z) = \{(q_0, Az)\}$$

$$\delta(q_0, a, A) = \{(q_0, A)\}$$

$$\delta(q_0, b, A) = \{(q_1, \lambda)\}$$

$$\delta(q_1, \lambda, z) = \{(q_2, \lambda)\}$$



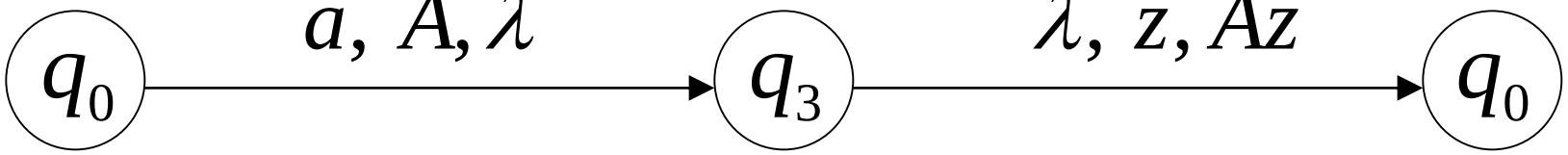
$$\delta(q_0, a, z) = \{(q_0, Az)\}$$

$$\delta(q_3, \lambda, z) = \{(q_0, Az)\}$$

$$\delta(q_0, a, A) = \{(q_3, \lambda)\}$$

$$\delta(q_0, b, A) = \{(q_1, \lambda)\}$$

$$\delta(q_1, \lambda, z) = \{(q_2, \lambda)\}$$



Example 7.8

$$\delta(q_0, a, z) = \{(q_0, Az)\}$$

$$\delta(q_3, \lambda, z) = \{(q_0, Az)\}$$

$$\delta(q_0, a, A) = \{(q_3, \lambda)\}$$

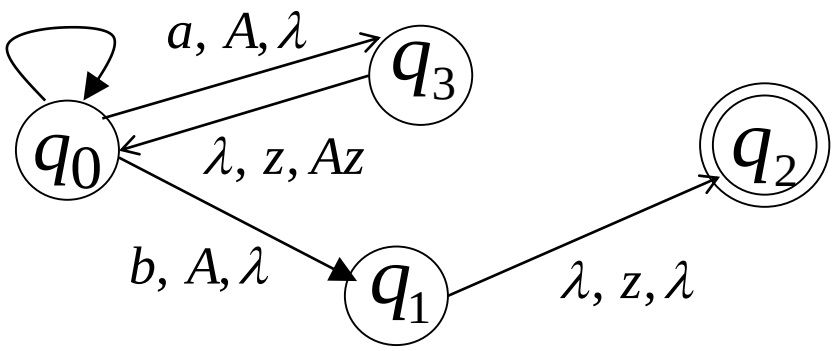
$$\delta(q_0, b, A) = \{(q_1, \lambda)\}$$

$$\delta(q_1, \lambda, z) = \{(q_2, \lambda)\}$$

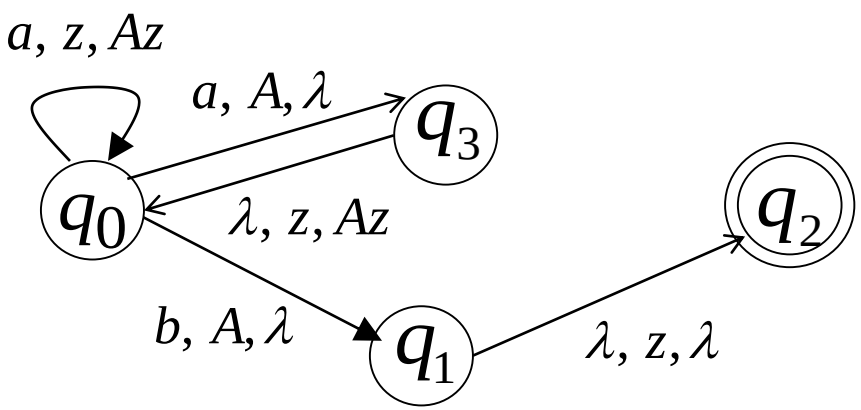
$$(q_0 A q_3) \rightarrow a$$

$$(q_0 A q_1) \rightarrow b$$

$$(q_1 z q_2) \rightarrow \lambda$$



$$\delta(q_0, a, z) = \{(q_0, Az)\}$$



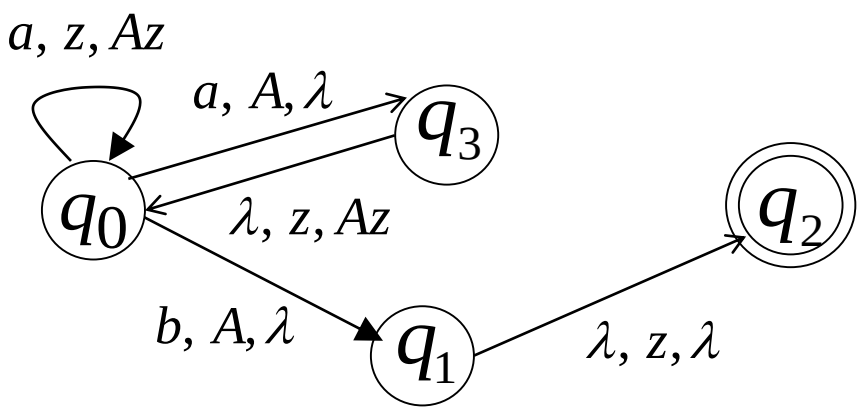
$$(q_0 z q_0) \rightarrow a(q_0 A q_0)(q_0 z q_0) \mid a(q_0 A q_1)(q_1 z q_0)$$
$$a(q_0 A q_2)(q_2 z q_0) \mid a(q_0 A q_3)(q_3 z q_0)$$

$$(q_0 z q_1) \rightarrow a(q_0 A q_0)(q_0 z q_1) \mid a(q_0 A q_1)(q_1 z q_1)$$
$$a(q_0 A q_2)(q_2 z q_1) \mid a(q_0 A q_3)(q_3 z q_1)$$

$$(q_0 z q_2) \rightarrow a(q_0 A q_0)(q_0 z q_2) \mid a(q_0 A q_1)(q_1 z q_2)$$
$$a(q_0 A q_2)(q_2 z q_2) \mid a(q_0 A q_3)(q_3 z q_2)$$

$$(q_0 z q_3) \rightarrow a(q_0 A q_0)(q_0 z q_3) \mid a(q_0 A q_1)(q_1 z q_3)$$
$$a(q_0 A q_2)(q_2 z q_3) \mid a(q_0 A q_3)(q_3 z q_3)$$

$$\delta(q_3, \lambda, z) = \{(q_0, Az)\}$$



$$(q_3 z q_0) \rightarrow (q_0 A q_0)(q_0 z q_0) \mid (q_0 A q_1)(q_1 z q_0)$$
$$(q_0 A q_2)(q_2 z q_0) \mid (q_0 A q_3)(q_3 z q_0)$$

$$(q_3 z q_1) \rightarrow (q_0 A q_0)(q_0 z q_1) \mid (q_0 A q_1)(q_1 z q_1)$$
$$(q_0 A q_2)(q_2 z q_1) \mid (q_0 A q_3)(q_3 z q_1)$$

$$(q_3 z q_2) \rightarrow (q_0 A q_0)(q_0 z q_2) \mid (q_0 A q_1)(q_1 z q_2)$$
$$(q_0 A q_2)(q_2 z q_2) \mid (q_0 A q_3)(q_3 z q_2)$$

$$(q_3 z q_3) \rightarrow (q_0 A q_0)(q_0 z q_3) \mid (q_0 A q_1)(q_1 z q_3)$$
$$(q_0 A q_2)(q_2 z q_3) \mid (q_0 A q_3)(q_3 z q_3)$$

In general:

$$(q_i A q_j) \overset{*}{\Rightarrow} w$$

if and only if

the NPDA goes from q_i to q_j
by reading string w and
the stack doesn't change below A
and then A is removed from stack

Therefore:

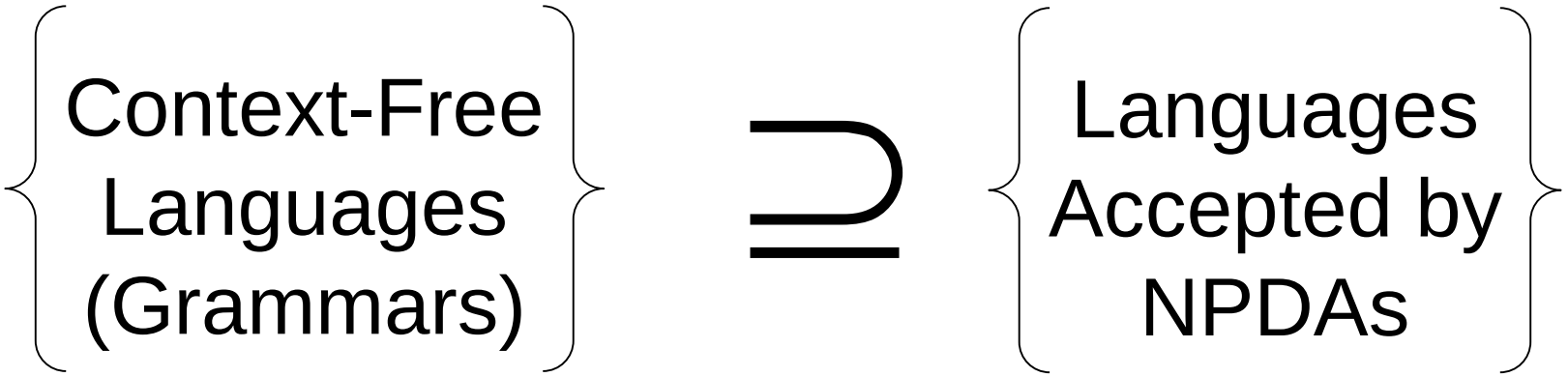
$$(q_0 z q_f) \overset{*}{\Rightarrow} w$$

if and only if

w is accepted by the NPDA

Therefore:

For any NPDA
there is a context-free grammar
that accepts the same language



Outline



Nondeterministic Pushdown Automata



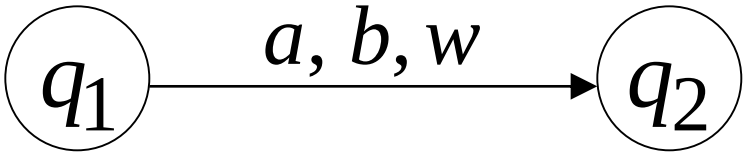
Pushdown Automata and Context-Free Languages

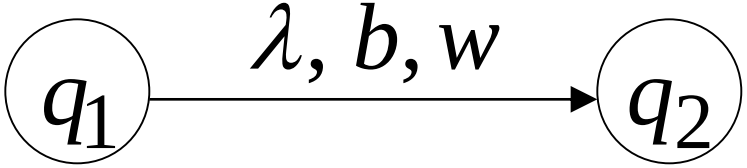


Deterministic Pushdown Automata and Deterministic CFLs

Deterministic PDA: DPDA

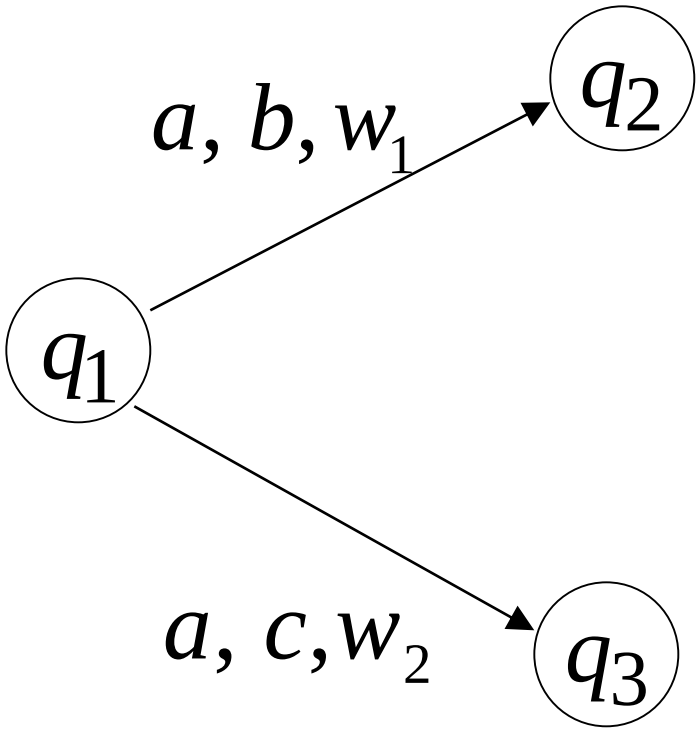
Allowed transitions:

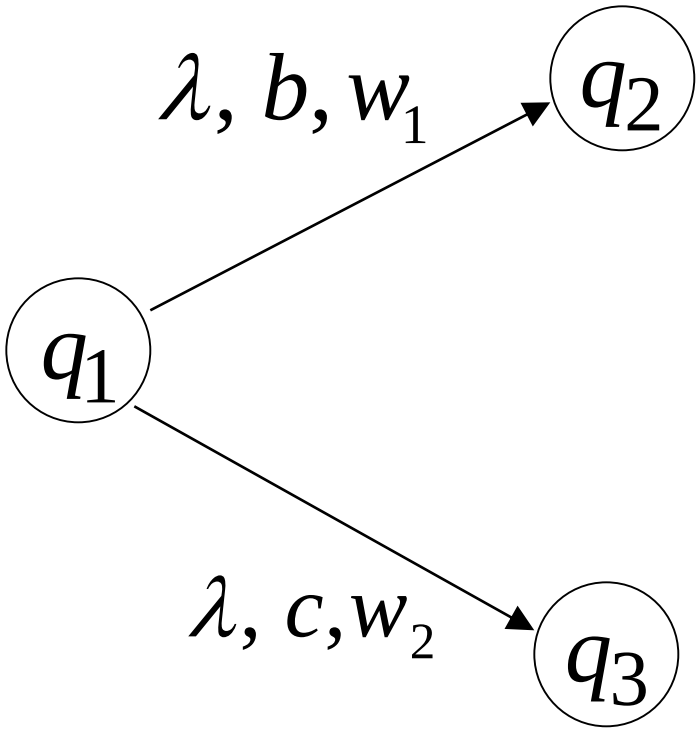




(deterministic choices)

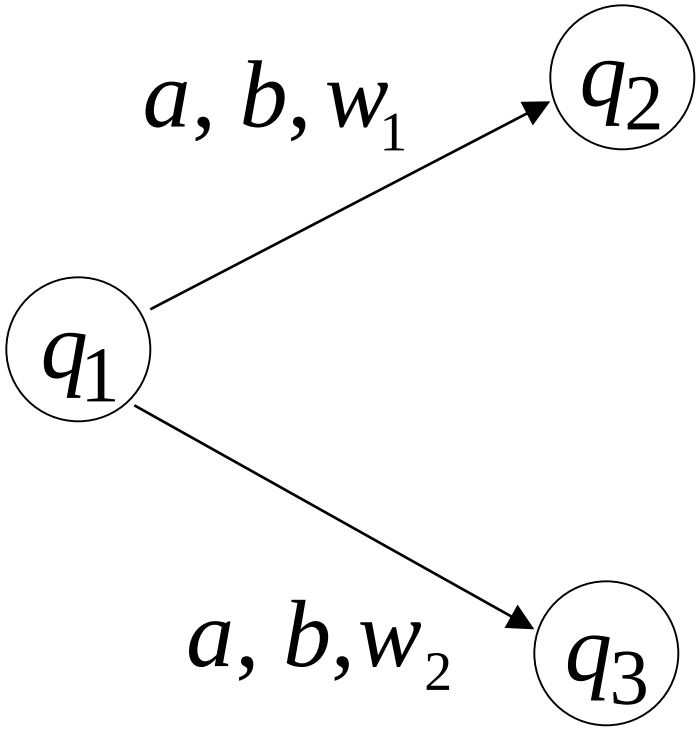
Allowed transitions:

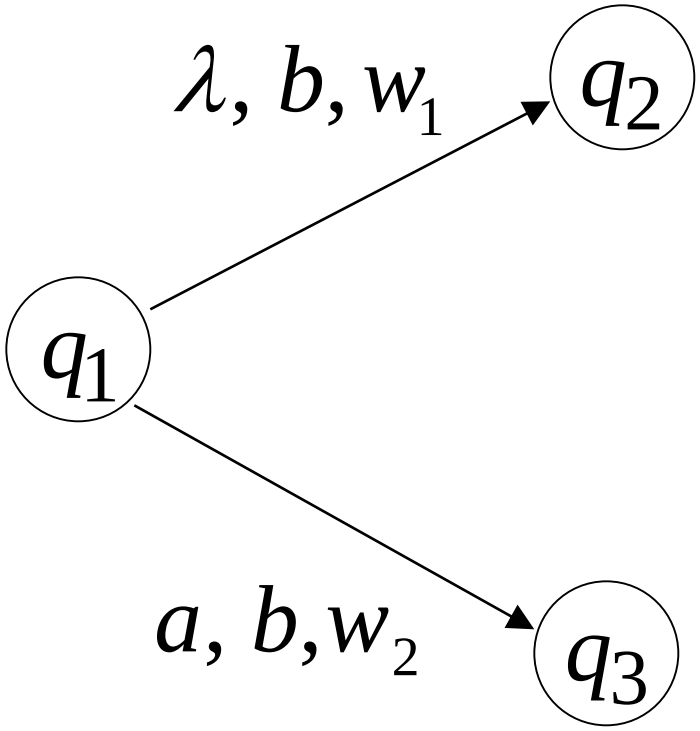




(deterministic choices)

Not allowed:





(non deterministic choices)

Deterministic PDA

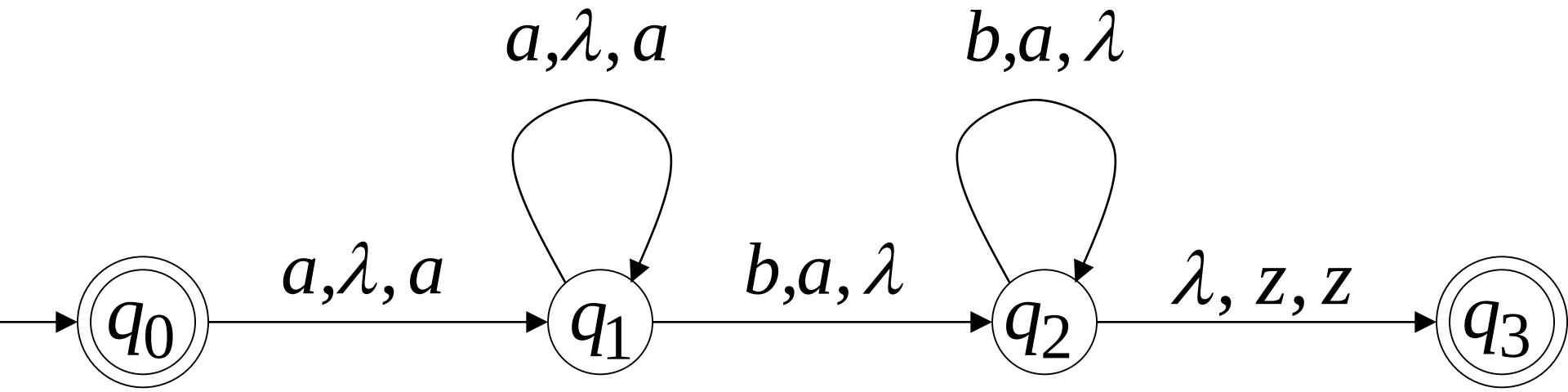
- $\delta(q, a, b)$ contains at most one element
- If $\delta(q, \lambda, b)$ is not empty, then $\delta(q, c, b)$ must be empty for every $c \in \Sigma$

- λ transition is possible
- Some transitions may be to the empty set

- At all times at most one possible move

DPDA example

$$L(M) = \{a^n b^n : n \geq 0\}$$



The language $L(M) = \{a^n b^n : n \geq 0\}$

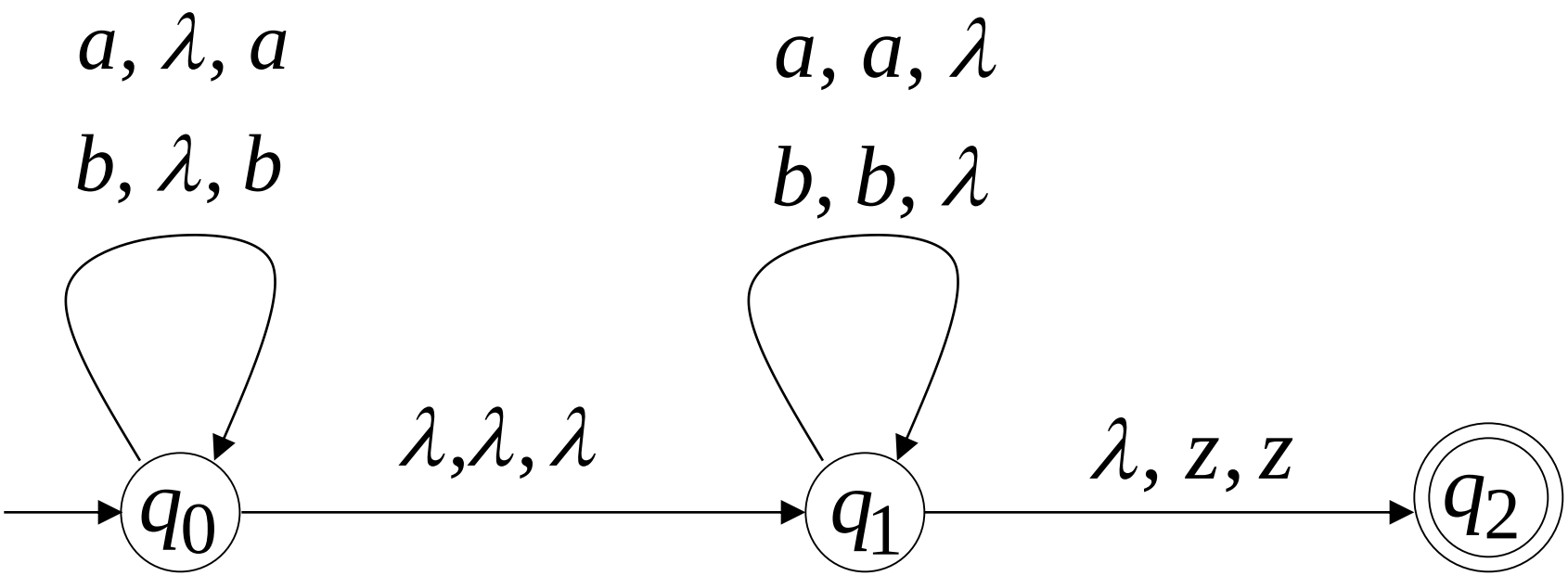
is **deterministic context-free**

Definition:

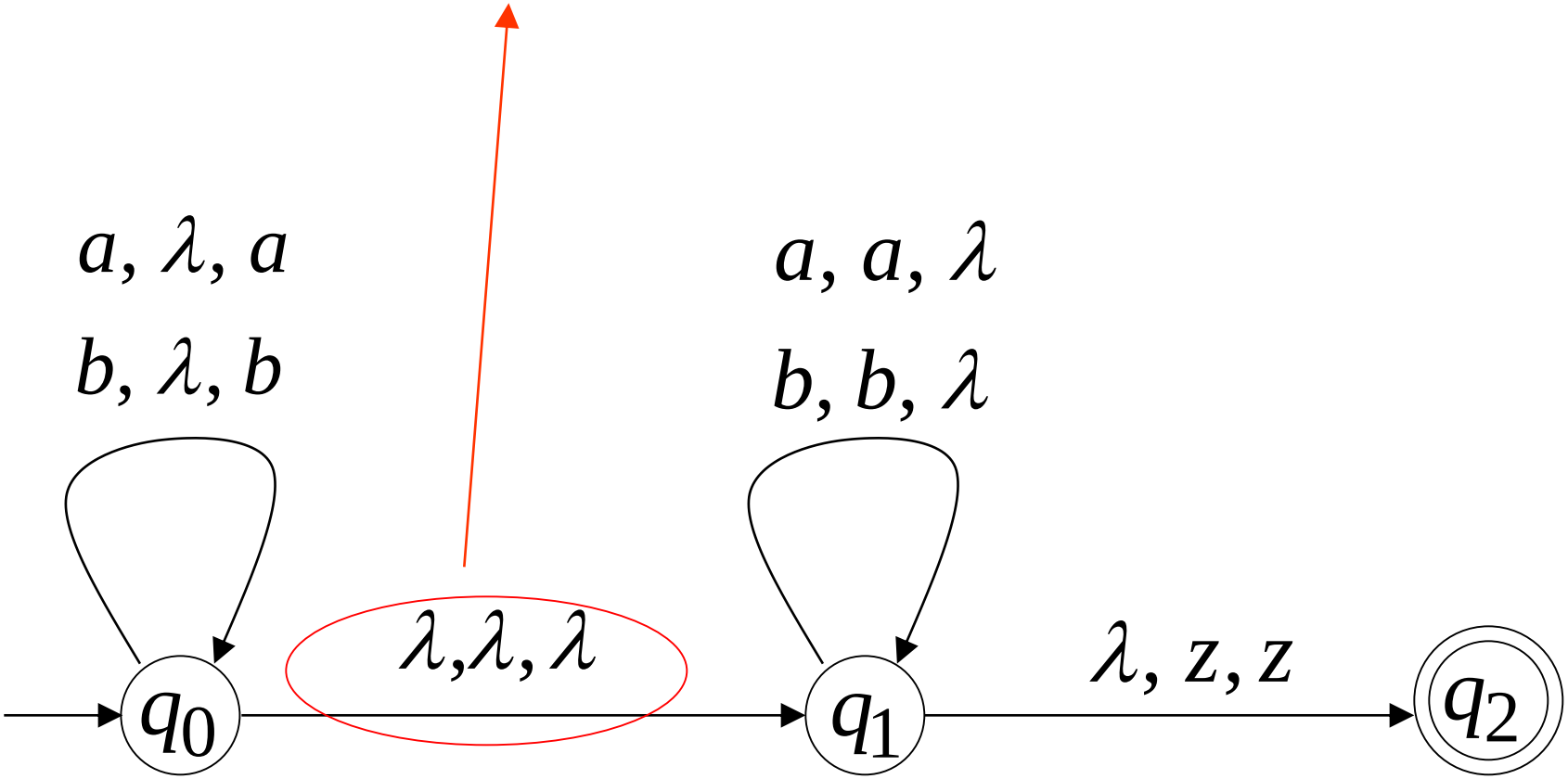
A language L is **deterministic context-free** if there exists some DPDA that accepts it

Example of Non-DPDA (NPDA)

$$L(M) = \{ww^R\}$$



Not allowed in DPDAs



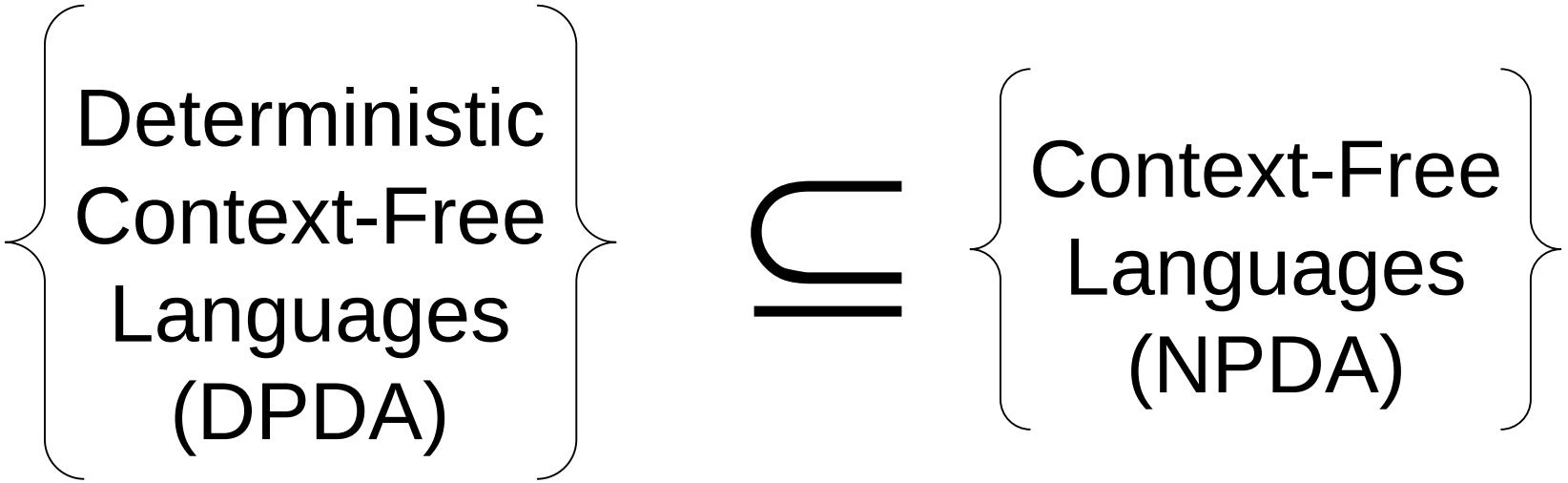
NPDAs

Have More Power than

DPDAs

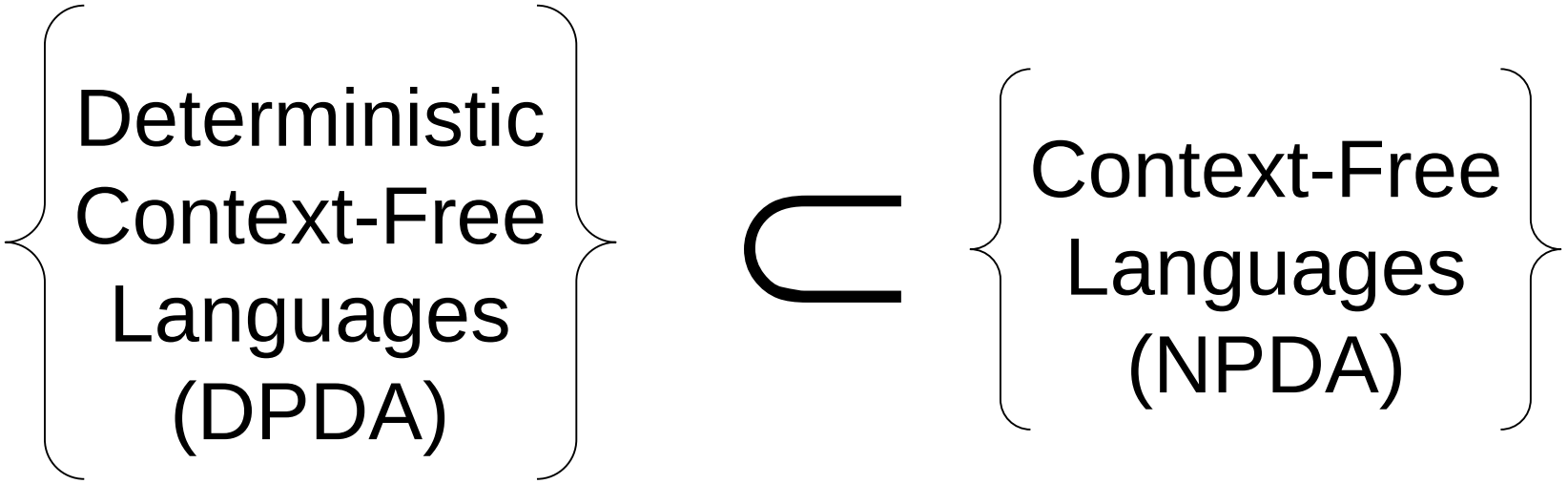
There are context-free languages that are not deterministic

It holds that:



Since every DPDA is also an NPDA

We will actually show:



We will show that there exists a context-free language L which is not accepted by any DPDA

The language is:

$$L = \{a^n b^n\} \cup \{a^n b^{2n}\} \qquad n \geq 0$$

We will show:

- L is context-free
- L is **not** deterministic context-free

$$L = \{a^n b^n\} \cup \{a^n b^{2n}\}$$

Language L is context-free

Context-free grammar for L :

$S \to S_1 \mid S_2$ $\qquad \{a^n b^n\} \cup \{a^n b^{2n}\}$

$S_1 \to aS_1 b \mid \lambda$ $\qquad \{a^n b^n\}$

$S_2 \to aS_2 bb \mid \lambda$ $\qquad \{a^n b^{2n}\}$

Theorem:

The language $L = \{a^n b^n\} \cup \{a^n b^{2n}\}$

is **not** deterministic context-free

(there is **no** DPDA that accepts L)

Proof: Assume for contradiction that

$$L = \{a^n b^n\} \cup \{a^n b^{2n}\}$$

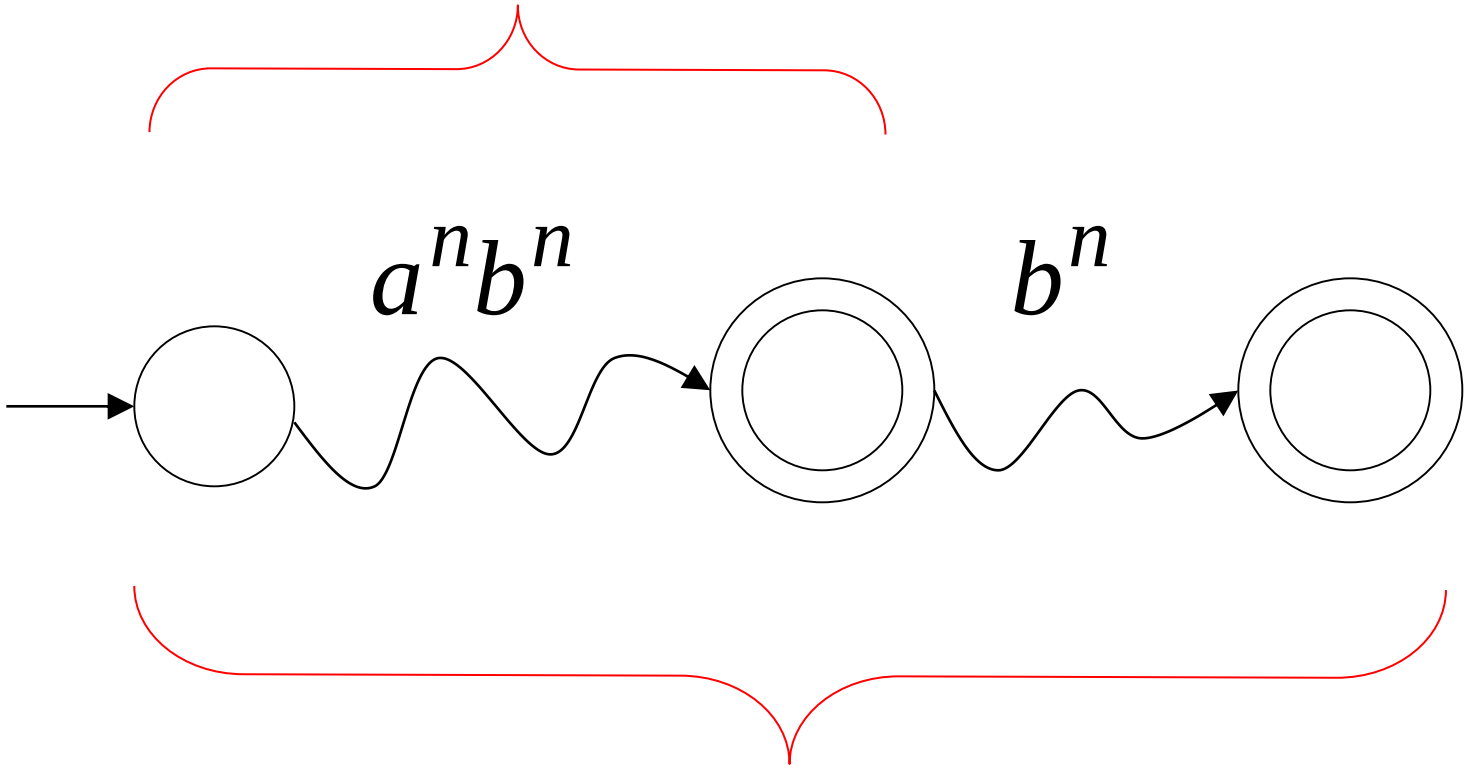
is deterministic context free

Therefore:

there is a DPDA M that accepts L

DPDA M with $L(M) = \{a^n b^n\} \cup \{a^n b^{2n}\}$

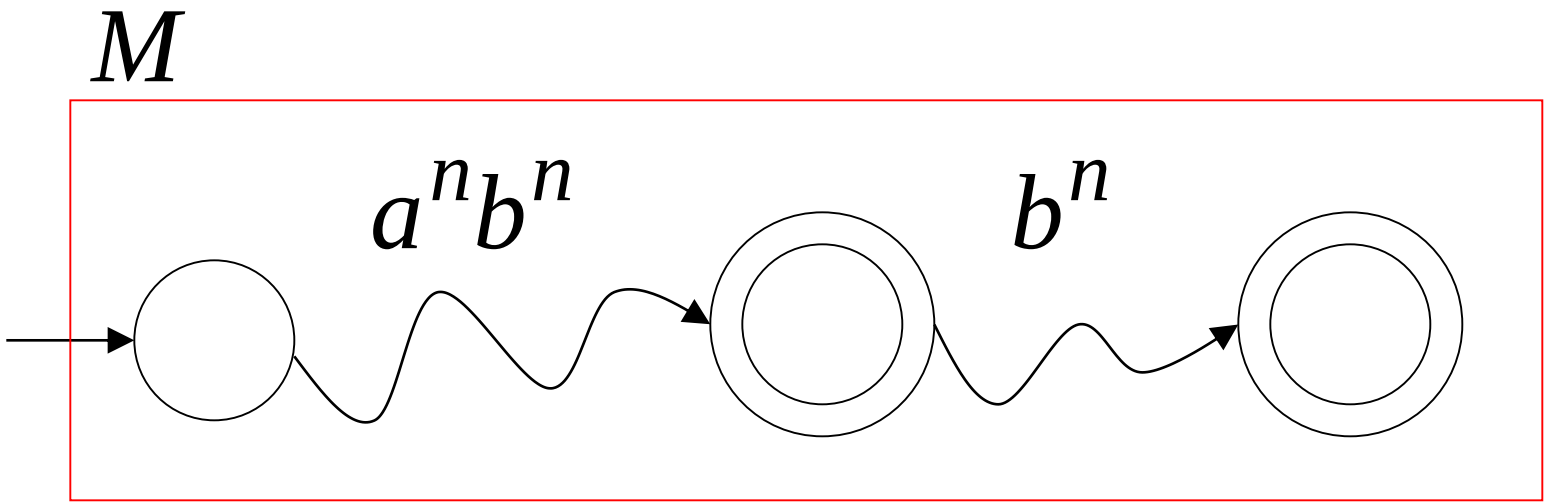
accepts $a^n b^n$



accepts $a^n b^{2n}$

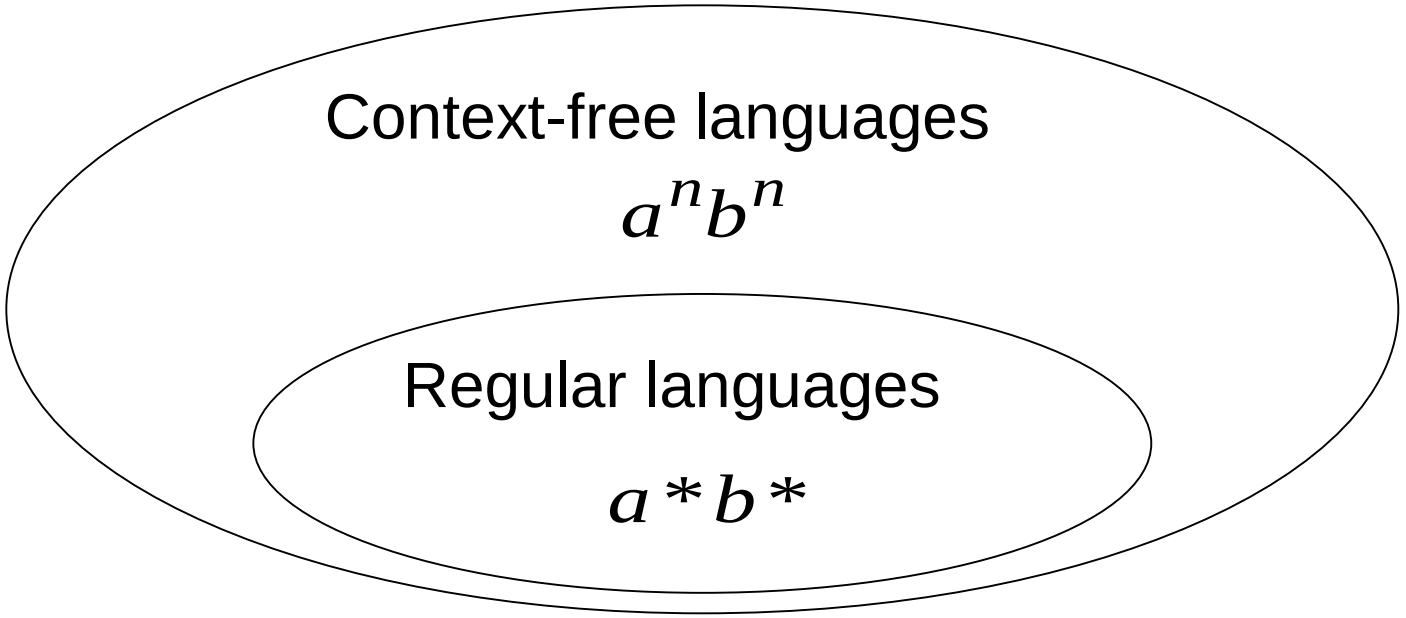
DPDA M with $L(M) = \{a^n b^n\} \cup \{a^n b^{2n}\}$

Such a path exists because of the determinism



Fact 1:

The language $\{a^n b^n c^n\}$ is not context-free



(we will prove this at a later class using pumping lemma for context-free languages)

Fact 2:

The language
is not context-free

$$L \cup \{a^n b^n c^n\}$$

$$(L = \{a^n b^n\} \cup \{a^n b^{2n}\})$$

(we can prove this using pumping lemma
for context-free languages)

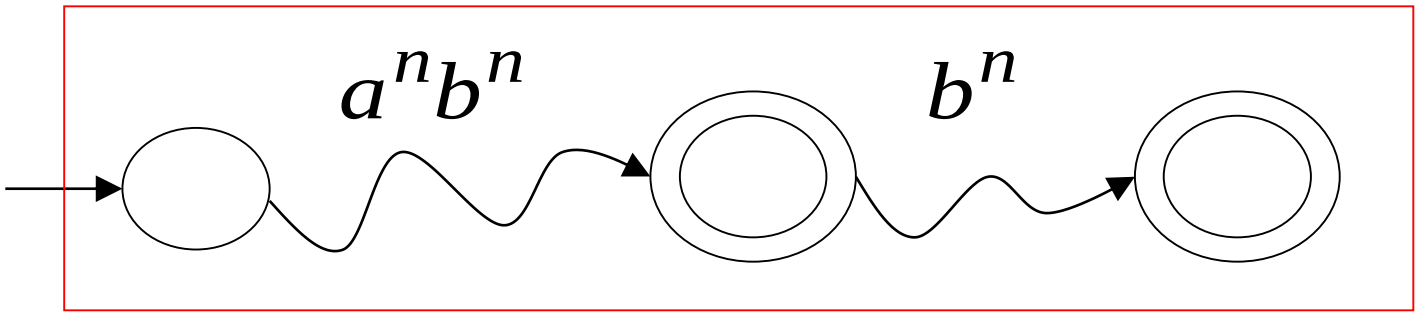
We will construct a DPDA that accepts:

$$L \cup \{a^n b^n c^n\}$$

$$(L = \{a^n b^n\} \cup \{a^n b^{2n}\})$$

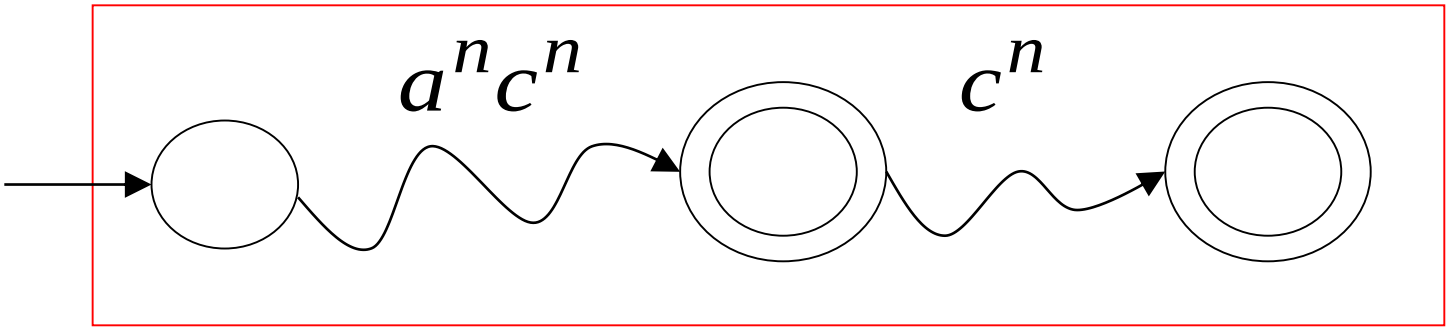
which is a contradiction!

M $L(M) = \{a^n b^n\} \cup \{a^n b^{2n}\}$



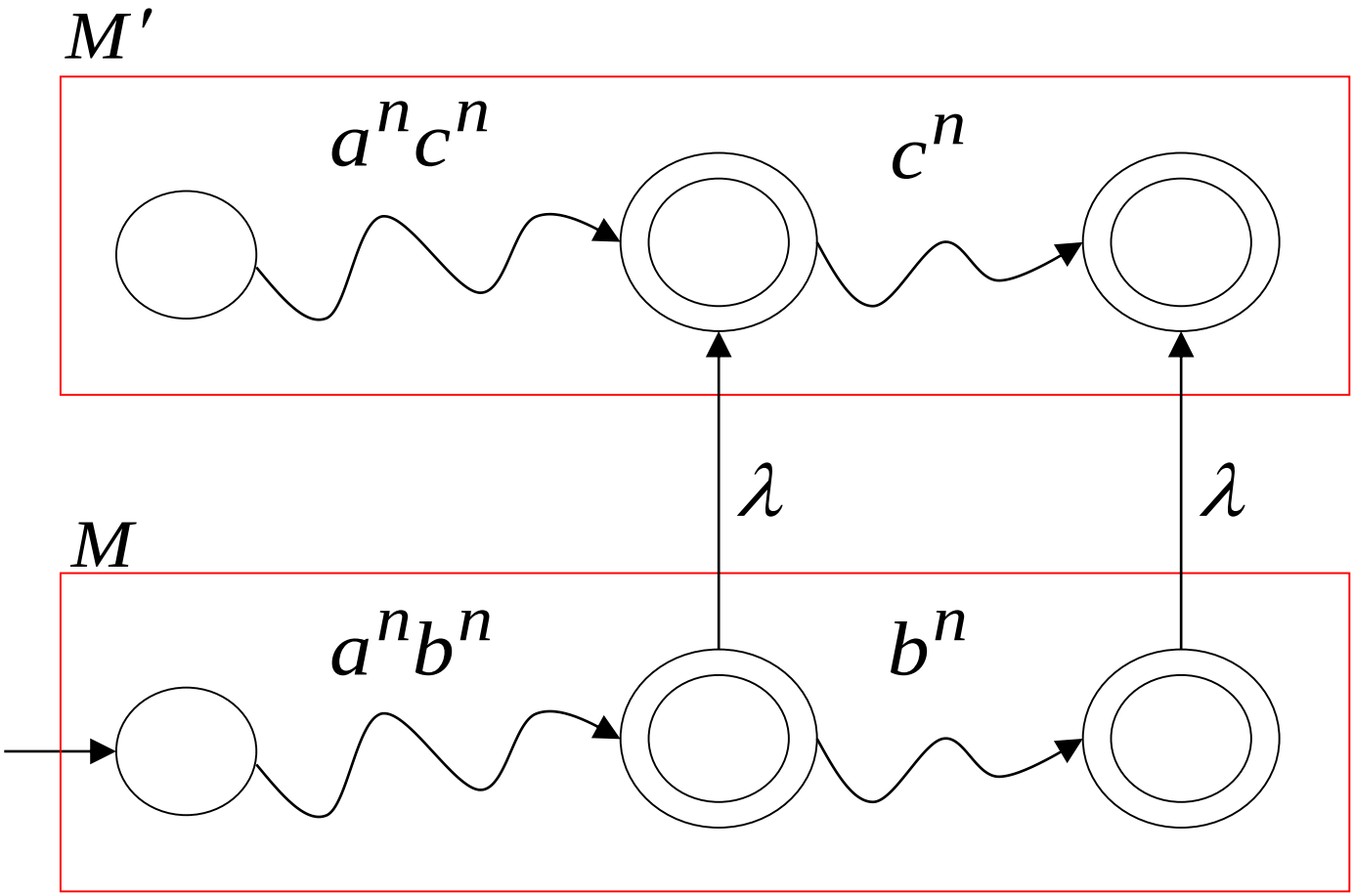
Modify M Replace b with c

M' $L(M') = \{a^n c^n\} \cup \{a^n c^{2n}\}$



The DPDA that accepts $L \cup \{a^n b^n c^n\}$

Connect final states of M'
with final states of M



Since $L \cup \{a^n b^n c^n\}$ is accepted by a DPDA

it is context-free

Contradiction!

(since $L \cup \{a^n b^n c^n\}$ is not context-free)

Therefore:

Not deterministic context free

$$L = \{a^n b^n\} \cup \{a^n b^{2n}\}$$

There is **no** DPDA that accepts it

End of Proof

Quiz

Show that $L = \{a^n b^m, n > m\}$ is a deterministic context-free language.

$$\begin{aligned}
\delta(q_0, a, z) &= \{(q_1, Yz)\}, \\
\delta(q_1, a, Y) &= \{(q_1, A)\}, \\
\delta(q_1, a, A) &= \{(q_1, AA)\}, \\
\delta(q_1, b, A) &= \{(q_2, \lambda)\}, \\
\delta(q_2, b, A) &= \{(q_2, \lambda)\}, \qquad \text{where } F = \{q_1, q_2\}.
\end{aligned}$$

Show that $L = \{a^n b^m, n < m\}$ is a deterministic context-free language.

$$\begin{aligned}
\delta(q_0, b, z) &= \{(q_2, z)\}, \\
\delta(q_0, a, z) &= \{(q_0, Az)\}, \\
\delta(q_0, a, A) &= \{(q_0, AA)\}, \\
\delta(q_0, b, A) &= \{(q_1, \lambda)\}, \\
\delta(q_1, b, A) &= \{(q_1, \lambda)\}, \\
\delta(q_1, b, z) &= \{(q_2, z)\}, \quad \text{where } F = \{q_2\}. \\
\delta(q_2, b, z) &= \{(q_2, z)\},
\end{aligned}$$